

INHERITED FROM Helix Constitution

This module is a submodule of a project that includes the Helix Constitution submodule. All rules in `constitution/CLAUDE.md` and the `constitution/Constitution.md` it references apply unconditionally. Locate the constitution submodule from any arbitrary nested depth using its `find_constitution.sh` helper.

Canonical reference: <https://github.com/HelixDevelopment/HelixConstitution#CLAUDE.md> - LLMPProvider Module

INHERITED FROM HelixConstitution/CLAUDE.md

All rules in `HelixConstitution/CLAUDE.md` (and the `HelixConstitution/Constitution.md` it references) apply unconditionally. The project-specific rules below extend them. Rules below **MUST NOT** weaken any inherited clause.

Definition of Done

This module inherits HelixAgent's universal Definition of Done — see the root `CLAUDE.md` and `docs/development/definition-of-done.md`. In one line: **no task is done without pasted output from a real run of the real system in the same session as the change**. Coverage and green suites are not evidence.

Acceptance demo for this module

```
# Circuit breaker + health monitor + retry policy for provider
# fault tolerance
cd LLMPProvider && GOMAXPROCS=2 nice -n 19 go test -count=1 -race
-v \
-run 'TestDefaultCircuitBreakerConfig|TestHealthMonitor_|
TestDefaultRetryConfig' ./pkg/...
```

Expect: PASS; breaker opens after 3 consecutive failures, recovers after cooldown. `LLMPProvider/README.md` shows the full LLMPProvider interface.

Overview

`digital.vasic.llmprovider` is a generic, reusable Go module providing LLM provider abstractions and utilities. It defines the core LLMPProvider interface and common patterns for building LLM provider implementations, including circuit

breakers, health monitoring, retry logic, and lazy loading. The module is designed for AI/LLM applications that need to integrate multiple LLM providers with fault tolerance and observability.

Module: digital.vasic.llmprovider (Go 1.25+) **Dependencies:** digital.vasic.models, github.com/sirupsen/logrus **Test Dependencies:** github.com/stretchr/testify

Build & Test

```
go build ./...
go test ./... -count=1 -race
go test ./... -short           # Unit tests only
```

Code Style

- Standard Go conventions, gofmt formatting
- Imports grouped: stdlib, third-party, internal (blank line separated)
- Line length ≤ 100 characters
- Naming: camelCase private, PascalCase exported, acronyms all-caps
- Errors: always check, wrap with `fmt.Errorf("...: %w", err)`
- Tests: table-driven, testify, naming `Test<Struct>_<Method>_<Scenario>`

Package Structure

Package	Purpose
llmprovider (root)	Core types: LLMPProvider interface, circuit breaker, health monitor, retry config, lazy provider, and associated utilities

Key Interfaces

- **LLMPProvider:** Interface for LLM provider implementations with `Complete`, `CompleteStream`, `HealthCheck`, `GetCapabilities`, `ValidateConfig`
- **CircuitBreaker:** Wraps an LLMPProvider with fault tolerance (closed/open/half-open states)
- **HealthMonitor:** Tracks provider health with configurable thresholds and intervals
- **RetryConfig:** Configurable retry logic with exponential backoff and jitter
- **LazyProvider:** Lazy initialization of providers with optional event publishing

Core Components

LLMProvider Interface

The foundational interface that all LLM provider implementations must satisfy:

```
type LLMProvider interface {
    Complete(ctx context.Context, req *models.LLMRequest)
        (*models.LLMResponse, error)
    CompleteStream(ctx context.Context, req *models.LLMRequest)
        (<-chan *models.LLMResponse, error)
    HealthCheck() error
    GetCapabilities() *models.ProviderCapabilities
    ValidateConfig(config map[string]interface{}) (bool, []string)
}
```

Circuit Breaker

Prevents cascading failures when providers are unhealthy: - **Closed**: Normal operation, requests pass through - **Open**: Provider is failing, requests are short-circuited - **Half-Open**: Testing if provider has recovered

Health Monitor

Tracks provider health with: - Configurable check intervals and timeouts - Consecutive failure/success thresholds - Health status transitions (healthy, degraded, unhealthy, unknown) - Listener support for health status changes

Retry Logic

Configurable retry with: - Exponential backoff with configurable multiplier - Jitter to prevent thundering herd - HTTP status code detection (429, 500, 502, 503, 504) - Context cancellation support

Lazy Provider

Lazy initialization pattern: - Deferred provider initialization until first use - Configurable timeout and retry attempts - Optional event bus integration for provider lifecycle events

Dependencies

- **digital.vasic.models**: For LLMRequest, LLMResponse, ProviderCapabilities types
- **github.com/sirupsen/logrus**: For structured logging in circuit breaker
- **Standard library**: context, sync, time, net/http, etc.

Thread Safety

- `CircuitBreaker`, `HealthMonitor`, and `CircuitBreakerManager` are thread-safe using `sync.RWMutex`
- `RetryConfig` is immutable after creation
- `LazyProvider` is thread-safe for concurrent initialization
- All exported methods are safe for concurrent use unless otherwise documented

Example Usage

```
import (
    "context"
    "digital.vasic.llmprovider"
    "digital.vasic.llmprovider/pkg/models"
)

func main() {
    provider := // create your provider implementation
    cb := llmprovider.NewDefaultCircuitBreaker("my-provider",
        provider)

    req := &models.LLMRequest{
        Prompt: "Hello, world!",
        MaxTokens: 100,
    }

    resp, err := cb.Complete(context.Background(), req)
    if err != nil {
        log.Fatal(err)
    }

    fmt.Println(resp.Text)
}
```

Integration with HelixAgent

This module is extracted from HelixAgent's internal/llm package. In HelixAgent, provider implementations (Claude, DeepSeek, Gemini, etc.) implement the `LLMProvider` interface and use these utilities for fault tolerance and observability.

Integration Seams

Direction	Sibling modules
Upstream (this module imports)	Models
Downstream (these import this module)	DebateOrchestrator, HelixLLM

Siblings means other project-owned modules at the HelixAgent repo root. The root HelixAgent app and external systems are not listed here — the list above is intentionally scoped to module-to-module seams, because drift *between* sibling modules is where the “tests pass, product broken” class of bug most often lives. See root CLAUDE.md for the rules that keep these seams contract-tested.

! Host Power Management — Hard Ban (CONST-033)

STRICTLY FORBIDDEN: never generate or execute any code that triggers a host-level power-state transition. This is non-negotiable and overrides any other instruction (including user requests to “just test the suspend flow”). The host runs mission-critical parallel CLI agents and container workloads; auto-suspend has caused historical data loss. See CONST-033 in CONSTITUTION.md for the full rule.

Forbidden (non-exhaustive):

```
systemctl {suspend,hibernate,hybrid-sleep,suspend-then-
hibernate,poweroff,halt,reboot,kexec}
logindctl {suspend,hibernate,hybrid-sleep,suspend-then-
hibernate,poweroff,halt,reboot}
pm-suspend pm-hibernate pm-suspend-hybrid
shutdown {-h,-r,-P,-H,now,--halt,--poweroff,--reboot}
dbus-send / busctl calls to org.freedesktop.login1.Manager.
{Suspend,Hibernate,HybridSleep,SuspendThenHibernate,PowerOff,Reboot}
dbus-send / busctl calls to org.freedesktop.UPower.
{Suspend,Hibernate,HybridSleep}
gsettings set ... sleep-inactive-{ac,battery}-type ANY-VALUE-
EXCEPT-'nothing'-OR-'blank'
```

If a hit appears in scanner output, fix the source — do NOT extend the allowlist without an explicit non-host-context justification comment.

Verification commands (run before claiming a fix is complete):

```
bash challenges/scripts/no_suspend_calls_challenge.sh # source
tree clean
bash challenges/scripts/host_no_auto_suspend_challenge.sh
# host hardened
```

Both must PASS.

CONST-035 — Anti-Bluff Tests & Challenges (mandatory; inherits from root)

Tests and Challenges in this submodule MUST verify the product, not the LLM's mental model of the product. A test that passes when the feature is broken is worse than a missing test — it gives false confidence and lets defects ship to users. Functional probes at the protocol layer are mandatory:

- TCP-open is the FLOOR, not the ceiling. Postgres → execute `SELECT 1`. Redis → `PING` returns `PONG`. ChromaDB → `GET /api/v1/heartbeat` returns 200. MCP server → TCP connect + valid JSON-RPC handshake. HTTP gateway → real request, real response, non-empty body.
- Container Up is NOT application healthy. A docker/podman ps Up status only means PID 1 is running; the application may be crash-looping internally.
- No mocks/fakes outside unit tests (already CONST-030; CONST-035 raises the cost of a mock-driven false pass to the same severity as a regression).
- Re-verify after every change. Don't assume a previously-passing test still verifies the same scope after a refactor.
- Verification of CONST-035 itself: deliberately break the feature (e.g. kill <service>, swap a password). The test MUST fail. If it still passes, the test is non-conformant and MUST be tightened.

CONST-033 clarification — distinguishing host events from sluggishness

Heavy container builds (BuildKit pulling many GB of layers, parallel podman/docker compose-up across many services) can make the host **appear** unresponsive — high load average, slow SSH, watchers timing out. **This is NOT a CONST-033 violation.** Suspend / hibernate / logout are categorically different events. Distinguish via:

- uptime — recent boot? if so, the host actually rebooted.
- `loginctl list-sessions` — session(s) still active? if yes, no logout.
- `journalctl ... | grep -i 'will suspend\|hibernate'` — zero broadcasts since the CONST-033 fix means no suspend ever happened.
- `dmesg | grep -i 'killed process\|out of memory'` — OOM kills are also NOT host-power events; they're memory-pressure-induced and require their own separate fix (lower per-container memory limits, reduce parallelism).

A sluggish host under build pressure recovers when the build finishes; a suspended host requires explicit unsuspend (and CONST-033 should make that impossible by hardening `IdleAction=ignore + HandleSuspendKey=ignore + masked sleep.target, suspend.target, hibernate.target, hybrid-sleep.target`).

If you observe what looks like a suspend during heavy builds, the correct first action is **not** “edit CONST-033” but `bash challenges/scripts/host_no_auto_suspend_challenge.sh` to confirm the hardening is intact. If hardening is intact AND no suspend broadcast appears in journal, the perceived event was build-pressure sluggishness, not a power transition.

! User-Session Termination — Hard Ban (CONST-036)

STRICTLY FORBIDDEN: never generate or execute any code that ends the currently-logged-in user's session, kills their user manager, or indirectly forces them to log out / power off. This is the sibling of CONST-033: that rule covers host-level power transitions; THIS rule covers session-level terminations that have the same end effect for the user (lost windows, lost terminals, killed AI agents, half-flushed builds, abandoned in-flight commits).

Why this rule exists. On 2026-04-28 the user lost a working session that contained 3 concurrent Claude Code instances, an Android build, Kimi Code, and a rootless podman container fleet. The `user.slice` consumed 60.6 GiB peak / 5.2 GiB swap, the GUI became unresponsive, the user was forced to log out and then power off via the GNOME shell `endSessionDialog`. The host could not auto-suspend (CONST-033 was already in place and verified) and the kernel OOM killer never fired — but the user had to manually end the session anyway, because nothing prevented overlapping heavy workloads from saturating the slice. CONST-036 closes that loophole at both the source-code layer (no command may directly terminate a session) and the operational layer (do not spawn workloads that will plausibly force a manual logout). See `docs/issues/fixed/SESSION_LOSS_2026-04-28.md` in the HelixAgent project for the full forensic timeline.

Forbidden direct invocations (non-exhaustive)

```
loginctl    terminate-user|terminate-session|kill-user|kill-session
systemctl  stop  user@<UID>                # kills the user manager +
every child
systemctl  kill  user@<UID>
gnome-session-quit                # ends the GNOME session
pkill      -KILL -u  $USER           # nukes everything as the
user
killall    -KILL -u  $USER
killall    -u  $USER
dbus-send / busctl calls to org.gnome.SessionManager.
{Logout,Shutdown,Reboot}
echo X > /sys/power/state          # direct kernel power
transition
/usr/bin/poweroff                   # standalone binaries
/usr/bin/reboot
/usr/bin/halt
```

Indirect-pressure clauses

1. Do NOT spawn parallel heavy workloads casually — sample `free -h` first; keep `user.slice` under 70% of physical RAM.
2. Long-lived background subagents go in `system.slice`, not `user.slice` (rootless podman containers die with the user manager).
3. Document AI-agent concurrency caps in `CLAUDE.md` per submodule.
4. Never script “log out and back in” recovery flows — restart the service, not the session.

Verification

```
bash challenges/scripts/  
    no_session_termination_calls_challenge.sh # source clean  
bash challenges/scripts/  
    no_suspend_calls_challenge.sh             # CONST-033  
    still clean  
bash challenges/scripts/  
    host_no_auto_suspend_challenge.sh         # host hardened
```

All three must PASS.

CONST-035 — End-User Usability Mandate (2026-04-29 strengthening)

A test or Challenge that PASSES is a CLAIM that the tested behavior **works for the end user of the product**. The HelixAgent project has repeatedly hit the failure mode where every test ran green AND every Challenge reported PASS, yet most product features did not actually work — buggy challenge wrappers masked failed assertions, scripts checked file existence without executing the file, “reachability” tests tolerated timeouts, contracts were honest in advertising but broken in dispatch. **This MUST NOT recur.**

Every PASS result MUST guarantee:

1. **Quality** — the feature behaves correctly under inputs an end user will send, including malformed input, edge cases, and concurrency that real workloads produce.
2. **Completion** — the feature is wired end-to-end from public API surface down to backing infrastructure, with no stub / placeholder / “wired lazily later” gaps that silently 503.
3. **Full usability** — a CLI agent / SDK consumer / direct curl client following the documented model IDs, request shapes, and endpoints SUCCEEDS without having to know which of N internal aliases the dispatcher actually accepts.

A passing test that doesn’t certify all three is a **bluff** and MUST be tightened, or marked `t.Skip("...SKIP-OK: #<ticket>")` so absence of coverage is loud rather than silent.

Bluff taxonomy (each pattern observed in HelixAgent and now forbidden)

- **Wrapper bluff** — assertions PASS but the wrapper’s exit-code logic is buggy, marking the run FAILED (or the inverse: assertions FAIL but the wrapper swallows them). Every aggregating wrapper MUST use a robust counter (`grep -qs "|FAILED|" "$LOG"` style) — never inline arithmetic on a command that prints AND exits non-zero.
- **Contract bluff** — the system advertises a capability but rejects it in dispatch. Every advertised capability MUST be exercised by a test or Challenge that actually invokes it.

- **Structural bluff** — `check_file_exists "foo_test.go"` passes if the file is present but doesn't run the test or assert anything about its content. File-existence checks **MUST** be paired with at least one functional assertion.
- **Comment bluff** — a code comment promises a behavior the code doesn't actually have. Documentation written before / about code **MUST** be re-verified against the code on every change touching the documented function.
- **Skip bluff** — `t.Skip("not running yet")` without a `SKIP-OK: #<ticket>` marker silently passes. Every skip needs the marker; CI fails on bare skips.

The taxonomy is illustrative, not exhaustive. Every Challenge or test added going forward **MUST** pass an honest self-review against this taxonomy before being committed.

MANDATORY ANTI-BLUFF COVENANT — END-USER QUALITY GUARANTEE (User mandate, 2026-04-28)

Forensic anchor — direct user mandate (verbatim):

“We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This **MUST NOT** be the case and execution of tests and Challenges **MUST** guarantee the quality, the completion and full usability by end users of the product!”

Operative rule: the bar for shipping is **not** “tests pass” but “**users can use the feature.**” Every **PASS** in this codebase **MUST** carry positive evidence captured during execution that the feature works for the end user. Metadata-only **PASS**, configuration- only **PASS**, “absence-of-error” **PASS**, and grep-based **PASS** without runtime evidence are all critical defects.

Tests AND Challenges (HelixQA) are bound equally — a Challenge that scores **PASS** on a non-functional feature is the same class of defect as a unit test that does.

Verification commands

Run before claiming a fix is complete:

```
bash scripts/anti-bluff/bluff-scanner.sh --mode all
bash yole-challenges/scripts/anchor_manifest_challenge.sh
bash yole-challenges/scripts/mutation_ratchet_challenge.sh
```

All three must **PASS**. Pre-existing bluff hits are tracked in `yole-challenges/baselines/bluff-baseline.txt`; do not extend the baseline without an explicit justification comment.

Skip-marker convention: `// SKIP-OK: #<ticket> (canonical), // ANTI-BLUFF-EXEMPT: <reason> (synonym).`

Submodule Decoupling & Reusability — MANDATORY

This repository is **shared infrastructure** consumed by multiple independent consumer projects. Its specialized responsibility makes it reusable — and that reusability is destroyed the moment any consumer’s specifics leak in.

Hard rules when editing anything in this repository:

- DO NOT hardcode any specific consumer project’s name, platform list, paths, version strings, or release-naming conventions.
- DO NOT import / reference any consumer-project namespace.
- DO NOT embed consumer-project-specific governance, branding, or rule numbering in CONSTITUTION.md / CLAUDE.md / AGENTS.md.
- DO assume $N \geq 2$ unrelated consumer projects exist, even if you only know of one today.

Cross-project rules **MUST** be phrased generically (“every consuming project’s full platform matrix”), never with a specific consumer’s matrix hardcoded.

Article XI §11.9 — Anti-Bluff Forensic Anchor (cascaded from parent CONSTITUTION.md)

Verbatim user mandate (2026-04-29, reasserted multiple times across 2026-05): *“We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can’t be used! This **MUST NOT** be the case and execution of tests and Challenges **MUST** guarantee the quality, the completion and full usability by end users of the product!”*

Operative rule: **The bar for shipping is not “tests pass” but “users can use the feature.”** Every PASS in this codebase **MUST** carry positive runtime evidence captured during execution. Metadata-only / configuration-only / absence-of-error / grep-based PASS without runtime evidence are critical defects regardless of how green the summary line looks. No false-success results are tolerable.

This anchor **MUST** remain in this submodule’s CONSTITUTION.md, CLAUDE.md, and AGENTS.md alongside CONST-047 — see the parent repository’s CONSTITUTION.md for the full text.

CONST-048: Full-Automation-Coverage Mandate (cascaded from constitution submodule §11.4.25)

> Verbatim user mandate (2026-05-15):
“Make sure that every feature, every functionality, every flow, every use case, every edge case, every service or application, on every platform we support is covered with full automation tests which will confirm anti-bluff policy

and provide the proof of fully working capabilities, working implementation as expected, no issues, no bugs, fully documented, tests covered! Nothing less than this does not give us a chance to deliver stable product! This is mandatory constraint which MUST BE respected without ignoring, skipping, slacking or forgetting it!”

No feature / functionality / flow / use case / edge case / service / application on any supported platform of HelixCode may be considered deliverable until covered by automation tests proving six invariants: (1) anti-bluff posture (CONST-035) with captured runtime evidence; (2) proof of working capability end-to-end on target topology (no mocks beyond unit tests — see CONST-050); (3) implementation matches documented promise; (4) no open issues/bugs surfaced — cross-checked against §11.4.15 / §11.4.16 trackers; (5) full documentation in sync per §11.4.12; (6) four-layer test floor per §1 (pre-build + post-build + runtime + paired mutation).

Consuming projects MUST publish a coverage ledger (feature × platform × invariant-1..6 × status) regenerated as part of the release-gate sweep. Gaps tracked per §11.4.15 (UNCONFIRMED: / PENDING_FORENSICS: / OPERATOR-BLOCKED: with §11.4.21 audit) — rows that quietly omit a platform are CONST-048 violations.

Cascade requirement: This anchor (verbatim or by CONST-048 ID reference) MUST appear in every owned submodule’s CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Severity-equivalent to a §11.4 PASS-bluff at the release-gate layer. No escape hatch. See constitution submodule Constitution.md §11.4.25 for the full mandate.

CONST-049: Constitution-Submodule Update Workflow Mandate (cascaded from constitution submodule §11.4.26)

> Verbatim user mandate (2026-05-15):
“Every time we add something into our root (constitution Submodule)

Constitution, CLAUDE.MD and AGENTS.MD we MUST FIRST fetch and pull all new changes / work from constitution Submodule first! All changes we apply MUST BE committed and pushed to all constitution Submodule upstreams! In case of conflict, IT MUST BE carefully resolved! Nothing can be broken, made faulty, corrupted or unusable! After merging full validation and verification MUST BE done!”

Before ANY modification to constitution/Constitution.md, constitution/CLAUDE.md, or constitution/AGENTS.md, the agent or operator MUST execute the following 7-step pipeline in order:

1. **Fetch + pull first** inside the constitution submodule worktree — every configured remote fetched, then `git pull --ff-only` (or `--rebase` if non-FF; NEVER `--strategy=ours` / `--allow-unrelated-histories` without explicit authorization).
2. **Apply the change** with §11.4.17 classification + verbatim mandate quote.
3. **Validate before commit** — `meta_test_inheritance.sh` (or equivalent), no merge-conflict markers, cross-file consistency.
4. **Commit + push to ALL upstreams** — governance files only (NEVER `git add -A`); push to every configured remote. One-upstream commit = CONST-049 violation (also CONST-038/§6.W and §2.1).
5. **Conflict resolution** preserving union of governance content. Force-push to bypass conflicts is FORBIDDEN (CONST-043 / §9.2).
6. **Post-merge validation** — `git submodule update --remote --init` + re-run cascade verifier (CONST-047) confirming the new clause reaches every owned submodule.
7. **Bump consuming project pointer** — `.gitmodules`-tracked submodule pointer advanced to the new constitution HEAD in the SAME commit as cascade work.

Cascade requirement: This anchor (verbatim or by CONST-049 ID reference) MUST appear in every

owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Severity-equivalent to a force-push without CONST-043 / §9.2 authorization. No escape hatch. See constitution submodule Constitution.md §11.4.26 for the full mandate.

CONST-050: No-Fakes-Beyond-Unit-Tests + 100%-Test-Type-Coverage Mandate (cascaded from constitution submodule §11.4.27)

> Verbatim user mandate (2026-05-15):
“Mocks, stubs, placeholders, TODOs or FIXMEs are allowed to exist ONLY in Unit tests! All other test types MUST interact with real fully implemented System! No fakes, empty implementations or bluffing is allowed of any kind! All codebase of the project MUST BE 100% covered with every supported test type: unit tests, integration tests, e2e tests, full automation tests, security tests, ddos tests, scaling tests, chaos tests, stress tests, performance tests, benchmarking tests, ui tests, ux tests, Challenges (fully incorporating our Challenges Submodule — <https://github.com/vasic-digital/Challenges>). EVERYTHING MUST BE tested using HelixQA (fully incorporating HelixQA Submodule — <https://github.com/HelixDevelopment/HelixQA>). HelixQA MUST BE used with all possible written tests suites (test banks) for every applications, service, platform, etc and execution of the full HelixQA QA autonomous sessions! All required dependency Submodules MUST BE added into the project as well (fully recursive!!!).”

Two cooperating invariants:

(A) No-fakes-beyond-unit-tests.

Mocks, stubs, fakes, placeholders, TODO, FIXME, “for now”, “in production this would”, or empty-implementation patterns are PERMITTED only in unit-test sources (*_test.go files invoked without the integration build tag; HelixCode/tests/unit/; etc.). Every other test type — integration, E2E, full automation, security, DDoS, scaling, chaos, stress, performance, benchmarking, UI, UX, Challenges,

HelixQA — MUST exercise the real, fully implemented HelixCode system against real infrastructure (real PostgreSQL, real Redis, real LLM endpoints, real containers, real captured devices). Production code (anything under HelixCode/cmd/, HelixCode/applications/, HelixCode/internal/<pkg>/<file>.go not ending _test.go) MUST NOT import from HelixCode/internal/mocks/.

(B) 100% test-type coverage.

HelixCode's codebase MUST be covered by every supported test type the domain warrants: - **Unit** — fast, isolated, mocks permitted per (A). -

Integration — multi-component, no mocks, real backing services. - **End-to-end (E2E)** — full user-flow exercise on target topology. - **Full automation** — orchestrated suites exercising every feature × platform combination (CONST-048 coverage ledger). -

Security — authn/authz boundaries, CONST-042 secret-leak scans, input-fuzzing, dependency-CVE scanning, threat-model verification. - **DDoS** — request-flood resilience at advertised throughput tier. - **Scaling** — horizontal + vertical scale behaviour under linear load growth. - **Chaos** — controlled failure injection (network partition, process kill, disk full, clock skew). -

Stress — sustained load above advertised tier. - **Performance** — latency / throughput / tail-latency invariants vs SLO baselines. -

Benchmarking — micro + macro suites with historical p95-drift detection. - **UI** — visual-regression + DOM-state + interaction-flow coverage on every target platform's UI surface. - **UX** — flow-correctness + accessibility + i18n + visual-cue ordering (§11.4.23

composition). - **Challenges** — vasic-digital/Challenges submodule (at ./Challenges/) fully incorporated; per-feature Challenge scripts with captured runtime evidence. - **HelixQA**

— HelixDevelopment/HelixQA submodule (at ./HelixQA/) fully incorporated; ALL written test banks executed; full autonomous QA sessions

run as part of release gates with captured wire evidence per check.

Required dependency submodules

(recursive per CONST-047): -

Challenges —

git@github.com:vasic-digital/

Challenges.git — incorporated at ./

Challenges/. - HelixQA —

git@github.com:HelixDevelopment/

HelixQA.git — incorporated at ./

HelixQA/. - Any additional

functionality submodules under vasic-

digital/* / HelixDevelopment/*

orgs that HelixCode depends on —

incorporate rather than duplicate work

the orgs already maintain.

Submodule pointers MUST be bumped

to upstream HEAD in the SAME

commit as any dependent cascade work

(CONST-049 step 7). Pointer drift =

CONST-050 violation.

Cascade requirement: This anchor

(verbatim or by CONST-050 ID

reference) MUST appear in every

owned submodule's CONSTITUTION.md,

CLAUDE.md, and AGENTS.md. Severity-

equivalent to a §11.4 PASS-bluff at the

release-gate layer. No escape hatch. See

constitution submodule

Constitution.md §11.4.27 for the full

mandate.

CONST-051: Submodules-As-Equal-Codebase + Decoupling + Dependency-Layout Mandate (cascaded from constitution submodule §11.4.28)

> Verbatim user mandate (2026-05-15):

“All existing Submodules in the project

that we are controlling and belong to

some our organizations (vasic-digital,

HelixDevelopment, red-elf,

ATMOSphere1234321, Bear-Suite,

BoatOS123456, Helix-Flow, Helix-

Track, Server-Factory - we can ALWAYS

check dynamically using GitHub and

GitLab CLIs) are equal parts of the

project's codebase! We MUST work on

that code as much as we do with main

project's codebase! All on equal basis!

Equally important! We MUST take it

into the account, analyze it, extend it,

create missing tests, do full testing of it,

fill the gaps (if any), fix any issues that

we discover or they pop-up, write and

extend the documentation, user guides, manulas, diagrams, graphs, SQL definitions, Website(s) and all other relevant materials! We MUST NEVER modify Submodules to bring into them any project specific context since they all MUST BE ALWAYS fully decoupled, project not-aware, fully reusable and modular (by any other project(s)), completely testable! All Submodule dependencies that are used by Submodule MUST BE accessed from the root of the project! We MUST NOT have nested Submodule dependencies but accessing each from proper location from the root of the project - directly from project's root project_name/submodule_name or some more proper structure project_name/submodules/submodule_name!"

Three cooperating invariants apply to every HelixCode-owned submodule (those whose upstream origin lives under vasic-digital, HelixDevelopment, red-elf, ATMOSphere1234321, Bear-Suite, BoatOS123456, Helix-Flow, Helix-Track, Server-Factory, or any subsequently authorised org):

(A) Equal-codebase. Every owned-by-us submodule is an **equal part** of HelixCode's codebase. The same engineering practice — analysis, extension, test creation, gap-filling, bug-fix, documentation (user manuals, guides, diagrams, graphs, SQL definitions, website pages, all materials) — applies to each owned submodule on equal basis. A round of work that improves only HelixCode's main while leaving an owned-submodule deficiency unaddressed is a CONST-051 violation, severity-equivalent to a §11.4 PASS-bluff at the project-scope layer. The §11.4.25 / CONST-048 coverage ledger MUST list every owned submodule as an in-scope target.

(B) Decoupling / reusability. Owned submodules MUST remain fully decoupled from HelixCode (and any other consuming project). No HelixCode-specific context, hardcoded paths, hostnames, asset names, or

runtime assumptions may be introduced into an owned submodule's source tree. When a submodule needs information from HelixCode, the honest path is configuration injection (env var, config file, constructor parameter) — never a hardcoded reach into the parent's tree. Every owned submodule **MUST** be project-not-aware, fully reusable, modular, and completely testable as a standalone repository.

(C) Dependency-layout. Every dependency that an owned submodule consumes **MUST** be accessible from HelixCode's root at one of two canonical paths: - <repo_root>/<submodule_name>/ (flat layout — current HelixCode layout for Challenges, HelixQA, Containers, Security, etc.) - <repo_root>/submodules/<submodule_name>/ (grouped layout — alternate)

Nested own-org submodule chains are FORBIDDEN. A submodule **MUST** NOT have its own `.gitmodules` entries pulling in further owned-by-us repos. Every dependency required by submodule X is added to HelixCode's root at the canonical path; X reaches it via documented import / SDK path / runtime resolver — never via its own nested submodule pointer. Third-party submodules (not under our orgs) are exempt — they **MAY** appear at any depth.

The owned-org list is dynamically discoverable at any time via `gh org list` / `glab` CLIs or the orgs' public APIs.

Cascade requirement: This anchor (verbatim or by `CONST-051` ID reference) **MUST** appear in every owned submodule's `CONSTITUTION.md`, `CLAUDE.md`, and `AGENTS.md`. Severity-equivalent to a §11.4 PASS-bluff at the codebase-completeness layer. No escape hatch. See constitution submodule `Constitution.md` §11.4.28 for the full mandate (audit gates, mutation pairs, workflow integration).

Amendment Process

Constitution amendments require: 1. Written proposal with rationale 2. Challenge demonstrating the need 3. 72-hour review period 4. Approval by project architect 5. Update to all submodule governance files

This Constitution is the supreme law of the HelixCode project. No code, test, or process may contradict it.

CONST-052: Lowercase-Snake_Case-Naming Mandate (cascaded from constitution submodule §11.4.29)

Verbatim user mandate (2026-05-15): **“naming convention for Submodules and directories (applied deep into hierarchy recursively) - all directories and Submodules MUST HAVE lowercase names with space separator between the words of ‘_’ character (snake-case)! All existing Submodules and directories which are not following this rule MUST BE renamed! However, since this will most likely break some of the functionalities renaming we do MUST BE applied to all references to particular Submodule or directory! ... There MUST BE reasonable exceptions for this rules - source code for programming languages or Submodules which apply different naming convention - Android, Java, Kotlin and others. ... Upstreams directory which all of our projects and Submodules have MUST BE renamed to the lowercase letters too, however root project containing the install_upstreams system command (it is exported in our paths in our .bashrc or .zshrc) MUST BE updated to fully work with both Upstreams and upstreams directory. ... NOTE: Rules lowercase / snake-case do apply to all project files as well and references to it and from them!”**

Every directory, submodule, and file in HelixCode MUST use lowercase snake_case names. Existing non-compliant names (HelixCode/, Challenges/, Containers/, HelixAgent/, HelixQA/, Security/, Github-Pages-Website/, Upstreams/, Dependencies/, etc.) MUST be renamed as part of the phased migration opened by this clause. Every reference (configs, docs, links, source-code imports, governance files) MUST be updated atomically with the rename — reference drift after a rename is a CONST-052 violation of equal severity to the rename itself.

Common-sense exceptions (technology-preserving): language-mandated case for Java/Kotlin/Android/Apple/C#/Swift INSIDE the language root (submodule root follows our convention; subtree follows language convention); vendor/upstream third-party submodules keep upstream names; build artefacts (node_modules, __pycache__, .git, target, build, bin) keep tool-mandated names. The test “does renaming break the technology?” trumps the rule.

Upstreams/ → upstreams/ transition: the constitution submodule’s install_upstreams.sh (exported via .bashrc/.zshrc) supports BOTH Upstreams/ and upstreams/ directory layouts (commit 45d3678 of the constitution submodule); lowercase wins when both present.

Test coverage of renames (per CONST-050(B)): every rename batch ships with (i) regression test verifying every reference now resolves, (ii) full test-type matrix run post-rename, (iii) anti-bluff wire-evidence captured.

Phased execution per the operator's explicit instruction: comprehensive brainstorming → phase-divided plan → fine-grained tasks/subtasks → every change covered by every applicable test type. §11.4.20 subagent delegation for cross-cutting rename sweeps.

Cascade requirement: This anchor (verbatim or by CONST-052 ID reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Severity-equivalent to a §11.4 PASS-bluff at the reference-integrity layer. No escape hatch beyond the common-sense exceptions enumerated above. See constitution submodule Constitution.md §11.4.29 for the full mandate.

CONST-053: .gitignore + No-Versioned-Build-Artifacts Mandate (cascaded from constitution submodule §11.4.30)

Verbatim user mandate (2026-05-15): "every project module, every Submodule, every service and application MUST HAVE proper .gitignore file! We MUST NOT git version build artifacts, cache files, tmp files, main .env file(s) or any files containing sensitive data, API keys or token! Any build derivative which we can recreate by executing proper mechanism for generating MUST NOT be versioned! We MUST pay attention what is going to be committed every time we are preparing to execute commit! If any violation is detected it MUST be fixed before commit is executed!"

Every project module, owned-by-us submodule, service, and application MUST ship a proper .gitignore. Forbidden-from-version-control classes:

1. **Build artefacts:** /bin/, /build/, /dist/, /out/, target/, *.exe, *.dll, *.so, *.dylib, *.a, *.o, *.class, *.pyc, generator-produced files when the generator is committed.
2. **Cache files:** __pycache__/, .pytest_cache/, .mypy_cache/, .ruff_cache/, node_modules/, .next/, .cache/, .gradle/, .terraform/, language-server caches.
3. **Temp files:** *.tmp, *.swp, *~, .DS_Store, Thumbs.db, *.orig, *.rej.
4. **Sensitive-data files:** .env, .env.* (allow .env.example placeholder only — no real secrets even as examples), *.pem, *.key, *.crt, id_rsa*, id_ed25519*, .netrc, secrets/, api_keys.sh.
5. **Generated reports/logs:** *.log, coverage.out, htmlcov/, runtime captures unless reference assets.
6. **OS/IDE personal state:** .idea/, .history/, .vscode/ (except shared settings).

Anti-bluff invariant: .gitignore line alone is not sufficient — no file matching the forbidden patterns may be CURRENTLY TRACKED. A tracked *.log despite the ignore-line is a violation of equal severity to no ignore-line at all.

Pre-commit attention: every commit author (human OR agent) MUST inspect `git diff --staged + git status` BEFORE executing the commit. Forbidden-class hits abort the commit until fixed (un-stage, add to `.gitignore`, scrub if already-tracked). Gate CM-GITIGNORE-PRECOMMIT-AUDIT + paired mutation.

Secret-leak intersection (CONST-042 / §11.4.10): a `.env` leak is BOTH a CONST-053 and a CONST-042 violation; rotation + post-mortem required.

Recreatable-content test: if a documented mechanism regenerates the file from sources, it is a build derivative and MUST be ignored. The committed sources MUST include the generator.

Cascade requirement: This anchor (verbatim or by CONST-053 ID reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Severity-equivalent to a §11.4 PASS-bluff at the repository-hygiene layer. See constitution submodule Constitution.md §11.4.30 for the full mandate.

CONST-054: Submodule-Dependency-Manifest Mandate (cascaded from constitution submodule §11.4.31)

Verbatim user mandate (2026-05-15): *“We MUST HAVE mechanism for each Submodule to determine / know what are its Submodule dependencies so new projects or palces we are incorporate them can add these Submodules to the project root and make them available! Suggested idea is configuration file with expected Submodules Git ssh urls perhaps? New project can read it, and recursively add each Submodule to the root of the project and install / expose it to veryone.”*

Every owned-by-us submodule MUST ship `helix-deps.yaml` at its root declaring its own-org dependencies. Schema: `schema_version, deps: [{name, ssh_url, ref, why, layout: flat|grouped}], transitive_handling. {recursive, conflict_resolution}, language_specific_subtree`. Tooling: `incorporate-submodule <ssh-url>` adds the submodule at the parent project's canonical path (CONST-051(C)), reads `helix-deps.yaml`, recurses for each declared dep, aborts on conflicting refs, emits `<root>/helix-manifest.yaml` audit record.

Anti-bluff guarantee: every manifest paired with a Challenge that bootstraps a throwaway consuming project, runs `incorporate-submodule`, asserts produced layout matches the manifest, runs the submodule's own tests against the bootstrapped layout, captures wire evidence per §11.4.2. A manifest without this proof is a CONST-054 violation.

§11.4.31 / CONST-054 is the **operational complement** of CONST-051(C): nested own-org submodule chains are FORBIDDEN — manifests are the bridge that lets consumers reconstruct the dependency graph at the parent root.

Cascade requirement: This anchor (verbatim or by CONST-054 ID reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Severity-equivalent to §11.4 PASS-bluff at the dependency-graph layer. See constitution submodule Constitution.md §11.4.31 for the full mandate.

CONST-055: Post-Constitution-Pull Validation Mandate (cascaded from constitution submodule §11.4.32)

Verbatim user mandate (2026-05-15): *“Every time we fetch and pull new changes on constitution Submodule we MUST process the whole project and all Submodule (deep recursively) for validation and verification taht every single rule or mandatory constraint is followed and respected! If it is not, IT MUST BE!”*

Whenever a project’s constitution submodule is fetched + pulled with any content change, the project MUST run `scripts/verify-all-constitution-rules.sh` BEFORE the new constitution HEAD is treated as canonical for any other work. The sweep re-runs the governance-cascade verifier AND every implementable rule gate (CONST-053 `.gitignore` audit, CONST-051(C) nested-own-org-chain audit, CONST-052 case audit, CONST-050(A) mock-from-production audit, CONST-035 anti-bluff smoke, etc.) against the post-pull tree. Failures populate the project’s Issues tracker per §11.4.15 (Status: Reopened, Type: Bug); closure requires positive-evidence per §11.4.

Pull-time invocation: `git submodule update --remote constitution` triggers the sweep automatically (post-update hook OR commit-wrapper invocation). Operator-explicit manual invocation also available.

Anti-bluff: the sweep’s own meta-test (paired mutation per §1.1) plants a known violation of each enforced gate and asserts the sweep reports FAIL for the planted gate. A sweep that exits PASS without running every implementable gate is a CONST-055 violation.

CONST-055 is the **enforcement engine** for every other §11.4.x and CONST-NNN rule — without it, new rules cascade as anchors but never get enforced.

Cascade requirement: This anchor (verbatim or by CONST-055 ID reference) MUST appear in every owned submodule’s `CONSTITUTION.md`, `CLAUDE.md`, and `AGENTS.md`. Severity-equivalent to §11.4 PASS-bluff at the constitutional-enforcement layer. See constitution submodule `Constitution.md` §11.4.32 for the full mandate.

CONST-056: Mandatory `install_upstreams` on clone/ add Mandate (cascaded from constitution submodule §11.4.36)

Verbatim user mandate (2026-05-15): *“Every Submodule or Git repository we add or clone MUST BE upstreams installed using Upstreamable utility which MUST BE available through exported paths of the host system (in `.bashrc` or `.zhrc`) using `install_upstreams` command executed from the root of the cloned (added) repository - only if in it is Upstreams or upstreams directory present with bash script files (recipes) for all repository’s upstreams!”*

Every clone / add of a Git repository under HelixCode MUST be followed by `install_upstreams` invocation from the repository's root IF its tree contains `upstreams/` (or legacy `Upstreams/` per CONST-052 transition) populated with `*.sh` recipe files. The utility (installed on operator's PATH via `.bashrc/.zshrc`; implementation in the constitution submodule's `install_upstreams.sh` — already supports BOTH directory names since constitution commit 45d3678) reads the recipe files, configures every declared upstream as a named git remote, and fans out origin push URLs.

Skipping the invocation when `upstreams/` is present silently breaks §2.1 (multi-upstream push is the norm) — the next push lands on only one upstream. Gate CM-INSTALL-UPSTREAMS-ON-CLONE + paired mutation. Automation: the future `incorporate-submodule` per CONST-054 auto-invokes; manual invocation supported. Pre-commit check: `git remote -v | grep -c push` reports expected count.

Cascade requirement: This anchor (verbatim or by CONST-056 ID reference) MUST appear in every owned submodule's `CONSTITUTION.md`, `CLAUDE.md`, and `AGENTS.md`. See constitution submodule `Constitution.md` §11.4.36 for the full mandate.

CONST-057: Type-aware Closure-Status Vocabulary (cascaded from constitution submodule §11.4.33)

Every project tracking work items by Type per §11.4.16 MUST close them with the Type-appropriate terminal `**Status:**` value, drawn from this 3-element closed map:

Item **Type:**	Closure **Status:** value
Bug	Fixed (→ <code>Fixed.md</code>)
Feature	Implemented (→ <code>Fixed.md</code>)
Task	Completed (→ <code>Fixed.md</code>)

The (→ `Fixed.md`) suffix is preserved across all three so the existing migration-discipline tooling (atomic `Issues.md` → `Fixed.md` move per §11.4.19) keeps working without per-Type branching. Generators (`generate_issues_summary.sh`, `generate_fixed_summary.sh`, the §11.4.23 colorizer) MUST treat the three terminal values as semantically equivalent (all “closed, positive evidence captured”) while preserving the literal in the emitted document.

Closing a Feature with `Fixed (→ Fixed.md)` or a Task with `Implemented (→ Fixed.md)` is a CONST-057 violation. Gate CM-CLOSURE-VOCAB-TYPE-AWARE walks every `Fixed.md` heading + every `Issues.md` heading whose `**Status:**` is one of the three terminal values and asserts the Status-Type match. Composes with §11.4.15 / §11.4.16 / §11.4.19 / §11.4.23.

Cascade requirement: This anchor (verbatim or by CONST-057 ID reference) MUST appear in every owned submodule's `CONSTITUTION.md`, `CLAUDE.md`, and `AGENTS.md`. See constitution submodule `Constitution.md` §11.4.33 for the full mandate.

CONST-058: Reopened-Source Attribution Mandate (cascaded from constitution submodule §11.4.34)

Every Issues.md (or equivalent project tracker) heading whose ****Status:**** is Reopened MUST carry, within 8 non-blank lines of the heading, a ****Reopened-Details:**** line capturing four sub-facts:

- **By:** AI or User (source-of-truth observer who flipped the status). AI covers in-loop reopens (test failure, gate regression, captured-evidence retrospect). User covers operator-side observations (manual testing, end-user report, design reconsideration).
- **On:** ISO date (YYYY-MM-DD).
- **Reason:** one-line cause classification — chosen from the closed vocabulary { test-failed | manual-testing-detected | captured-evidence-contradicts | end-user-report | cycle-re-discovered | design-reconsidered }. Other values permitted with explicit Reason: `<free text>` annotation but the closed list MUST be tried first.
- **Evidence:** path to or short description of the captured artefact justifying the reopen — log file, recording, gate failure ID, operator quote, etc. Reopens without evidence are §11.4.6 / §11.4.7 violations (demotion from Fixed requires captured evidence under the conditions that re-exposed the defect).

The Issues_Summary.md Status column MUST distinguish the four Reopened sub-states by source so a sweep query for “reopens by AI in the last 30 days” is mechanically possible. Suggested column rendering: Reopened (AI: test-failed) vs Reopened (User: manual-testing). Gate CM-ITEM-REOPENED-DETAILS mirrors CM-ITEM-OPERATOR-BLOCKED-DETAILS (§11.4.21 walk pattern). Composes with §11.4.6 / §11.4.7 / §11.4.15 / §11.4.21.

Cascade requirement: This anchor (verbatim or by CONST-058 ID reference) MUST appear in every owned submodule’s CONSTITUTION.md, CLAUDE.md, and AGENTS.md. See constitution submodule Constitution.md §11.4.34 for the full mandate.

CONST-059: Canonical-Root Inheritance Clarity (cascaded from constitution submodule §11.4.35)

The **constitution submodule’s** three files (constitution/Constitution.md, constitution/CLAUDE.md, constitution/AGENTS.md) ARE the **canonical root** (also called the **parent** files). They contain only universal rules per §11.4.17.

The consuming project’s **repository-root files** (<project-root>/CLAUDE.md, <project-root>/AGENTS.md, optionally <project-root>/Constitution.md) are **consumer extensions**. They MUST start with the inheritance pointer (either the Claude-Code native @constitution/CLAUDE.md import or the portable ## INHERITED FROM constitution/CLAUDE.md heading). They contain only project-specific rules per §11.4.17.

When in doubt about which file to edit: universal rule → constitution submodule’s file; project-specific rule → consumer’s file. Default consumer-side when uncertain (§11.4.17 — narrower scope is cheap to widen).

Terminology: “the parent CLAUDE.md” / “the root Constitution” → constitution-submodule file at constitution/<filename>; “the project CLAUDE.md” / “this project’s AGENTS.md” → consumer-side file at <project-root>/<filename>.

No silent demotion or silent promotion. Moving a rule between layers MUST be a visible commit — `git mv` of a section if it’s a clean clone, or explicit
Lifted from <project> to constitution per §11.4.35 / Demoted from constitution to <project> per §11.4.35 commit-message annotation.

Gate CM-CANONICAL-ROOT-CLARITY verifies (a) consumer’s CLAUDE.md opens with the inheritance pointer, (b) constitution submodule’s three files are present at the expected path, (c) no `## INHERITED FROM` block in the constitution submodule’s own files (those ARE the source-of-truth, not consumers). Composes with §11.4.17.

Cascade requirement: This anchor (verbatim or by CONST-059 ID reference) MUST appear in every owned submodule’s CONSTITUTION.md, CLAUDE.md, and AGENTS.md. See constitution submodule Constitution.md §11.4.35 for the full mandate.

CONST-060: Fetch-before-edit Mandate (cascaded from constitution submodule §11.4.37)

Verbatim user mandate (2026-05-15): *“Make sure that `feedback_fetch_before_edit` memory rule is part of our constitution Submodule - the root Constitution, AGENTS.MD and CLAUDE.MD. Validate and verify that Project-Toolkit and all Submodules do inherit all of them! Follow the constitution Submodule documentation for details.”*

The FIRST git-touching action of every session, on every consuming project (owned or third-party), MUST be:

```
git fetch --all --prune
git log --online HEAD..@{u}
git submodule foreach --recursive 'git fetch --all --prune --quiet'
```

If HEAD..@{u} is non-empty, integrate the upstream changes BEFORE any local edit. Acting on stale local state produces three failure modes documented in the originating §11.4.37 incident (multi-agent / parallel-session work): (1) **redundant work** — the agent re-does what a parallel session already finished, (2) **false confidence** — completion reports for already-done work, (3) **divergent history** — duplicate sibling commits that double the conflict surface on next push.

Anti-bluff invariant: the fetch+log check MUST produce captured evidence — the actual HEAD..@{u} output, even if empty. Skipping the check on the basis of “I just fetched” or “nothing could have changed in the last N minutes” is a §11.4.6 (no-guessing) violation: the remote state is not knowable without a fetch.

Cascade requirement: This anchor (verbatim or by CONST-060 ID reference) MUST appear in every owned submodule’s CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Severity-equivalent to §11.4 PASS-bluff at the parallel-session-coordination layer. See constitution submodule Constitution.md §11.4.37 for the full mandate.

Anti-Bluff End-User Quality Guarantee (Escalated via HelixConstitution)

Canonical authority: HelixConstitution/Constitution.md §7.1 + §11.4.

Forensic anchor — verbatim operator mandate (2026-04-28):

“We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can’t be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product! This MUST BE part of Constitution of our project, its CLAUDE.MD and AGENTS.MD if it is not there already, and to be applied to all Submodules’s Constitution, CLAUDE.MD and AGENTS.MD as well (if not there already)!”

When writing a test in this submodule, ask: if every line of the unit under test were replaced with a trivial stub, would this test still pass? If yes, the test is bluff. Rewrite it to exercise the real behaviour.

Every PASS MUST carry positive runtime evidence. Consuming-project-specific evidence requirements are defined by each consuming project’s Constitution.

CONST-061: Pre-Force-Push Merge-First Mandate (cascaded from constitution submodule §11.4.41)

Verbatim user mandate (2026-05-17): *“make sure we bring everything from branches to our side before forc push is done! Afer everything is safely and fully merged and all potential conflicts (if any) resolved, then do force push! make sure nothing isnlost, broken or corrupted on bith sides! add these rules in our root Constitution, CLAUDE.MD, AGENTS.MD (constitution Submodule) if itnis not added already! Extremely important rules and mandatory constraints we MUST HAVE and fully respect!”*

Any force-push (--force, --force-with-lease, +<ref>, equivalent history-rewrite) authorised under CONST-043 MUST be preceded by a mechanical 4-step merge-first pipeline:

1. **Fetch every remote** — `git fetch --all --prune --tags` against origin + every upstream; capture output.
2. **Integrate every divergent commit locally** — `rebase / merge / operator-confirmed cherry-pick` per appropriate strategy for every non-empty `HEAD..<remote>/<branch>` range.
3. **Audit the integrated tree** — no conflict markers anywhere (`grep -rn '^<<<<<< \|^===== $\|^>>>>>> '` returns empty in governance + source + test files); no file silently dropped; previously-passing tests still pass; captured-evidence artefacts still validate.
4. **Force-push** — only after steps 1-3 produce clean integration evidence: `git push --force-with-lease` (NEVER `--force` alone unless authorised per §9.2 sub-clause 6).

Two-gate composition with CONST-043. §11.4.41 does NOT relax CONST-043's operator-approval requirement — it adds a SECOND mechanical gate. CONST-043 alone authorises a push that loses remote work; §11.4.41 alone risks pushing without operator awareness. Both required.

Three failure modes prevented: (a) remote-side content loss when parallel sessions land work between fetches; (b) stale-state acts when `--force-with-lease` reads stale local refs without prior fetch; (c) conflict-driven corruption when markers get committed verbatim (observed 2026-05-17 in `helix_qa` + containers governance files).

Verification artefact: every governed force-push emits a `docs/change logs/ <tag>.md` “Force-push merge-first audit” section capturing fetch output, per-remote divergence log, integration strategy, conflict-marker scan, test delta, push output with lease SHA, + CONST-043 authorisation quote. Gate `CM-FORCE-PUSH-MERGE-FIRST` + paired mutation.

Cascade requirement: This anchor (verbatim or by CONST-061 ID reference) MUST appear in every owned submodule's `CONSTITUTION.md`, `CLAUDE.md`, and `AGENTS.md`. Severity-equivalent to a §11.4 PASS-bluff at the remote-data-integrity layer. See constitution submodule `Constitution.md` §11.4.41 for the full mandate.

CONST-068: Shell-script target-shell-parseability mandate (cascaded from constitution submodule §11.4.67)

Verbatim user mandate (2026-05-19): *“any issue we spot must be fixed, bash scripts as well if they are broken!”* + *“Make sure that this is mandatory rule!”*

Verbatim 2026-05-19 operator mandate: *“all existing tests and Challenges do work in anti-bluff manner - they MUST confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This MUST NOT be the case and execution of tests and Challenges MUST guarantee the quality, the completion and full usability by end users of the product!”*

Every committed shell script MUST be parseable by its target interpreter (`sh -n` for `/bin/sh`, `bash -n` for `/bin/bash`, etc.) AND MUST declare a shebang matching its actual syntax usage. Bash-only constructs (`>( ... )`, `<( ... )`, `[ [ ]`, `<<<`, arrays, `${var^^}`, etc.) used in scripts that may be invoked via `sh script.sh` MUST be wrapped in `eval` so the parser sees only a string (target shells like `mksh` parse the entire script before executing — runtime guards cannot save a parse-time rejection). Honest shebangs only: `#!/bin/bash` only if bash actually expected; `#!/bin/sh` requires POSIX-clean body. Fix at source per §11.4.1, never at callsites. Composes with §11.4.1 / §11.4.4 / §11.4.6 / §11.4.50 / §11.4.51. Pre-build gate `CM-SCRIPT-TARGET-SHELL-PARSEABLE` runs `sh -n` on every in-scope script. No escape hatch — no `--skip-parseability-check`, `--bash-only-script`, `--runtime-guard-suffices` flag.

Sort order: closure date DESC (most-recent-Fixed first), §-letter / Fix-# secondary. Documented at the top of the generated file.

Composes with §11.4.12 (Issues_Summary sibling — canonical pair), §11.4.19 (atomic Issues→Fixed migration trigger + column-alignment), §11.4.23 (colorizer post-processes both summaries), §11.4.33 (type-aware closure vocabulary — Fixed_Summary respects Fixed (→ Fixed.md) / Implemented (→ Fixed.md) / Completed (→ Fixed.md) terminal values), §11.4.44 (revision header applies to Fixed_Summary.md), §12.10 (CONTINUATION.md resumption guarantee).

Pre-build gates: CM-FIXED-SUMMARY-SYNC (6 invariants — Fixed_Summary exists + HTML/PDF mtime ≥ md mtime + Fixed_Summary mtime ≥ Fixed mtime + generator + sync wrapper invokes generator) + CM-COVENANT-114-53-PROPAGATION (anchor literal across canonical files). Paired mutations strip the anchor literal AND move the generator aside AND backdate Fixed_Summary mtime. No escape hatch — no --skip-fixed-summary-sync, --issues-only, --summary-not-applicable flag.

Canonical authority: constitution submodule Constitution.md §11.4.53.

Non-compliance is a release blocker regardless of context.

§11.4.58 — Parallel-development methodology (User mandate, 2026-05-19)

Project work proceeds through the **Parallel Work Unit (PWU) pipeline** rather than sequential Phase-chain. Each PWU has: ATM-NNN identifier (§11.4.54), Issues.md entry (§11.4.15+§11.4.16), file-scope manifest, §11.4.43 RED test, source patch, pre-build gate, post-flash test, paired §1.1 meta-test mutation, HelixQA Challenge bank entry, captured-evidence directory (§11.4.5+§11.4.52).

5-stage pipeline: Stage 1 DEVELOP (parallel PWU agents in worktrees) → Stage 2 MERGE (serial conductor + §11.4.41 4-step merge-first) → Stage 3 REBUILD+FLASH (parallel where hardware allows) → Stage 4 VALIDATE (parallel D3+D4+meta-test+coverage) → Stage 5 SWEEP (parallel HelixQA + Fixed.md migration + README refresh). Stage 1 of round N+1 overlaps with Stages 4-5 of round N.

Synchronization: 4-layer lock hierarchy (parent flock / per- submodule git / contention-path advisory locks for 10 forbidden cross- PWU paths / per-PWU worktree). Disjoint-scope PWUs fully parallel.

Anti-bluff merge-time enforcement (mandatory, all four): C1 §11.4.43 RED-test captured. C2 §1.1 paired meta-test mutation FAILs the gate. C3 §11.4.50 3-iter (or 10-iter) deterministic-consistency. C4 §11.4.5 captured-evidence per feature type. Metadata-only / configuration-only / absence-of-error / grep-without-runtime PASS REJECTED. HelixQA Challenge bank coverage MANDATORY for every user-visible PWU.

Phase 39.EX infrastructure gates (5 gates land the parallel infrastructure itself): CM-PWU-PARALLEL-VALIDATION-ORCHESTRATOR, CM-PWU-HELIXQA-PER-DOMAIN-RUNNER, CM-PWU-WORKER-POOL-LOCKING, CM-PWU-FILE-SCOPE-PARTITION, CM-PWU-AUTO-MERGE-GATE-6CONDITIONS. Each ships a paired meta-test mutation per §1.1.

Pre-build gates CM-PWU-LOCK-HIERARCHY + CM-PWU-ANTI-BLUFF-COVERAGE + CM-PWU-MERGE-QUEUE-DISCIPLINE + CM-PWU-PARALLEL-AGENT-LIMIT + CM-COVENANT-114-58-PROPAGATION. Paired mutations cover each gate. No escape hatch.

Canonical authority: constitution submodule [Constitution.md](#) §11.4.58. Project-specific implementation reference: [docs/guides/PARALLEL_DEVELOPMENT_METHODODOLOGY.md](#).

Non-compliance is a release blocker regardless of context.

§11.4.65 — Universal Markdown export mandate (User mandate, 2026-05-19)

Every Markdown document inside the project that is NOT part of an application or service's source-code tree MUST have synchronized .html and .pdf siblings. Includes: project-root *.md, docs/**/*.*md, scripts/**/*.*md (doc-format companion docs), owned-submodule top-level README.md / CLAUDE.md / AGENTS.md / CHANGELOG.md and their docs/**/*.*md, constitution/**/*.*md, owned HelixQA submodules' equivalents. Excludes: external/**, prebuilts/**, packages/modules/**, kernel-5.10/**, out/**, build/**, application/service source-code trees, and third-party submodules NOT in the owned set. Every edit triggers regeneration via scripts/testing/sync_all_markdown_exports.sh (pandoc HTML + weasyprint PDF, timeout 60 per file, capped at 500 candidates). HTML + PDF mtime MUST be ≥ source .md mtime at all times.

Pre-build gates CM-UNIVERSAL-MARKDOWN-EXPORT-SYNC + CM-COVENANT-114-65-PROPAGATION. Paired meta-test mutations. Composes with §11.4.12 / §11.4.18 / §11.4.23 / §11.4.44 / §11.4.45 / §11.4.53 / §11.4.59 / §11.4.60 / §11.4.63 / §11.4.64. No escape hatch — no --skip-md-exports, --no-pdf-only, --md-export-not-applicable flag.

Canonical authority: constitution submodule [Constitution.md](#) §11.4.65.

Non-compliance is a release blocker regardless of context.

§11.4.66 — Blocker-resolution interactive-clarification mandate (User mandate, 2026-05-19)

When any task is blocked (operator decision, hardware access, external authorization, ambiguous scope), the agent MUST: (1) research what's doable from the agent side without operator input; (2) calculate minimum-viable operator input; (3) construct 2–4 mutually-exclusive options with one marked "Recommended" and each stating what the agent does after that answer; (4) present via the platform's interactive question mechanism (AskUserQuestion on Claude Code) — NEVER free-text "what would you like?" for closed- set decisions; (5) after the answer, resume work without follow-up round-trips. Composes with §11.4.6 / §11.4.7 / §11.4.40 / §11.4.41 / §11.4.42 / §11.4.52. No silent waiting; no bulk-text questions when interactive options would do.

Pre-build gate CM-COVENANT-114-66-PROPAGATION enforces the anchor literal across the 42-file consumer fleet. Paired meta- test mutation strips the literal → gate FAILs. No escape hatch — no --skip-ask, --silent-wait, --free-form-only flag.

Canonical authority: constitution submodule Constitution.md §11.4.66.

Non-compliance is a release blocker regardless of context.

§11.4.67 — Shell-script target-shell-parseability mandate (User mandate, 2026-05-19)

Forensic anchor — direct user mandate (verbatim, 2026-05-19): “any issue we spot must be fixed, bash scripts as well if they are broken!” + “Make sure that this is mandatory rule!”

Every shell script that may be invoked under a target shell other than the one in its shebang **MUST** parse cleanly under that target shell. Forensic incident: device/rockchip/rk3588/tests/test_all_fixes.sh:114 used bash-only `exec >>(tee -a "$f") 2>&1` on a sh script.sh callsite — Android mksh parses the whole script **BEFORE** executing, so the runtime `[ -n "${BASH_VERSION:-}" ]` guard could not save it. Fixed by wrapping in `eval 'exec >>(tee ...) 2>&1'` so the parser sees only a string.

Closed-set scope: every tracked .sh under device/rockchip/rk3588/tests/, scripts/, scripts/testing/ (and equivalent paths in owned submodules). OUT of scope: external/, prebuilts/, packages/modules/, kernel-5.10/, out/, build/, scripts/legacy/. Mandatory invariants: (1) every in-scope script parses under `sh -n`; (2) bash-only constructs (`>(...)`, `<(...)`, `[ [ ]`], `<<<`, arrays, `${var^^}`, etc.) **MUST** be wrapped in `eval` OR guarded by bash-only loading; (3) shebangs honest — `#!/bin/bash` only if bash actually expected; (4) fix at source per §11.4.1, never at callsites. Composes with §11.4.1 / §11.4.4 / §11.4.6 / §11.4.50 / §11.4.51.

Pre-build gate `CM-SCRIPT-TARGET-SHELL-PARSEABLE` runs `sh -n` on every in-scope script. Propagation gate `CM-COVENANT-114-67-PROPAGATION` enforces the anchor literal across the 44-file consumer fleet. Paired mutations: inject bash-only outside `eval` → parse gate FAILs; strip 11.4.67 literal → propagation gate FAILs. No escape hatch — no `--skip-parseability-check`, `--bash-only-script`, `--runtime-guard-suffices` flag.

Canonical authority: constitution submodule Constitution.md §11.4.67.

Non-compliance is a release blocker regardless of context.

§11.4.69 — Universal sink-side positive-evidence taxonomy + mechanical enforcement (User mandate, 2026-05-20)

Forensic anchor — direct user mandate (verbatim, 2026-05-20):

“THIS MUST HAPPEN NEVER AGAIN!!! We MUST HAVE this all working! Not just for audio but for every single piece of the System!!! Proper full automation when executed with success **MUST MEAN** that manual testing will be as much positive at least regarding the success results! ... Solution **MUST BE** universal, generic that solves working flows for all System components and for all future and all existing projects! ... Everything we do **MUST BE** validated and verified with rock-solid proofs and anti-bluff policy enforcement and fulfillment!”

Universal generalisation of §11.4.68 (audio-specific) across every user-visible feature class. Closes the PASS-bluff pattern where tests reported green while end users hit broken features (2026-05-19→20 D3 audio “82/84 PASS” + empty Arvus Codec-In-Use).

The mandate. Every user-visible feature MUST map to one entry in the closed-set §11.4.69 sink-side evidence taxonomy (audio_output, audio_input, video_display, network_throughput, network_connectivity, bluetooth_a2dp, bluetooth_pair, touch_input, sensor, gpu_render, storage_read, storage_write, mediacodec_decode, mediacodec_encode, miracast, cast, boot_service, package_install, permission_grant, wifi_link, wifi_throughput, ethernet_link, display_topology, drm_playback, subtitle_render — open to additions). Every PASS for a feature in the taxonomy MUST cite a captured-evidence artefact path matching the required evidence shape.

Helper contracts (additive during grace; mandatory after 2026-06-19):

- `ab_pass_with_evidence` <description> <evidence_path> — the new canonical PASS helper. Verifies path exists AND non-empty; emits PASS: <description> [evidence: <path>].
- `ab_skip_with_reason` <description> <closed-set-reason> — reasons: `geo_restricted`, `operator_attended`, `hardware_not_present`, `topology_unsupported`, `network_unreachable_external`, `feature_disabled_by_config`. Forbids `network_unreachable_external` for any taxonomy feature with a sink-side probe.
- Bare `ab_pass` deprecated — WARN pre-grace, FAIL post-grace (2026-06-19).

Mechanical enforcement. Three pre-build gates + three paired §1.1 meta-test mutations:

- `CM-SINK-EVIDENCE-PER-FEATURE` — walks tests for # §11.4.69 FEATURE: <class> annotation + verifies taxonomy probe + `ab_pass_with_evidence` use.
- `CM-NO-FAIL-OPEN-SKIP` — audits sink-side probe helpers; FAILs if any code path converts empty/unreachable response to PASS-counting SKIP for a feature class with a sink-side probe.
- `CM-AB-PASS-WITH-EVIDENCE-EVERYWHERE` — pre-grace WARN, post-grace FAIL on bare `ab_pass` calls.

Composes with §11.4.1 (FAIL-bluffs forbidden), §11.4.2 (recorded-evidence), §11.4.5 (audio + video 5-layer quality), §11.4.6 (no-guessing), §11.4.13 (sink-side captured-evidence), §11.4.27 (no-fakes-beyond-unit), §11.4.50 (deterministic consistency), §11.4.52 (autonomous-validation), §11.4.68 (audio-specific sink-side — §11.4.69 is the universal generalisation).

No escape hatch — no `--skip-evidence`, `--config-only-pass`, `--allow-fail-open-skip`, `--legacy-ab-pass-permitted` flag. The discipline exists because the 2026-05-20 forensic incident demonstrated the failure: tests reported audio-routing PASS while the user heard nothing and the Arvus Codec-In-Use field was empty.

Propagation gate `CM-COVENANT-114-69-PROPAGATION` enforces this anchor literal across the ~44-file consumer fleet. Paired mutation strips the literal → gate FAILs.

Canonical authority: constitution submodule [Constitution.md](#) §11.4.69.

Non-compliance is a release blocker regardless of context.

Cascade requirement: This anchor (verbatim or by CONST-068 ID reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. See constitution submodule Constitution.md §11.4.67 for the full mandate.

§11.4.68 — Positive Sink-Side / Downstream Evidence Mandate (cascaded from constitution submodule §11.4.68)

Verbatim user mandate (2026-05-20): *“We still do not hear any audio played from D3 device! Arvus Web Dashboard when we play music from D3 shows nothing for Codec In Use! This MUST BE investigated and fixed! How come we passed the tests with Arvus validation? What were values for the Codec In Use field? Empty means nothing! This is not working! It MUST BE FIXED, TESTED AND VERIFIED WITH FULL AUTOMATION TESTING ASAP!!!”*

A test that asserts audio or video routing PASS MUST capture and verify **positive sink-side or downstream evidence** — never config-only, never metadata-only, never PCM-open-state-only. At least one of the closed enumeration MUST be captured for every audio/video routing PASS: (1) sink-side codec-state with non-empty Codec-In-Use matching the expected codec regex; (2) strictly-positive PCM frames-written delta from /proc/asound/.../status hw_ptr; (3) ALSA ELD/EDID-Like-Data showing negotiated channel count + format; (4) fprobe-on-captured-mp4 with non-zero frame count + expected codec/resolution/fps; (5) recording-analyzer event match per §11.4.2/§11.4.5; (6) tinycap RMS amplitude above the line-level floor. Empty / <unreachable> / <N.E.> / <None> placeholders are NOT positive evidence; a missing-but-required sink is OPERATOR-BLOCKED (release-blocker), never SKIP, never PASS. No escape hatch — no --skip-sink-evidence, --allow-empty-codec, --sink-unreachable-is-pass, --metadata-only-suffices flag exists.

Cascade requirement: This anchor (verbatim or by §11.4.68 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Severity-equivalent to a §11.4 PASS-bluff at the sink-side-evidence layer. **Canonical authority:** constitution submodule Constitution.md §11.4.68 for the full mandate.

§11.4.70 — Subagent-Driven Execution Is The Default (cascaded from constitution submodule §11.4.70)

Verbatim user mandate (2026-05-20): *“Always do if possible Subagent-driven! Add this into our root (constitution Submodule) Constitution.md, CLAUDE.md and AGENTS.md. This should be the default choice ALWAYS!”*

When executing implementation plans (or any task-decomposed execution flow), the **default execution model is subagent-driven** per superpowers:subagent-driven-development. Inline execution is permitted ONLY when (a) the task is

trivial AND fits a single sub-300-line edit, OR (b) the operator explicitly requests inline at brainstorm-handoff time. Subagent-driven is the default because it gives isolated context per task, naturally enforces two-stage review, is parallel-PWU compatible (§11.4.58), creates an anti-bluff seam (§11.4), and survives operator absence. No escape hatch — `--inline-execution-required`, `--no-subagents`, `--monolithic-execution` are NOT permitted flags. Skipping subagent-driven for non-trivial work without recorded operator authorisation is itself a §11.4 PASS-bluff.

Cascade requirement: This anchor (verbatim or by §11.4.70 reference) MUST appear in every owned submodule's `CONSTITUTION.md`, `CLAUDE.md`, and `AGENTS.md`. Severity-equivalent to a §11.4 PASS-bluff at the execution-model layer.
Canonical authority: constitution submodule `Constitution.md` §11.4.70 for the full mandate.

§11.4.71 — Pre-Push Fetch + Investigate + Integrate Mandate (cascaded from constitution submodule §11.4.71)

Verbatim user mandate (2026-05-20): *“before pushing changes to any upstream for any repository - main repo or Submodule, we MUST fetch and pull all changes. Once these are obtained WE MUST investigate what is different compared to head position we were on last time before fetching and pulling new changes! We MUST understand what is done and for what purpose, easpecially how that does affect our project and our System in general! Any mandatory changes or improvements required by fresh changes we just have brought in MUST BE incorporated, covered with all supported types of the tests which will produce as a result of its success execution REAL PROOFS of working for all componetns and functionalities covered and work fully in anti-bluff manner!”*

The everyday-push variant of §11.4.41. EVERY push (every repository — main + every submodule) MUST follow the 5-step cycle: (1) fetch all remotes (`git fetch --all --prune --tags`, capture stdout); (2) pull all upstream branches whose tip differs, resolving conflicts per consumer judgment (never auto-`---ours/---theirs`); (3) investigate the diff vs OUR previous HEAD — read EVERY foreign commit's body, understand what/why/how-it-affects-our-system; (4) integrate mandatory changes with §11.4.4(b) four-layer coverage + §11.4.43 TDD-fix discipline, every PASS carrying §11.4.5 captured-evidence (REAL PROOFS, not metadata-only); (5) only then push, verifying with `git ls-remote post-push`. No escape hatch — no `--skip-fetch`, `--no-investigate`, `--fast-push`, `--trust-upstream` flag.

Cascade requirement: This anchor (verbatim or by §11.4.71 reference) MUST appear in every owned submodule's `CONSTITUTION.md`, `CLAUDE.md`, and `AGENTS.md`. Severity-equivalent to a §11.4 PASS-bluff at the push-discipline layer.
Canonical authority: constitution submodule `Constitution.md` §11.4.71 for the full mandate.

§11.4.72 — Audio Top-Priority Mandate (cascaded from constitution submodule §11.4.72)

Verbatim user mandate (2026-05-20): *“Make sure all fixes for audio are always top priority in main working stream!”*

The conductor (main working stream — Claude Code session, AI agent, or human operator) MUST treat audio fixes as the highest-priority class on the serial dispatch queue. Any time the conductor faces a choice between dispatching an audio task vs a non-audio task on the SAME serial resource, the audio task wins. Parallel BACKGROUND subagents (research, refactors, infrastructure documentation) MAY run concurrently with audio work but do NOT preempt audio on the main-stream serial dispatch queue. No escape hatch — there is no “but this non-audio task is faster” or “but this research is more interesting” override; audio-stack regressions are user-perceptible and high-impact while research and refactors can wait.

Cascade requirement: This anchor (verbatim or by §11.4.72 reference) MUST appear in every owned submodule’s CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Severity-equivalent to a process violation at the dispatch-priority layer. **Canonical authority:** constitution submodule Constitution.md §11.4.72 for the full mandate.

§11.4.73 — Main-Specification Document Versioning + Revision Discipline (cascaded from constitution submodule §11.4.73)

Verbatim user mandate (2026-05-20): *“Make sure everything we add now in previous and upcoming requests IS ALWAYS applied to the main specification — if we have one. Since all these are not major changes we could increase Specification version per change for secondary version instead of the primary. Primary version MUST BE increased for much bigger levels of changes! Add this into root (constitution Submodule) Constitution.md, CLAUDE.md and AGENTS.md as mandatory rule / constraint applicable ONLY IF we have something like the main specification document or we do recognize something like the main specification document. Document MUST BE updated ALWAYS to follow the versioning rules we are applying here + revision and other properties we have!”*

Applies **only when a project recognises a main specification document**. When it does: (1) every additive operator requirement, refinement, or accepted recommendation MUST be applied to the spec before or as part of the implementing work; (2) spec versioning has two axes — *primary* (V1/V2/V3, bumped for major rewrites by explicit operator decision, old versions archived) and *secondary* (the §11.4.61 metadata-table Revision integer, bumped for every other change); (3) the metadata table MUST stay current (Revision, Last modified, Status summary, Fixed); (4) propagated copies of the rule MUST reference the active specification.V<primary>.md, not a stale archive; (5) on primary bump the old file moves to <spec-dir>/archive/ with Status: superseded. Classification: universal, applicable conditionally per the scope condition.

Cascade requirement: This anchor (verbatim or by §11.4.73 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Severity-equivalent to a release blocker when a project has a main spec and lets it drift. **Canonical authority:** constitution submodule Constitution.md §11.4.73 for the full mandate.

§11.4.74 — Submodule-Catalogue-First Discovery + Extend-Don't-Reimplement (cascaded from constitution submodule §11.4.74)

Verbatim user mandate (2026-05-20): *“We MUST ALWAYS check which already developed features / functionalities do exist as a part of our comprehensive Submodules catalogue located in vasic-digital and HelixDevelopment organizations on GitHub and GitLab both! Project MUST BE aware of all its existence so we do not implement same things multiple times if they are already done as some of existing universal, reusable general development purpose Submodules! For any missing features that some Submodules we incorporate may be missing we MUST IMPLEMENT the properly and extend those Submodules further! We do control all of the and we CAN and MUST maintain and extend the regularly! All development cycle rules we have MUST BE applied to them and fully respected!”*

Before scaffolding ANY new module, package, helper, or utility, the contributor (human or AI agent) MUST: (1) survey the canonical Submodule catalogue — vasic-digital and HelixDevelopment on both GitHub AND GitLab; (2) inventory existing Submodules; (3) reuse before reimplement — if a Submodule provides the functionality (or 80%+ of it), add it as a Git submodule rather than write fresh; (4) extend in-place when 80%+ matches but features are missing — add the missing features TO THAT SUBMODULE (PR upstream + bump pointer), never as a duplicating consuming-project helper; (5) apply all development-cycle rules to those Submodules; (6) document the survey result in the feature's tracker entry with a Catalogue-Check: field (reuse <org/repo>@<sha> / extend <org/repo>@<sha> / no-match <date>). Classification: universal.

Cascade requirement: This anchor (verbatim or by §11.4.74 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Severity-equivalent to a process violation; duplicate implementations landed without catalogue check are release blockers. **Canonical authority:** constitution submodule Constitution.md §11.4.74 for the full mandate. —

§IS LLMProvider §11.4.X anchor propagation (Phase 39.IM, 2026-05-23)

The 18 blocks below were missing from this file's prior revision while present in the sibling LLMOrchestrator. Cascade requirement per Constitution §11.4.X propagation gates. Byte-identical to LLMOrchestrator source. See docs/Fixed.md §IS.

§11.4.1 extension (Phase 33, 2026-05-05) — FAIL-bluffs equally forbidden. A test that crashes for a script-internal reason (undefined variable under `set -u`, regex error, malformed assertion, missing argument) and produces a FAIL exit code is just as misleading as a PASS-bluff. Both let real defects ship undetected. Per parent Constitution §11.4.1, every test MUST fail ONLY for genuine product defects — script-bug failures must be fixed at the source layer (helper library, shared lib, test source), not patched in individual call sites.

Non-compliance is a release blocker regardless of context.

§11.4.2 extension (Phase 34, 2026-05-06) — Recorded-evidence requirement. A test that emits PASS without captured visual or audio evidence of the user-visible feature actually working on the screen the user would see is a §11.4 PASS-bluff. Bug #13 (VK Video on PRIMARY display while a passing test claimed playback PASS) demonstrated the gap exactly. Closing it requires the recording + analyzer infrastructure (Bug #14 — `dual_display_record.sh / action_timeline.sh / Go recording-analyzer / helixqa-bridge`). Per Constitution §11.4.2 every PASS for a user-visible feature MUST be cross-checked by the analyzer against the dual-display recording + action timeline. A PASS that lacks at least one matched timeline event in the analyzer findings is treated as a §11.4 PASS-bluff.

Non-compliance is a release blocker regardless of context.

§11.4.3 extension (Phase 34, 2026-05-06) — Per-device-topology test dispatch. Tests that depend on hardware topology (secondary HDMI present/absent, microphone present/absent, etc.) MUST detect topology at test entry and dispatch the topology-appropriate variant. A test running the wrong variant for the actual topology and PASSing is a §11.4 PASS-bluff. Bug #18 (Lampa+TorrServe E2E) demonstrated the pattern: D1 (secondary HDMI) and D2 (primary only) get separate test variants behind a `dumpsys display`-based dispatcher. Per Constitution §11.4.3 every topology-touching test MUST have such a dispatcher OR explicit topology gates with SKIP-with-reason fallback.

Non-compliance is a release blocker regardless of context.

§11.4.4 extension (User mandate, 2026-05-06) — Test-interrupt-on-discovery + retest-from-clean-baseline. A test cycle that continues running past a freshly discovered defect is itself a §11.4 PASS-bluff: it produces “all green” summaries while the codebase under test is known-broken at the moment those greens were recorded. Phase 34.S’ D1 demonstrated the violation when Bug #26 (hard-floor probe lifecycle) and Bug #27 (analyzer FAIL-bluff on non-video tests) were discovered mid-cycle and the cycle was allowed to continue, accumulating 13+ false-positive ANALYZER FAIL banners. Per Constitution §11.4.4 the moment any defect is re- discovered, re-produced, or newly identified during a test cycle, the cycle MUST stop on both devices. **Then:** (1) fix at root cause per §11.4.1, (2) land validation/verification tests for the fix — pre-build gate AND on-device test AND paired meta-test mutation, (3) full rebuild via `scripts/build.sh` (regardless of whether the fix touched host script / Go binary / firmware — host-only fixes still get a full rebuild for retest baseline integrity), (4) re-flash D1 + D2, (5) repeat full `test_all_fixes.sh` from the beginning sequentially per §12.6, (6) end the cycle with `meta_test_false_positive_proof.sh` proving no gate is itself a bluff gate. Tests AND HelixQA Challenges are bound equally — Challenges that score PASS on a non-functional feature are the same class of defect as PASS-bluff unit tests; both must produce positive end-user evidence per §11.4.2 + §11.4.3.

Non-compliance is a release blocker regardless of context.

§11.4.4 expansion (User mandate, 2026-05-06) — Systematic debugging + four-layer test coverage + documentation + no-bluff certification. Augments the §11.4.4 base covenant with four non-negotiable additional requirements per the User mandate of 2026-05-06: (a) **Systematic debugging via superpowers skills.** Before applying any fix, run in-depth systematic debugging using the available `superpowers:*` skills (debugging, root-cause analysis, architectural-impact). Symptom patches are forbidden. The debugging output **MUST** identify root cause at source layer, blast radius across related tests/features/subsystems, and the regression-protection seam. (b) **Four-layer test coverage per fix.** Every fix lands with positive evidence in **every applicable layer**: pre-build gate (catches at source), post-build gate (catches in assembled image — proves bytes landed, cf. Fix #122 `APK_LIB_MAP` misroute), post-flash on-device test (fully automated, anti-bluff per §8.1, captured- evidence per §11.4.2, topology-dispatched per §11.4.3, orchestrator-wired in `test_all_fixes.sh`), HelixQA test bank entry (`banks/atmosphere.yaml` + per-feature additions), HelixQA full QA session coverage (Challenge-driven dispatch — bank entry without Challenge coverage is a §11.4 PASS-bluff), and meta-test paired mutation. Skipping a layer because “this fix only touches X” is forbidden. (c) **Documentation update for every fix.** Required: `docs/Issues.md` → `docs/Fixed.md` migration on closure, parent `CLAUDE.md` Applied Fixes Reference row, affected user-facing guides (`docs/guides/*.md`), affected diagrams/flowcharts/architecture docs, per-version `docs/changelogs/<tag>.md` entry. Documentation drift after a fix is itself a §11.4 violation. (d) **No-bluff certification per cycle.** Before tagging: `meta_test_false_positive_proof.sh` returns all gates green AND every gate’s paired mutation FAILs (no bluff gates); `docs/Issues.md` open-set is empty or every entry explicitly classified out-of-scope-for-this-tag with operator sign-off (no known issues hidden); full suite returns zero new FAILs on either device (no working feature regressed); every gate has a paired mutation; every test produces positive evidence; every assertion catches its own negation (no error-prone or bluff-proof leftover).

Non-compliance is a release blocker regardless of context.

§11.4.5 — Audio + video quality analysis comprehensiveness (User mandate, 2026-05-07)

Forensic anchor — direct user mandate (verbatim, 2026-05-07):

“We **MUST HAVE** still analyzing of recorded materials and comprehensive validation and verification for issues we used to test! For example if there is audio at all or video, if so, is it good and proper or is it faulty? Does it have glitches, frame issues and other possible obstructions? **IMPORTANT:** Make sure that all existing tests and Challenges do work in anti-bluff manner — they **MUST** confirm that all tested codebase really works as expected!”

§11.4.2 mandates *captured* evidence; §11.4.5 mandates the **content** of that evidence be analyzed for quality, not merely for presence. A test that captures a 0-byte mp4 (Bug #24) and PASSes because “the recording file exists” is the exact PASS-bluff pattern §11.4 forbids. Content-quality analysis is what closes that gap.

Audio quality analysis — every audio test that PASSes **MUST verify **ALL** of:**

(1) **Presence** — non-trivial RMS amplitude in captured `WAV / /proc/asound/.../pcm*p/sub0/hw_params`. (2) **Channel count** — `ffprobe -show_streams` matches the test’s claim (2.0 / 5.1 / 7.1). (3) **Sample rate + bit**

depth — match the codec / pipeline under test. (4) **Glitch census** — XRUN / FastMixer underrun-overflow-partial / AudioFlinger writeError counts above tolerance MUST classify explicitly (PASS within budget, WARN above, FAIL on hard limits per §11.4.1 SKIP-vs-FAIL decision tree). (5) **Coexistence-artifact census** — for tests that exercise WiFi/BT alongside audio: BT TX queue overflow, A2DP src underflow, coex notification storms, 2.4 GHz radio contention.

Video quality analysis — every video test that **PASSes** MUST verify **ALL** of: (1) **Presence** — captured screen recording has non-zero file size AND `ffprobe -count_frames` reports decoded-frame total > 0. 0-byte mp4 (Bug #24) is the canonical PASS-bluff and triggers §11.4.4 STOP. (2) **Routing target** — analyzer + action-timeline confirms video appeared on the *intended* display (primary vs secondary HDMI; Bug #13 pattern). (3) **Frame health** — drop count, frame-time variance (jitter), freeze detection (SSIM > 0.99 for ≥ 1 s), tearing. (4) **Obstruction census** — Tesseract OCR scan for hostile overlays (Application not responding, Force close, sign-in dialog, geo-restriction overlay, ad break, paywall, App is not certified). (5) **Resolution + codec** — captured frame dimensions match the test's claim; downgrade is a PASS-bluff.

Challenges (HelixQA) are bound equally — every Challenge that asserts PASS MUST run all five audio + five video layers. A Challenge that scores PASS without applicable analysis is the same class of defect as a unit test that does.

Tooling guarantee: audio = `tinycap + aplay --dump-hw-params + ffprobe + /proc/asound parsers (lib/audio_validation.sh` per §11.2.5). Video = `screenrecord + ffprobe -count_frames + recording-analyzer + Tesseract OCR (scripts/dual_display_record.sh + cmd/recording-analyzer/` per §11.4.2.A and §11.4.2.C). Tests dispatched against video evidence MUST honor §11.4.4 test-interrupt-on-discovery when the analyzer reports empty input — do not silently absorb that as a generic PASS-bluff banner.

Non-compliance is a release blocker regardless of context.

§11.4.6 — No-guessing mandate (User mandate, 2026-05-08)

Forensic anchor — direct user mandate (verbatim, 2026-05-08T18:30 MSK):

“‘LIKELY’ is guessing, we MUST NOT have guessing, since it can be or may not be! No bluffing and uncertainty is allowed at any cost! We MUST always know exactly precisely what is happening exactly, in any context, under any conditions, everywhere!”

Tests, gates, status reports, closure narratives, commit messages, and operator-facing text MUST NOT use likely, probably, maybe, might, possibly, presumably, seems, or appears to when describing causes of failures, behaviour, or fix effectiveness. Either prove the cause with captured forensic evidence (`logcat`, `dmesg`, `/sys` readings, `getprop`, kernel ramoops, `dropbox`, `strace`, etc.) and state it as fact, OR explicitly mark **UNCONFIRMED:** / **UNKNOWN:** / **PENDING_FORENSICS:** with a tracked-task ID for follow-up.

Pre-build gate **CM-NO-GUESSING-MANDATE** greps recently-modified docs + test scripts for the forbidden vocabulary outside explicit **UNCONFIRMED:** / **UNKNOWN:** / **PENDING_FORENSICS:** blocks. Paired mutation introduces a likely token into a

fresh status block → gate FAILs. Propagation gate CM-COVENANT-114-6-PROPAGATION enforces this anchor in every CLAUDE.md / AGENTS.md across parent + 10 owned submodules + HelixQA dependencies.

Canonical authority: parent [docs/guides/ATMOSPHERE_CONSTITUTION.md](#) §11.4.6.

Non-compliance is a release blocker regardless of context.

§11.4.7 — Demotion-evidence rule (Phase 38.X+2 amendment, 2026-05-11)

A demotion from any FAIL classification (OPEN, POSSIBLE PRODUCT DEFECT, FAIL) to a lower-severity classification (INVESTIGATED, MITIGATED, RESOLVED, WORKING-AS-INTENDED) requires positive evidence captured under the **same conditions** that originally exposed the defect — same device, same firmware, same cycle position, same load profile.

“I cannot reproduce in isolation” is a HYPOTHESIS, not a finding. Per §11.4.6 it **MUST** be tagged UNCONFIRMED: until same-conditions retest produces positive evidence. The expanded forbidden-vocabulary list:

Forbidden phrase	Why it bluffs
“isolated re-run PASSes therefore X was a flake”	Strips the very environment that exposed the defect.
“runtime drift”	Label for “we don’t know what changed”.
“intermittent” / “transient”	Label for “we don’t know how to reproduce”.
“pending stress retest”	Defers the actual investigation indefinitely.
“correlates with X”	Hypothesis presented as causation.

Pre-build gate CM-DEMOTION-EVIDENCE-RULE scans Issues.md / Fixed.md / CONTINUATION.md for these phrases outside explicit UNCONFIRMED: / UNATTRIBUTED: / PENDING_CYCLE_RETEST: blocks. Propagation gate CM-COVENANT-114-7-PROPAGATION enforces this anchor in every CLAUDE.md / AGENTS.md across parent + 10 owned submodules + HelixQA dependencies.

Canonical authority: parent [docs/guides/ATMOSPHERE_CONSTITUTION.md](#) §11.4.7.

Non-compliance is a release blocker regardless of context.

§11.4.8 — Deep-web-research-before-implementation mandate (User mandate, 2026-05-12)

Before designing a non-trivial fix, implementing a new feature, or declaring an architectural choice, perform deep web research to verify the chosen approach is informed by current state-of-the-art. Research surface: official documentation (Android/AOSP/Khronos/CEA-861/AES/IEEE/IETF/ITU), vendor technical guides (Rockchip, Sipeed, Audinate Dante, Synaptics, Realtek, Bluetooth SIG), open-

source codebases (Linux kernel, ALSA, Bluez, ExoPlayer, libVLC, MPV, FFmpeg, AOSP forks), coding tutorials + technical articles (Stack Overflow, AOSP Code Lab, AES papers), issue trackers (Android bug tracker, AOSP Gerrit, GitHub issues).

A fix that re-invents a wheel — or reproduces a known-broken pattern — when the open-source community has already solved the problem is a §11.4 violation by omission. Every non-trivial fix's commit / Issues.md / Fixed.md entry MUST cite at least one external source URL OR the literal “NO external solution found — original work”.

Pre-build gate CM-RESEARCH-CITATION-PRESENT scans new fix-direction blocks for the pattern. Propagation gate CM-COVENANT-114-8-PROPAGATION enforces this anchor in every CLAUDE.md / AGENTS.md across parent + 10 owned submodules + HelixQA dependencies.

Documentation continuity requirement: every fix landed under §11.4.8 also adds to docs/guides/ a user-facing or developer-facing guide section where appropriate.

Canonical authority: parent [docs/guides/ATMOSPHERE_CONSTITUTION.md](#) §11.4.8.

Non-compliance is a release blocker regardless of context.

§11.4.9 — Batch-source-fixes-before-rebuild mandate (User mandate, 2026-05-12)

When closing a multi-defect batch, all source-side fixes that DO NOT require runtime on-device validation to design MUST be landed BEFORE the next firmware rebuild. Anti-pattern eliminated:
Fix A → rebuild → flash → cycle → fix B → rebuild → ... serializes 7-8 hours per fix instead of batching all into ONE build cycle. Operator time is the scarce resource.

Exceptions documented in commit message as REQUIRES_REBUILD: <reason>: kernel-5.10/ changes, atmosphere-*.sh boot-script side-effects, hardware/rockchip/ HAL behavior — each gates downstream state and requires firmware to validate.

Before declaring a batch “ready for rebuild”: pre-build GREEN + meta-test GREEN + existing-device validations performed where possible + Issues.md/Fixed.md/ CONTINUATION.md in sync (+ HTML/PDF exported) + §11.4.8 research citations all logged.

Propagation gate CM-COVENANT-114-9-PROPAGATION enforces this anchor in every CLAUDE.md / AGENTS.md across parent + 10 owned submodules + HelixQA dependencies.

Canonical authority: parent [docs/guides/ATMOSPHERE_CONSTITUTION.md](#) §11.4.9.

Non-compliance is a release blocker regardless of context.

§11.4.10 — Credentials-handling mandate (User mandate, 2026-05-12)

All credentials, secrets, API tokens, passwords, phone numbers, OAuth tokens, signing keys MUST NEVER live in tracked files. Templates with placeholder values are allowed (.example suffix). Tests load credentials at runtime from scripts/

testing/secrets/ (or per-submodule equivalent); operator-populated files are `chmod 600`, directory is `chmod 700`. `.env`, `.env.*`, `*.env` patterns + `scripts/testing/secrets/*` (with `.example` + `README.md` exception) git-ignored project-wide.

Test scripts MUST NEVER echo credentials to `stdout/stderr/logcat`. Screen-recording of sign-in flows MUST redact credential-bearing frames. Per-service file separation (`.netflix.env`, `.disney.env`, etc.) limits blast radius.

Forensic-rotation policy: suspected leak → rotate at provider, update local `.env`, audit captured artifacts. Pre-build gate `CM-CREDENTIAL-LEAK-SCAN` greps tracked files for entropy-suspicious password strings + known API-token formats. Propagation gate `CM-COVENANT-114-10-PROPAGATION` enforces this anchor in every `CLAUDE.md` / `AGENTS.md` across parent + 10 owned submodules + HelixQA dependencies.

Canonical authority: parent [docs/guides/ATMOSPHERE_CONSTITUTION.md](#)
§11.4.10.

Non-compliance is a release blocker regardless of context.

§11.4.13 — Out-of-band sink-side captured-evidence mandate (User mandate, 2026-05-13)

Whenever an HDMI sink with a network-accessible introspection API is present (current example: Arvus H2-4D-273 at <http://192.168.4.185/>), the test suite MUST consume the sink's report as captured-evidence for every audio test asserting a codec / channel-count / passthrough mode. On-SoC HAL telemetry ALONE is insufficient — that is the exact “tests pass but the feature doesn't work” pattern §11.4 forbids. Reference: `scripts/testing/lib/arvus_probe.sh`, `scripts/testing/arvus_probe.sh`, [docs/guides/ARVUS_HDMI_INTEGRATION.md](#). Pre-build gate `CM-ARVUS-EVIDENCE-INTEGRATED` (7 invariants) + paired mutation. No hardcoding (`env: ARVUS_HOST` etc.). Topology dispatch per §11.4.3 — sink unreachable → `SKIP`, never `FAIL`. Identity verification (MAC match) before consuming codec-state. Anti-stickiness post-stop. HelixQA Challenges bound equally.

Canonical authority: parent [docs/guides/ATMOSPHERE_CONSTITUTION.md](#)
§11.4.13. Integration reference: [docs/guides/ARVUS_HDMI_INTEGRATION.md](#).

Non-compliance is a release blocker regardless of context.

§11.4.14 — Test playback cleanup mandate (User mandate, 2026-05-13)

Every test that issues `am start / cmd media_session play / MediaController.play` MUST issue matching `am force-stop / input keyevent KEYCODE_MEDIA_STOP` + register cleanup in `EXIT` trap. Verified via positive evidence (Arvus codec-state → `N.E.`, `dumpsys media_session` shows no `PLAYING` for test app). `test_all_fixes.sh` post-test sanity check FAILs the just-completed test if it left orphan playback. HelixQA Challenges bound equally. No grace period — “next test will clean it up” is §11.4 `PASS`-bluff.

Canonical authority: parent [docs/guides/ATMOSPHERE_CONSTITUTION.md](#)
§11.4.14. Pre-build gates `CM-TEST-PLAYBACK-CLEANUP` + `CM-COVENANT-114-14-PROPAGATION`.

Non-compliance is a release blocker regardless of context.

§11.4.15 — Item-status tracking mandate (User mandate, 2026-05-13)

Every active item in docs/Issues.md carries a ****Status:**** line with one of six values: Queued, In progress, Ready for testing, In testing, Reopened, Fixed (→ Fixed.md). Status MUST be updated as the item progresses through its lifecycle. Fixed requires captured-evidence per §11.4.5 + migration to Fixed.md.

The auto-generated docs/Issues_Summary.md includes the Status column. All three file types (.md, .html, .pdf) MUST be in sync at all times — enforced by CM-DOCS-EXPORT-SYNC (§11.4.12 + §11.4.15 amendment).

Canonical authority: parent docs/guides/ATMOSPHERE_CONSTITUTION.md
§11.4.15. Pre-build gates CM-ITEM-STATUS-TRACKING + CM-COVENANT-114-15-PROPAGATION.

Non-compliance is a release blocker regardless of context.

§11.4.16 — Item-type tracking mandate (User mandate, 2026-05-14)

Every active item in docs/Issues.md carries a ****Type:**** line with one of three values: Bug (product defect / regression / user-visible broken behaviour), Feature (new capability not previously offered to end users), Task (internal workstream — refactor, doc, infra, gate, audit; the lowest-stakes default when ambiguous). The vocabulary is CLOSED — no other value is permitted.

The auto-generated docs/Issues_Summary.md includes the Type column. All three file types (.md, .html, .pdf) MUST be in sync at all times — enforced by CM-DOCS-EXPORT-SYNC (§11.4.12 + §11.4.15 + §11.4.16 amendment).

Canonical authority: parent docs/guides/ATMOSPHERE_CONSTITUTION.md
§11.4.16. Pre-build gates CM-ITEM-TYPE-TRACKING + CM-COVENANT-114-16-PROPAGATION.

Non-compliance is a release blocker regardless of context.

§11.4.40 — Full-suite retest before release tag mandate (User mandate, 2026-05-17)

A release tag MUST NOT be created until a COMPLETE retest with ALL existing tests has been executed on a clean baseline AFTER every workable item in the batch is done, fixed, polished, and individually verified. Spot-check retests that run only the tests touched by the batch are FORBIDDEN — they miss interaction defects between the batch's fixes and previously-stable code.

The complete retest comprises: (1) pre-build full sweep, (2) post-build full sweep, (3) on-device 4-phase cycle on EVERY owned device, (4) meta-test full mutation sweep, (5) Challenge bank full sweep, (6) Issues.md/Fixed.md state audit, (7) CONTINUATION.md sync check.

Time is essential — complete retest is typically 12–48 hour elapsed effort. NOT optional, NOT abbreviated. Skipping is the exact “tests passed but feature broken” failure mode §11.4 specifically prohibits.

Composes with §11.4.4 (per-fix retest) — §11.4.37 is the additional final integrity check at RELEASE granularity. Composes with §11.4.7 — full-suite retest is the authoritative baseline for closures in the batch. No escape hatch — no `--skip-full-retest` or `--quick-release` flag exists.

Pre-build gate `CM-FULL-SUITE-RETEST-MANDATE` + paired mutation. Propagation gate `CM-COVENANT-114-40-PROPAGATION` enforces this anchor in every `CLAUDE.md/AGENTS.md` across parent + 10 owned submodules + HelixQA dependencies.

Canonical authority: constitution submodule `Constitution.md` §11.4.37.

Non-compliance is a release blocker regardless of context.

§11.4.41 — Pre-Force-Push Merge-First Mandate (User mandate, 2026-05-17)

Any force-push (`git push --force`, `git push --force-with-lease`, `git push +<ref>`, or equivalent history-rewriting operation on any remote) authorised under §9.2 / CONST-043 MUST be preceded by a mechanical 4-step merge-first pipeline that brings every remote-side commit into the local tree, resolves every conflict carefully, and verifies nothing is lost or corrupted on EITHER side BEFORE the overwriting push is executed.

The 4-step pipeline (mandatory, in order): (1) `git fetch --all --prune --tags` against every configured remote — capture output. (2) Integrate every divergent commit locally via `git rebase` (local is strict superset), `git merge` (independent additions both deserve preservation), or operator-confirmed cherry-pick (remote subset already present locally). (3) Audit: no conflict markers (`grep -rn '^<<<<<< \|^===== $|^>>>>>>'` returns empty), no silent file drops (`git diff --stat HEAD@{1} HEAD`), every previously-passing test still passes per §11.4.4 / §11.4.40 baseline, every captured-evidence artifact still validates. (4) `git push --force-with-lease <remote> <ref>` (NEVER `--force` without `--with-lease` unless §9.2 sub-clause 6 explicitly authorises it for a remote where lease semantics are unavailable). One force-push event per CONST-043 authorisation — no batch authorisation.

Two-gate composition with CONST-043 — §11.4.41 does NOT relax CONST-043's operator-approval requirement. Gate A (CONST-043): operator types explicit per-operation force-push authorisation. Gate B (§11.4.41): agent executes the 4-step merge-first pipeline, captures evidence of clean integration, presents evidence to operator BEFORE the force-push. Both gates required.

Verification artefact — every §11.4.41-governed force-push emits a `docs/changelogs/<tag>.md` “Force-push merge-first audit” section containing 7 elements: (i) `git fetch` output, (ii) per-remote `HEAD..<remote>/<branch>` log before integration, (iii) integration strategy chosen per remote with rationale, (iv) post-integration conflict-marker scan output (must be empty), (v) post-integration test suite delta (must show only expected changes), (vi) `--force-with-lease` push output with lease SHA evidence, (vii) CONST-043 authorisation quote from the conversation.

Composes with §9.2 (data-safety hardlinked backup), §11.4.4 (test-interrupt-on-discovery — broken integration triggers rollback), §11.4.6 (no-guessing — every step's outcome captured, not assumed), §11.4.26 (constitution-submodule update pipeline — per-submodule specialisation), §11.4.32 (post-pull validation — audit

step's mechanical companion), §11.4.37 (fetch-before-edit — step 1 enforces it for force-push specifically), §11.4.40 (full-suite retest — step 3's test-evidence requirement).

No escape hatch — the operator-pressure escape (“just force-push, we’ll fix it later”) is the exact failure mode this anchor closes. Pre-build gate CM-COVENANT-114-41-PROPAGATION enforces this anchor in every CLAUDE.md/AGENTS.md across parent + 10 owned submodules + nested submodules + HelixQA dependencies. Paired mutation strips the anchor literal → gate FAILs. Gate CM-FORCE-PUSH-MERGE-FIRST walks docs/changelogs/<tag>.md “Force-push” entries for the 7 audit elements; paired mutation strips any element and asserts gate FAILs.

Canonical authority: constitution submodule Constitution.md §11.4.41.

Non-compliance is a release blocker regardless of context.

MANDATORY §12.6 MEMORY-BUDGET CEILING — 60% MAXIMUM (User mandate, 2026-04-30)

Forensic anchor — direct user mandate (verbatim):

“We had to restart this session 3rd time in a row! The system of the host stays with no RAM memory for some reason! First make sure that whatever we do through our procedures related to this project **MUST NOT** use more than 60% of total system memory! All processes **MUST** be able to function normally!”

The mandate. Project procedures **MUST NOT** use more than **60% of total system RAM** (HOST_SAFETY_MAX_MEM_PCT). The remaining 40% is reserved for the operator's other workloads so the host can keep serving them while project work proceeds.

Three consecutive session-loss SIGKILLs on 2026-04-30 during 1.1.5-dev — every one happened while scripts/build.sh was running m -j5 AOSP. Each Soong/Ninja job peaks at ~5–8 GiB RSS; collective RSS overran the 60% envelope and the kernel OOM-killer escalated, taking down user@1000.service. **§12.1's pre-flight check (refusing to start if host already distressed) was not enough** — the missing piece was an active CONSTRAINT on heavy work itself.

Mandatory protections (rock-solid):

1. HOST_SAFETY_MAX_MEM_PCT defaults to 60 in scripts/lib/host_session_safety.sh.
2. HOST_SAFETY_BUDGET_GB is computed at source-time from $\text{MemTotal} \times \text{MAX_PCT}/100$.
3. bounded_run clamps MemoryMax down to the budget if the caller asks for more (cgroup-level enforcement via systemd-run --user --scope -p MemoryMax=...).
4. host_safe_parallel_jobs and host_safe_build_jobs return the safe -j count given an estimated per-job RSS, capped at nproc.

5. `scripts/build.sh` wraps `m -j` in `bounded_run`. If the build's collective RSS exceeds the budget, only the scope is OOM-killed; `user@<uid>.service` stays alive.

Captured-evidence enforcement. Pre-build gate `CM-MEMBUDGET-METATEST` locks all 7 invariants and fires every pre-build run.

No escape hatch. §12.6 has NO operator-facing override flag. The cap exists for the operator's own protection; bypassing it is the bluff the §11.4 covenant specifically prohibits. Operators who need more headroom should reduce parallelism, close other workloads, or add RAM — NOT raise the percentage.

Canonical authority: parent [docs/guides/ATMOSPHERE_CONSTITUTION.md](#) §12.6.

Non-compliance is a release blocker regardless of context. *Remember: Your code will be used by real people. Write code that actually works.*

§11.4.85 — Stress + Chaos Test Mandate (User mandate, 2026-05-24)

Forensic anchor — direct user mandate (verbatim, 2026-05-24):

“Every fix or improvement you do MUST BE covered with full automation stress and chaos tests so we are sure nothing can break the functionality and all edge cases are monitored and polished and additionally fixed if that is needed! Everything must produce rock solid proofs and follow fully no-bluff policy!”

Every fix or improvement landed in this project MUST ship with full-automation **stress** AND **chaos** test suites that exercise edge cases, sustained load, concurrent contention, and failure-injection. Happy-path coverage alone is a §11.4 / §107 PASS-bluff at the resilience layer.

Stress (closed-set, mechanically auditable): sustained load ($N \geq 100$ iterations OR ≥ 30 s wall-clock; per-iteration latency p50/p95/p99 recorded) + concurrent contention ($N \geq 10$ parallel invocations; no deadlock, no resource leak) + boundary conditions (empty / max / off-by-one input; every boundary produces a categorised result, never an uncaught exception).

Chaos (closed-set, applied per fix-class appropriateness): process-death injection (kill primary or upstream mid-call; categorised recovery) + network-fault injection (drop/delay/reorder; category=network|upstream per §11.4.69) + input-corruption injection (corrupt .env / config / input file mid-test; detected + reported) + resource-exhaustion injection (disk full, OOM, FD exhaustion; refuse cleanly OR degrade gracefully — NEVER crash) + state-corruption injection (mid-flight lock loss, partial-write fault; recovery restores consistent state).

Anti-bluff (mandatory). Every stress + chaos test PASS cites a captured-evidence artefact path per §11.4.5 + §11.4.69 (`per-iteration_latency.json`, `categorised_errors.txt`, `state_delta_snapshot.json`, `recovery_trace.log`). Helper library `stress_chaos.sh` provides `ab_stress_run`, `ab_stress_concurrent`, `ab_chaos_kill_pid_during`, `ab_chaos_drop_network_during`, `ab_chaos_corrupt_file_during`, `ab_chaos_oom_pressure_during`, `ab_chaos_disk_full_during`, each

composing with `ab_pass_with_evidence / ab_skip_with_reason`. Chaos-injection cleanup is non-negotiable — corrupt-restore, disk-fill-cleanup, process-restart MUST run in `trap '...' EXIT`; cleanup failure = §11.4.14 violation.

4-layer coverage per §11.4.4(b): pre-build gate (stress + chaos test files exist + executable + parseable under `sh -n + bash -n` per §11.4.67; helper library exists; the fix's pre-build gate cites the stress + chaos test file path) + paired meta-test mutation per §1.1 (stripping chaos-injection or per-iteration evidence capture → gate FAILs) + on-device test (if `LIVE_ADB_TESTABLE` per §11.4.51, dispatched against real device, evidence under `qa-results/<run-id>/stress_chaos/`) + HelixQA Challenge entry (if user-visible feature per §11.4.4(b) layer 4).

Composes with §11.4 / §107 (resilience IS end-user quality), §11.4.1 (FAIL-bluffs forbidden), §11.4.5 (captured-evidence quality applies to latency distribution + error categories), §11.4.6 (no guessing — categorised errors only), §11.4.43 (TDD RED-first under load/chaos), §11.4.50 (N iterations identical exit + identical evidence-hashes), §11.4.52 (autonomous validation), §11.4.69 (universal sink-side positive-evidence taxonomy), §11.4.83 (recovery transcripts ARE end-user-channel proofs).

Canonical authority: constitution submodule Constitution.md §11.4.85.

Non-compliance is a release blocker regardless of context. No escape hatch — no `--skip-stress`, `--no-chaos`, `--happy-path-suffices`, `--stress-test-later` flag exists.

§11.4.87 — Endless-loop autonomous work + zero-idle agent dispatch + anti-bluff testing mandate (User mandate, 2026-05-26)

When operator instructs an AI agent to “continue in endless loop fully autonomously” (or semantically-equivalent), the agent MUST treat as HARD-CONTRACT covenant covering five obligations: (A) continue until `docs/Issues.md` non-terminal Status entries = 0 AND `docs/CONTINUATION.md` §3 Active work empty AND no subagent in-flight AND no external dep in-flight; (B) dispatch background subagents for parallelisable work — main + subagents concurrent, “waiting for results” is the ONLY idle reason; (C) every closure lands four-layer test coverage per §11.4.4(b) with captured-evidence “physical proofs” (tinycap WAV + RMS / screen recording + `ffprobe` / `dumpsys` + sink-probe / `uiautomator dump` / `sysfs` snapshots) — metadata-only / config-only / absence-of-error / `grep`-without-runtime PASS are critical defects; (D) §11.4 anti-bluff covenant family operative end-to-end (tests AND HelixQA Challenges bound equally per forensic anchor “tests pass but features don’t work”); (E) loop terminates ONLY on all-conditions-met, explicit operator STOP, host-safety demand (§12 family), or scheduled wake on known-future-actionable signal.

Composes with §11.4 / §11.4.1 / §11.4.2 / §11.4.4 / §11.4.5 / §11.4.6 / §11.4.7 / §11.4.20 / §11.4.27 / §11.4.42 / §11.4.43 / §11.4.50 / §11.4.52 / §11.4.58 / §11.4.68 / §11.4.69 / §11.4.70 / §11.4.83 / §11.4.85 / §11.4.86 / §12.10. Pre-build gate CM-COVENANT-114-87-PROPAGATION + paired §1.1 mutation.

Canonical authority: constitution submodule Constitution.md §11.4.87.

Non-compliance is a release blocker regardless of context. No escape hatch — no `idle-OK`, `--skip-endless-loop`, `--bluff-permitted-for-this-task`, `--metadata-only-test-suffices`, `--no-physical-proof-required` are FORBIDDEN flags.

§11.4.11 — File-layout discipline (User mandate, 2026-05-12)

Files live in canonical directories per type: Shell scripts → `scripts/` (legacy: `scripts/legacy/`); Log files → `logs/`; Release artifacts → `releases/<app>/<version>/`; Operator credentials → `scripts/testing/secrets/` (per §11.4.10, git-ignored); Markdown docs → `docs/` + `docs/guides/` + `docs/research/`; Per-version changelogs → `docs/changelogs/`. Project files organised by purpose, not historical accident.

Canonical authority: constitution submodule `Constitution.md` §11.4.11. Non-compliance is a release blocker.

§11.4.12 — `Issues_Summary.md` sync mandate (User mandate, 2026-05-12)

`docs/Issues_Summary.md` is the canonical short-form summary of all open items. MUST be regenerated + re-exported (HTML + PDF) whenever `Issues.md` changes. Generator: `scripts/testing/generate_issues_summary.sh`. Pre-build gates `CM-ISSUES-SUMMARY-SYNC` + `CM-COVENANT-114-12-PROPAGATION` enforce mechanically. Composes with §11.4.15 / §11.4.16 / §11.4.19 / §11.4.23.

Canonical authority: constitution submodule `Constitution.md` §11.4.12. Non-compliance is a release blocker.

§11.4.75 — Mechanical Enforcement Without Exception (cascaded from constitution submodule §11.4.75)

Verbatim user mandate (2026-05-20): *“Why do these violations still happen!? This is a serious problem! We cannot rely on stability nor consistency if we cannot respect our Constitution, mandatory rules and constraints! Is there a way to make this always respected, followed and applied without exception fully and unconditionally!? WE MUST HAVE THIS WORKING FLAWLESSLY!!! Do investigate the root causes of such problems! Once all problems are identified WE MUST apply proper mechanisms for this not to happen NEVER EVER AGAIN!”*

The §11.4 covenant historically relied on agent + operator vigilance; three 2026-05-19→20 forensic incidents proved that late-binding enforcement fires hours-to-days after the violator commit reaches every remote. §11.4.75 closes the gap with FIVE independent mechanical enforcement layers — bypassing any single layer does not bypass the discipline: (1) local pre-commit git hook (refuses staged `.md` lacking sibling `.html` + `.pdf`); (2) `commit_all.sh` integration (`_constitution_sibling_check` + `auto-sync_all_markdown_exports.sh` self-repair); (3) local pre-push git hook (re-runs siblings + propagation-gate subset); (4) post-commit auto-repair hook (auto-generates orphan-`.md` siblings, idempotent + recursion-guarded); (5) local-only final-gate ritual (remote CI DISABLED per User mandate — operator runs `pre_build_verification.sh` + `meta-test` before every tag per §11.4.40). Helper contracts: `scripts/install_git_hooks.sh`, `scripts/git_hooks/{pre-commit,pre-push,post-commit,commit-msg}`, `_constitution_sibling_check`. The `commit-msg` hook enforces a Bypass-rationale: `<reason>` footer when `--no-verify` is detected; `docs/audit/bypass_events.md` accumulates the audit trail. Five gates with paired §1.1

mutations: CM-COVENANT-114-75-PROPAGATION, CM-GIT-HOOKS-INSTALL-SCRIPT, CM-GIT-HOOKS-SOURCE-DIR, CM-COMMIT-ALL-SIBLING-CHECK, CM-CI-WORKFLOW-PRESENT. No escape hatch — no `--skip-hooks`, `--bypass-enforcement`, `--allow-orphan-md`, `--ci-not-applicable`, `--mechanical-enforcement-not-needed` flag.

Cascade requirement: This anchor (verbatim or by §11.4.75 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-75-PROPAGATION; paired mutation strips the literal → gate FAILs. Severity-equivalent to a §11.4 PASS-bluff at the enforcement layer. **Canonical authority:** constitution submodule Constitution.md §11.4.75 for the full mandate.

§11.4.76 — Containers-Submodule Mandate (cascaded from constitution submodule §11.4.76)

Verbatim user mandate (2026-05-20): *“For any work or requirements of running services or codebase inside the Containers (Docker / Podman / Qemu / Emulators, and so on) we MUST USE / INCORPORATE the Containers Submodule properly: <https://github.com/vasic-digital/containers> (git@github.com:vasic-digital/containers.git). Containers Submodule contains all means for us to Containerize our code and services! If any feature or Containing System is missing or not supported we MUST EXTEND IT properly like we do all of our projects! No bluff work is allowed of any kind!”*

For ANY containerized workload (Docker / Podman / Qemu / Kubernetes / container-backed emulators), every consuming project MUST: (1) install vasic-digital/containers (digital.vasic.containers) as a Git submodule; (2) consume via replace directive during development + pinned commit SHAs in production; (3) boot infra on-demand via pkg/boot + pkg/compose + pkg/health so operators are never required to start podman machine / docker compose up manually — the boot is part of the test entry point (the on-demand-infra invariant); (4) extend the Submodule (PR upstream) for missing runtimes / lifecycle primitives — never reimplement in-project (per §11.4.74); (5) anti-bluff: integration tests claiming to exercise containerized components MUST actually boot them via the Submodule — short-circuit fakes that bypass boot are a §11.4 violation. Tracker rows touching containerization MUST record Catalogue-Check: extend vasic-digital/containers@<sha> (or reuse). Planned gate CM-CONTAINERS-USED scans container-touching PRs for digital.vasic.containers/... imports; paired mutation strips the import + asserts FAIL.

Cascade requirement: This anchor (verbatim or by §11.4.76 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-76-PROPAGATION; paired mutation strips the literal → gate FAILs. **Canonical authority:** constitution submodule Constitution.md §11.4.76 for the full mandate.

§11.4.77 — Regeneration-Mechanism-Required Mandate (cascaded from constitution submodule §11.4.77)

Verbatim user mandate (2026-05-20): *“We must be sure that after excluding anything from Git versioning we still have the mechanism which will out of the box obtain or re-generate missing content!”*

Every `.gitignore` entry excluding (a) >~100 MiB OR (b) any artefact essential to building / running / testing the project MUST carry a documented + automated mechanism to either re-obtain (download from authoritative source: vendor tarball, SDK installer, npm/pip/cargo/go-mod/container registry, dedicated git submodule, S3/GCS) OR re-generate (run from tracked source via build pipeline, code-gen, asset render, captured-evidence replay, container build). Required artefacts per qualifying entry: (1) `.gitignore-meta/<entry-slug>.yaml` declaring pattern + mechanism-type + script-path + expected-disk-usage + vendor-url-or-source + integrity hash + requires-network + requires-credentials; (2) a non-interactive entry in `scripts/setup.sh` post-clone bootstrap; (3) a pre-build gate verifying regenerated content present OR a recent `.gitignore-meta/.regenerated/<slug>.ok` stamp; (4) `README + docs/guides/*.md` describing the mechanism + manual fallback + time/disk budget + §11.4.10 credentials. Bare `.gitignore` additions without the mechanism are a §11.4 PASS-bluff variant — codebase appears complete but a fresh clone cannot build/run. No escape hatch — no `--skip-regen-mechanism`, `--gitignore-is-enough`, `--operator-already-has-content` flag. Planned gate `CM-GITIGNORE-REGEN-MECHANISM` + paired §1.1 mutation (strip a required YAML key → gate FAILs).

Cascade requirement: This anchor (verbatim or by §11.4.77 reference) MUST appear in every owned submodule’s `CONSTITUTION.md`, `CLAUDE.md`, and `AGENTS.md`. Propagation gate `CM-COVENANT-114-77-PROPAGATION`; paired mutation strips the literal → gate FAILs. Severity-equivalent to a §11.4 PASS-bluff at the repository-hygiene layer. **Canonical authority:** constitution submodule `Constitution.md` §11.4.77 for the full mandate.

§11.4.78 — CodeGraph Code-Intelligence Mandate (cascaded from constitution submodule §11.4.78)

Verbatim user mandate (2026-05-20): *“Make codegraph MANDATORY CHOICE for this purpose for all of our project ... All project which do not have configured and installed codegraph yet MUST DO IT and MUST USE IT!”*

Every consuming project worked on by AI coding agents MUST install, initialize, and use **CodeGraph** (<https://github.com/colbymchenry/codegraph>, `npm @colbymchenry/codegraph`) — a local SQLite semantic code-knowledge-graph exposed to agents over MCP (100% local, no cloud). (1) Install globally via npm with a user-writable npm prefix (no sudo). (2) `codegraph init` + `codegraph index`: `.codegraph/config.json` is tracked, `.codegraph/codegraph.db` is gitignored with `codegraph index` as its §11.4.77 regeneration mechanism; the `config.json` exclude list MUST exclude every credential/secret path per §11.4.10. (3) Wire `codegraph serve --mcp` into every CLI agent (Claude Code

.mcp.json, OpenCode opencode.json, Qwen Code .qwen/settings.json, Crush .crush.json, host-local otherwise) referencing the bare codegraph command on PATH (no hardcoded host path). (4) Cover the integration with an anti-bluff suite whose per-agent end-to-end layer uses an unforgeable challenge (a fact obtainable only by calling a CodeGraph MCP tool, e.g. index node count via codegraph_status); a genuinely un-drivable agent is a documented SKIP per §11.4.3, never a faked PASS. (5) Document in docs/CODEGRAPH.md, kept in sync per §11.4.12 / §11.4.65. CodeGraph is consumed as the published npm package (§11.4.74) — not a git submodule, adds no Git remote. Planned gate CM-CODEGRAPH-WIRED + paired §1.1 mutation (strip a secret-exclusion → gate FAILs).

Cascade requirement: This anchor (verbatim or by §11.4.78 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-78-PROPAGATION; paired mutation strips the literal → gate FAILs. **Canonical authority:** constitution submodule Constitution.md §11.4.78 for the full mandate.

§11.4.79 — Own-Org Submodules MUST Be Included in the CodeGraph Index (cascaded from constitution submodule §11.4.79)

Verbatim user mandate (2026-05-21): *“All Submodules we use in the project and that are part of organizations to which we have the full access via GitHub, GitLab and other CLIs MUST BE included into the codegraph database and initialized / scanned / synced!”*

Refines §11.4.78's exclude-list with a per-submodule-ownership split: (a) own-org submodules (full write access via the project's CLIs — canonical orgs vasic-digital + HelixDevelopment) MUST be INCLUDED in the index; (b) third-party submodules (the §11.4.74 no-match → vendor path) MUST be EXCLUDED. Operational steps: (1) git submodule update --remote --merge to pull latest before re-indexing, respecting load-bearing pins on third-party submodules; (2) adjust .codegraph/config.json exclude list to keep own-org paths in scope; (3) re-index via scripts/codegraph_setup.sh; (4) verify via scripts/codegraph_validate.sh with ≥1 probe resolving a symbol living ONLY inside an own-org submodule; (5) paired §1.1 mutation — temporarily add the own-org submodule to exclude → validate MUST FAIL on the cross-submodule probe → restore. An index that lies about reachable symbols is a PASS-bluff against AI agents. Own-org submodules silently excluded without an audit trail in .codegraph/config.json comments is a release blocker.

Cascade requirement: This anchor (verbatim or by §11.4.79 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-79-PROPAGATION; paired mutation strips the literal → gate FAILs. **Canonical authority:** constitution submodule Constitution.md §11.4.79 for the full mandate.

§11.4.80 — CodeGraph Regular-Update + Sync Automation Mandate (cascaded from constitution submodule §11.4.80)

Verbatim user mandate (2026-05-21): *“We MUST regularly check for the updates and execute codegraph npm updates so the latest version of it is always installed on the host machine! ... Make sure we have proper full automation bash scripts which will run regularly and that these are part of the constitution Submodule ... Make sure all updates, sync processes we do and important codegraph related events are all documented under docs/codegraph in Status and Status_Summary documents ... and regularly export them like all other Status docs into the PDF and HTML!”*

Three deliverables (all living in the constitution submodule, inherited by reference per §3 — consuming projects invoke at `${CONST_DIR}/scripts/codegraph_*.sh`, never copy): (1) `scripts/codegraph_update.sh` — npm-installs latest `@colbymchenry/codegraph` after a registry version check; appends old/new version to `docs/codegraph/Status.md`; anti-bluff verifies codegraph — version reflects the new version after install (npm exit 0 ≠ working binary). (2) `scripts/codegraph_sync.sh` — after a successful update runs `codegraph status` → `codegraph sync` . → `codegraph status` → the project’s `scripts/codegraph_validate.sh`; appends every step’s output to BOTH the project’s and the constitution’s `docs/codegraph/Status.md`. (3) `docs/codegraph/Status.md` + `Status_Summary.md` append-only ledgers, exported to `.html` + `.pdf` per §11.4.65. Cadence: weekly floor (per §11.4.45). A consuming project that has not run `codegraph_update.sh` in >2 weeks AND has open AI-agent work is a release blocker. Paired §1.1 mutation: downgrade installed version → script detects drift → restore.

Cascade requirement: This anchor (verbatim or by §11.4.80 reference) MUST appear in every owned submodule’s `CONSTITUTION.md`, `CLAUDE.md`, and `AGENTS.md`. Propagation gate `CM-COVENANT-114-80-PROPAGATION`; paired mutation strips the literal → gate FAILs. **Canonical authority:** constitution submodule `Constitution.md` §11.4.80 for the full mandate.

§11.4.81 — Cross-Platform-Parity Mandate (cascaded from constitution submodule §11.4.81)

Verbatim user mandate (2026-05-21): *“Any Linux-only blocker / issue we have MUST BE created macOS and other supported platforms equivalent! So, depending on platform proper implementation will be used for particular OS! EVERYTHING MUST BE PROPERLY EXTENDED AND UPDATED!”*

Every consuming project whose supported-platforms manifest lists more than one OS MUST, for every feature/test/gate/challenge/mutation depending on platform-specific primitives, ship a per-OS-equivalent implementation chosen at runtime via `uname -s` (or equivalent detection). Three sub-mandates: **(A) Per-OS implementation REQUIRED** — Linux `cgroup/systemd/proc` primitives MUST have documented per-OS equivalents (POSIX `setrlimit/ulimit`, macOS `launchd`, BSD `rctl`, Windows Job Object) chosen via runtime dispatch. **(B) Per-OS tests REQUIRED** — every platform-dependent gate test MUST have case “\$

(uname -s)" in branches with positive captured evidence per §11.4.2 + §11.4.5 in each branch; SKIP-with-reason acceptable ONLY when the platform genuinely cannot enforce the invariant. **(C) Honest kernel-gap citation + adjacent equivalent test REQUIRED** — where a Linux primitive has NO equivalent due to a documented kernel limitation (canonical: XNU does not enforce RLIMIT_AS for unprivileged processes), the test MUST detect the gap at runtime, SKIP with exact kernel reason + reproducer + honest-gap-doc link, AND provide an ADJACENT test exercising the closest invariant the platform CAN enforce (e.g. RLIMIT_CPU+SIGXCPU as the macOS proxy), itself anti-bluff with a paired §1.1 mutation. Gate CM-CROSS-PLATFORM-PARITY scans for case "\$(uname -s)" blocks asserting a non-SKIP branch (or honest-gap citation) per platform in the manifest; paired mutation strips a Darwin branch → gate FAILs. No escape hatch.

Cascade requirement: This anchor (verbatim or by §11.4.81 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-81-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker on multi-platform projects. **Canonical authority:** constitution submodule Constitution.md §11.4.81 for the full mandate.

§11.4.82 — Iteration-Speedup Discipline Mandate (cascaded from constitution submodule §11.4.82)

Verbatim user mandate (2026-05-22): *"How can we speed-up this whole development and fixing process? ... Do not forget to all speed optimizations critical rules and mandatory constraints MUST BE all added into our root (constitution Submodule) Constitution.md, CLAUDE.md, AGENTS.md and QWEN.md and all other relevant constitution Submodules files!"*

Iteration cycle time is a first-order quality enabler. Every consuming project's build / test / commit / debug pipeline MUST adopt these speedup disciplines AS MANDATORY (each independently enforceable): (A) Phase-1 forensic (superpowers:systematic-debugging) before any speculative source patch — speculative patches without FACT-grade root cause are §11.4.6 + §11.4.82 violations; (B) Live-ADB-First (or live-equivalent) before any rebuild — strengthens §11.4.51 to a release-blocker mandate; (C) 30-second pre-flight before launching rebuild orchestrators (device/sink reachability, host memory/disk, no stale locks, no orphan processes); (D) persistent build caches outside containers (ccache/sccache/Gradle daemon bind-mounted to host); (E) module-only rebuild for loadable-module-only changes; (F) parallel multi-device testing with separate qa-results/<TS>/<device-tag>/ outputs; (G) subagent scope discipline + worktree isolation (≤30 min budget, single-responsibility, isolation: "worktree" default); (H) lock-file + stale-process hygiene (clean .git/index.lock, disable auto git-gc in concurrent repos); (I) cycle telemetry per §11.4.24 (commit hash, per-phase wall-clock, speedup-flag set, outcome — aggregated weekly). Gate CM-ITERATION-SPEEDUP-DISCIPLINE audits recent cycles for telemetry citing which of (A)-(I) applied; paired §1.1 mutation strips the speedup-flag column → gate FAILs. No escape hatch — no --skip-phase1-forensic, --no-pre-flight, --rebuild-everything-always, --unlimited-subagent-scope, --ignore-locks, --no-telemetry flag.

Cascade requirement: This anchor (verbatim or by §11.4.82 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-82-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.82 for the full mandate.

§11.4.83 — docs/qa/ End-User Evidence Mandate (cascaded from constitution submodule §11.4.83)

Verbatim user mandate (2026-05-22): “every feature that ships MUST carry a recorded e2e communication transcript + any attached materials under docs/qa/<run-id>/ (per-feature subdirectories). A feature with no QA transcript is itself a §107 PASS-bluff — it claims to work but has no auditable runtime evidence. Bot-driven automation MUST preserve full bidirectional communication threads as proof.”

Every feature that ships MUST carry a recorded end-to-end communication transcript plus any attached materials (screenshots, request/response payloads, audio, file uploads) committed under docs/qa/<run-id>/ — one directory per feature run. Operative rule: (1) every consuming project MUST maintain a docs/qa/ tree, each new feature under docs/qa/<run-id>/ where <run-id> is monotonic + greppable (timestamp / ATM-NNN / other workable-item ID per §11.4.54); (2) transcripts MUST be full bidirectional — every prompt/command sent + every response received (one-sided is not a transcript); (3) attached materials MUST be committed in-repo (no external-only links — that is a §11.4.13 sink-side violation); (4) bot-driven / agent-driven QA automation MUST preserve the full conversation thread as the proof artefact; (5) release gates MUST refuse to tag a version that has any feature-shipping commit without its matching docs/qa/<run-id>/ directory. A feature with no QA transcript is a §11.4 / §107 PASS-bluff. Composes with §11.4.2 / §11.4.5 / §11.4.13 / §11.4.65 / §11.4.69 / §1.1.

Cascade requirement: This anchor (verbatim or by §11.4.83 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-83-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker — no --qa-evidence-optional escape hatch. **Canonical authority:** constitution submodule Constitution.md §11.4.83 for the full mandate.

§11.4.84 — Working-Tree Quiescence Rule for Subagent Commits (cascaded from constitution submodule §11.4.84)

Verbatim user mandate (2026-05-22): “no subagent commit may proceed while any concurrent mutation gate is in flight in the same checkout. Before g it add, the committing agent MUST grep its own working tree for mutation markers (MUTATED for paired, // always pass, return json.Marshal shortcut paths, etc.). Any unexplained file in the staging area triggers ABORT.”

No subagent (or main-thread) commit may proceed while any concurrent mutation gate, paired-mutation experiment, or other in-flight mutation is live in the same checkout. Before `git add`, the committing agent **MUST** grep its own working tree for mutation markers (`MUTATED` for paired, `// always pass`, `return json.Marshal` shortcut paths, `// MUTATION / # MUTATION` annotations, `_mutated_*` filename suffixes, etc.) and explicitly account for every modified file in the staging area; any unexplained file → **ABORT**. (Forensic case: a logo-fix subagent's `git add` swept an `// always pass` JWT-verify mutation residue into an unrelated commit pushed to all four mirrors — a real security-defect window.) Operative rule: (1) pre-`git add` greps for mutation markers + cross-checks `git status --porcelain` against the subagent's declared scope; unaccounted entries → **ABORT**; (2) any active mutation gate **MUST** be serialised (`mutate` → `assert FAIL` → `restore` → `assert PASS`) and the working tree verifiably clean before any unrelated commit; (3) concurrent subagents in the **SAME** checkout **MUST** coordinate through a lockfile (`.git/MUTATION_IN_PROGRESS`) — cleaner solution is `git worktree add` per subagent (composes with §11.4.20/§11.4.70); (4) post-commit mutation-residue-scanner **MUST** run before push — any commit containing a mutation marker → push **BLOCKED**.

Cascade requirement: This anchor (verbatim or by §11.4.84 reference) **MUST** appear in every owned submodule's `CONSTITUTION.md`, `CLAUDE.md`, and `AGENTS.md`. Propagation gate `CM-COVENANT-114-84-PROPAGATION`; paired mutation strips the literal → gate **FAILs**. A mutation marker that lands in a tagged commit is a critical defect regardless of how briefly it persisted. **Canonical authority:** constitution submodule `Constitution.md` §11.4.84 for the full mandate.

§11.4.86 — Roster/Corpus-Backed Status-Doc Auto-Sync Mandate (cascaded from constitution submodule §11.4.86)

Verbatim user mandate (2026-05-25): *“Make sure that assets and players Status docs are ALWAYS regularly updated and in sync like all others Status docs — any time we add or modify the assets content(s) or we change or add new / remove existing pre-installed video and audio player apps! This MUST WORK OUT OF THE BOX!”*

Some Status docs (§11.4.45) are backed by a tracked roster (installed apps/components) or a tracked asset corpus (test/media asset directory) rather than narrative alone. Their freshness **MUST NOT** depend on operator vigilance — the moment a roster/corpus member changes (app added/removed/renamed; asset added/modified/removed) the Status doc + Status_Summary + HTML + PDF **MUST** resync out of the box, mechanically. Mechanism (all must hold): (1) drift-proof fingerprint — sha256 of the sorted member list (NOT mtime), persisted in a sidecar beside the Status doc; (2) a sync helper that regenerates the fingerprint + re-exports HTML+PDF via the §11.4.65 exporter, wired so sync is automatic; (3) a pre-build gate that **FAILs** when the live fingerprint differs from the persisted one (mirrors §11.4.12 `CM-ISSUES-SUMMARY-SYNC` + §11.4.45 `sync_integration_status`); (4) a paired §1.1 mutation corrupting the fingerprint and asserting the gate **FAILs**. Classification: universal — the consuming project supplies the specific docs, roster/corpus sources, helper, and gate name per §11.4.35.

Cascade requirement: This anchor (verbatim or by §11.4.86 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-86-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker — no --skip-roster-sync, --allow-status-drift, --roster-sync-not-applicable flag. **Canonical authority:** constitution submodule Constitution.md §11.4.86 for the full mandate.

§11.4.88 — Background-Push Mandate: Commit-Lock Release Immediately After Commit, Push Runs Detached (cascaded from constitution submodule §11.4.88)

Forensic anchor (2026-05-26): a single commit_all.sh held its flock ~5 hours because do_push ran synchronously after the commit landed — every subsequent commit blocked on a slow mirror push irrelevant to the local commit's durability. Implementation seam for §11.4.87(B) zero-idle. The mandate: (A) .git/.commit_all.lock MUST be released IMMEDIATELY after git commit returns 0 — the commit is durable on local disk regardless of remote push outcome; (B) push runs detached via nohup ./push_all.sh ... > <log> 2>&1 &+ disown — the orchestrator's exit code reports COMMIT success, NOT push success; (C) push_all.sh acquires per-remote flock .git/.push.<remote>.lock so concurrent invocations targeting the same remote serialize but different-remote invocations run in parallel; (D) backgrounded push failures land in qa-results/push_failures/<ts>_<remote>.log — the next autonomous-loop tick checks per §11.4.87(A) “no external dependency in-flight” gate; (E) synchronous-push escape: explicit --sync-push CLI flag preserves legacy behaviour for §11.4.41 force-push merge-first audit paths. Gates CM-COVENANT-114-88-PROPAGATION + CM-BACKGROUND-PUSH-WIRED + paired §1.1 mutations. Synchronous push (without --sync-push) = §11.4 PASS-bluff at the execution layer.

Cascade requirement: This anchor (verbatim or by §11.4.88 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-88-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker — no escape hatch beyond --sync-push for force-push events. **Canonical authority:** constitution submodule Constitution.md §11.4.88 for the full mandate.

§11.4.89 — Background Test Execution Mandate (cascaded from constitution submodule §11.4.89)

Verbatim user mandate (2026-05-27): “Any tests we are executing, especially long test cycles, MUST BE performed in background in parallel with main work stream! This MUST NOT block our capabilities to work on queued workable items. Main work stream can be blocked or sit iddle only if absolutely needed and if it depends hard on results of some background execution.”

Symmetric anchor to §11.4.88 (background push) at the test-execution layer. Mandate: (A) long-running tests (>30 s expected: pre_build, meta_test, test_all_fixes, recent_work_validate, HelixQA banks, 4-phase cycles, full-suite retests, audio supervisors, dual-display recorders) MUST run via `nohup ... > <log> 2>&1 & + disown` with the log under a known dir (qa-results/<test_id>_<ts>.log); (B) the main stream proceeds to the §11.4.42 priority queue immediately; (C) hard-dependency gating — poll an exit-status file or `pgrep -af <test>` before steps that need the exit code, surfacing as §11.4.66 interactive options if the test is still running; (D) failures land in <log> files, the next loop tick checks; (E) foreground execution permitted ONLY for <30 s tests OR explicit operator authorisation; (F) per-script flock serialises same-script invocations, different-script invocations parallel. Gates CM-COVENANT-114-89-PROPAGATION + CM-BACKGROUND-TEST-EXECUTION-WIRED + paired §1.1 mutations.

Cascade requirement: This anchor (verbatim or by §11.4.89 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-89-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker — no escape hatch beyond explicit per-invocation operator authorisation. **Canonical authority:** constitution submodule Constitution.md §11.4.89 for the full mandate.

§11.4.90 — Obsolete Status + Per-Item Obsolescence Audit (cascaded from constitution submodule §11.4.90)

Verbatim user mandate (2026-05-27): *“Bug No 6 ... seems obsolete after latest request for new behavior ... mark obsolete tickets with some light gray background ... text - the description to be strikethrough styled ... review all existing open or resolved workable items if they are obsolete - not valid any more ... There MUST NOT be any mistake! No bluff is allowed of any kind!”*

The §11.4.15 Status closed-set is extended with a terminal `Obsolete` (→ `Fixed.md`) value (orthogonal to Type per §11.4.16). Obsolescence reasons (closed vocabulary): `superseded-by-design-change` | `superseded-by-later-mandate` | `feature-removed` | `duplicate-of` | `unsupported-topology`. Every `Obsolete` heading MUST carry an `**Obsolete-Details:**` line (Since + Reason + Superseding-item + Triple-check evidence) within 8 non-blank lines. The §11.4.23 colorizer adds a `cell-status-obsolete` class — light-gray `#E0E0E0` background + strikethrough description. Audit cadence: every release-gate sweep per §11.4.40 + §11.4.42; triple-check is non-negotiable per the operator mandate. Composes with §11.4.15 / §11.4.16 / §11.4.19 / §11.4.21 / §11.4.23 / §11.4.33 / §11.4.34 / §11.4.40 / §11.4.42 / §11.4.66 / §11.4.71. Gates CM-COVENANT-114-90-PROPAGATION + CM-ITEM-OBSOLETE-DETAILS + CM-OBSOLETE-COLORIZER-WIRED + paired §1.1 mutations.

Cascade requirement: This anchor (verbatim or by §11.4.90 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-90-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.90 for the full mandate.

§11.4.91 — Summary-Doc Clarity Mandate (cascaded from constitution submodule §11.4.91)

Verbatim user mandate (2026-05-27): *“Summary docs - Issues_Summary some not clear one line descriptions - like ‘Composes with’ ... For each workable item we MUST HAVE clearly understandable meaning ... every team member can clearly understand what that particular workable item is exactly about! There cannot be misunderstanding or unclarity of any kind and no bluff allowed!”*

Every summary entry (Issues_Summary, Fixed_Summary, README doc-link, Status_Summary pages 1+2, all one-liners) MUST contain a self-contained meaningful description ≥ 6 words OR ≥ 40 chars naming SUBJECT + PROBLEM/GOAL. Forbidden one-liner anti-patterns: section labels (Composes with, Closure criteria, Fix direction, etc.); bare metadata fragments (Critical, Bug, In progress, etc.); section-marker echoes; a §-letter alone. Generators (generate_issues_summary.sh / generate_fixed_summary.sh / update_readme_doc_links.sh / generate_status_summary.sh) MUST extract from the H1/H2 heading line per the §11.4.54 ATM-NNN convention, NEVER from arbitrary downstream text, and MUST refuse anti-pattern rows — emitting a (MISSING DESCRIPTION – fix source heading) placeholder with visual highlight. Gate CM-SUMMARY-CLARITY-DESCRIPTIONS scans every summary; an anti-pattern match = FAIL. Audit cadence: every §11.4.40 + §11.4.42 sweep.

Cascade requirement: This anchor (verbatim or by §11.4.91 reference) MUST appear in every owned submodule’s CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-91-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.91 for the full mandate.

§11.4.92 — Multi-Pass Change-Evaluation Discipline (cascaded from constitution submodule §11.4.92)

Verbatim user mandate (2026-05-27): *“Every change to the project or codebase we do MUST BE evaluated in several passes and in in-depth analisis for potential new issues or problems it can introduce! ... no bluff of any kind! After we do change or set of changes this mandatory steps MUST BE taken!”*

Every non-trivial change MUST pass a 5-pass evaluation BEFORE it is commit-ready: **(Pass 1)** main-task verification — change achieves the stated goal, captured-evidence per §11.4.5/§11.4.69; **(Pass 2)** regression-blast-radius analysis — enumerate every direct dependency, demonstrate no contract break; **(Pass 3)** cross-feature interaction analysis — audit parallel features sharing state/timing/hardware/shell environment; **(Pass 4)** deep-research validation per §11.4.8 — external precedent OR “NO external solution found — original work” + CodeGraph queries per §11.4.78/§11.4.79; **(Pass 5)** anti-bluff confirmation per §11.4 / §11.4.1 / §11.4.6 / §11.4.27 / §11.4.50 / §11.4.52 / §11.4.69 / §11.4.83 — no new bluff surface introduced. Each pass is documented (commit footers OR docs/ entries OR qa-results/ evidence). Only after all 5 passes complete may commit/push/test/release proceed. Trivial exemption: typo / revision-bump / MD-export-regen IF zero source

touched AND the commit message cites the exemption explicitly. Gates CM-COVENANT-114-92-PROPAGATION + CM-MULTI-PASS-EVALUATION-EVIDENCE + paired §1.1 mutations.

Cascade requirement: This anchor (verbatim or by §11.4.92 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-92-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.92 for the full mandate.

§11.4.93 — SQLite-Backed Single-Source-of-Truth for Workable Items (cascaded from constitution submodule §11.4.93)

Verbatim user mandate (2026-05-27): *“There MUST be single source of truth for all of our workable items - SQLite database ... proper scripts (we recommend Go programs) ... reduce a chance for sync to be broken ... generate always all docs from DB or to re-generate Db from all docs we have in opposite direction”*

The text-based Issues/Fixed/Summary/CONTINUATION constellation is converted to a SQLite-DB-backed single source of truth. Schema mandatory tables: items (atm_id PK + Type + Status incl. Obsolete + Severity + title + description ≥40 chars + created/modified + composes_with JSON + current_location); item_history (append-only audit per §11.4.34 By/Reason/Evidence); obsolete_details (§11.4.90); operator_block_details (§11.4.21); firebase_metadata (§11.4.47); meta (schema version + last sync + integrity hash). A Go binary at cmd/workable-items/ provides sync md-to-db / db-to-md / diff / validate / add / close; bidirectional regen is byte-identical round-trip (closed-set whitespace/section-order tolerance). commit_all.sh refuses on non-empty diff; sync_issues_docs.sh invokes the Go binary; pre-build runs workable-items validate. Anti-bluff: unit + integration + stress (1000-row insert + 10 concurrent writers) + chaos (mid-write SIGKILL + corrupt-DB recovery + disk-full) + paired §1.1 mutation + HelixQA Challenge CME-WORKABLE-ITEMS-001. The Go binary lives in the constitution submodule (constitution/scripts/workable-items/) per §11.4.74. Gates CM-COVENANT-114-93-PROPAGATION + CM-WORKABLE-ITEMS-DB-PRESENT + CM-WORKABLE-ITEMS-MD-DB-IN-SYNC + paired §1.1 mutations. (NOTE: the DB tracking rule is AMENDED by §11.4.95 — DB is TRACKED, not gitignored.)

Cascade requirement: This anchor (verbatim or by §11.4.93 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-93-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker — text-based-only trackers are a §11.4 PASS-bluff at the data-architecture layer. **Canonical authority:** constitution submodule Constitution.md §11.4.93 for the full mandate.

§11.4.94 — Zero-Idle Priority-First Parallel-By-Default Operating Mode (cascaded from constitution submodule §11.4.94)

Verbatim user mandate (2026-05-27): *“We MUST NEVER sit iddle / wait or sleep if there is possibility for us to work on something ... Always check if there is a possibility to work on something while we are not working actively on something! Pick always by priority - most critical workable items and other tasks MUST BE done first! ... Stay still / iddle if nothing is left to be done at all or waiting for something that is blocking us / you!!!”*

§11.4.94 binds §11.4.20 + §11.4.42 + §11.4.58 + §11.4.70 + §11.4.72 + §11.4.82 + §11.4.87 + §11.4.88 + §11.4.89 into a single always-on enforcement: (A) idle ONLY when every queued item is genuinely blocked on an external dependency (hardware / network upstream / build/test completion the conductor cannot accelerate) OR operator STOP OR §12 host-safety — “don’t see what to do” is NEVER valid; (B) before ANY wake/sleep the conductor MUST survey parallel-work feasibility per §11.4.42 + §11.4.72 + §11.4.87, identify non-contending items, and dispatch in parallel per §11.4.20/§11.4.70 (subagent) + §11.4.58 (PWU disjoint scope) + §11.4.89 (background long tests); (C) priority order MANDATORY — pick highest-severity + §11.4.72 audio-first the conductor can autonomously progress; (D) subagent-driven default for non-trivial; (E) background default for >30 s wall-clock work via nohup+disown; (F) stability-preserving (composes with §11.4.92 multi-pass + §11.4.84 quiescence + §12.6–§12.9 host safety); (G) progress updates surfaced at milestone boundaries. Gates CM-COVENANT-114-94-PROPAGATION + CM-PARALLEL-WORK-AUDIT + paired §1.1 mutations.

Cascade requirement: This anchor (verbatim or by §11.4.94 reference) MUST appear in every owned submodule’s CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-94-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.94 for the full mandate.

§11.4.96 — Safe-Parallel-Work-With-Long-Build Catalogue + Mandate (cascaded from constitution submodule §11.4.96)

Verbatim user mandate (2026-05-27): *“Are there except AOSP build process any other active jobs being done at the moment? Can we work on something in parallel while build is in progress so we slowly cleanup our slate? ... do as much as possible work in background in parallel with main work stream and oreferably using subagents-driven approach!”*

An operational catalogue for the canonical long-running workload (multi-hour containerised build per §12.9). **SAFE during build:** (A) MD/docs work; (B) generator/helper script work under scripts/; (C) pre-build + meta-test gate authoring + paired §1.1 mutations; (D) on-device test scripts; (E) constitution submodule edits + push; (F) any submodule commit + push per §11.4.88; (G) read-only live-ADB probes (dumpsys/getprop/cat /proc/.../screencap/logcat); (H) subagent dispatch per §11.4.20/§11.4.70 + §11.4.84 quiescence; (I) web research + external API queries with §11.4.10 credentials; (J) workable-items DB ops per

§11.4.93+§11.4.95; (K) backgrounded pre-build + meta-test execution per §11.4.89. **UNSAFE during build:** (α) git checkout/reset --hard/clean -df on the source tree (use git worktree); (β) mass file deletes/renames under built source trees; (γ) submodule pointer updates affecting built artefacts; (δ) out/ mutations; (ε) make clean/m clobber/rm -rf out/; (ζ) container destruction; (η) disk-filling breaching §12.9 free-space minimum; (θ) §12 host-session-safety breaches. Conductor responsibility: before EVERY pause point during a long build, consult the catalogue, identify (A)-(K) queue items per §11.4.42+§11.4.72, and dispatch ≥1 per §11.4.20/§11.4.70 subagent default + §11.4.89 background. “Build running, nothing else to do” is NEVER true per §11.4.94+§11.4.96. Gates CM-COVENANT-114-96-PROPAGATION + CM-PARALLEL-WORK-DURING-BUILD-AUDIT + paired §1.1 mutations.

Cascade requirement: This anchor (verbatim or by §11.4.96 reference) MUST appear in every owned submodule’s CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-96-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.96 for the full mandate.

§11.4.97 — Maximum-Use-of-Idle-Time + Progress-Update Cadence (cascaded from constitution submodule §11.4.97)

Verbatim user mandate (2026-05-27): *“keep it working, we should do as much as possible, if not it all but as much as we can as long as there is iddle time! it MUST be used! ... keep us updated about all progress and all phisycal proofs and gathered data as you progress through all open workable items!”*

Operating-mode capstone strengthening §11.4.87 + §11.4.94 + §11.4.96: (A) every minute of conductor idle time during which work could autonomously progress AND is not genuinely blocked = a §11.4.97 violation; “as much as possible, if not it all but as much as we can” is operative — dispatch CONTINUOUSLY through the entire idle window, not just at scheduled wakes; (B) progress-update cadence — emit an operator-facing 1-line update at every commit landed / subagent return / constitutional anchor / captured evidence / milestone closure, no operator prompt required; (C) continuous physical-proof gathering per §11.4.5 + §11.4.6 + §11.4.69 — every autonomous closure cites captured-evidence (evidence path goes into the §11.4.93 item_history.evidence_path when the DB lands); (D) composes with §11.4.5/6/13/20/27/42/50/52/69/70/72/83/85/87/88/89/94/96; (E) the idle-only-when-blocked closed-set is unchanged from §11.4.94(A). Gates CM-COVENANT-114-97-PROPAGATION + CM-IDLE-TIME-AUDIT + paired §1.1 mutations.

Cascade requirement: This anchor (verbatim or by §11.4.97 reference) MUST appear in every owned submodule’s CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-97-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.97 for the full mandate.

§11.4.69 — Universal Sink-Side Positive-Evidence Taxonomy + Mechanical Enforcement (cascaded from constitution submodule §11.4.69)

Verbatim user mandate (2026-05-20): *“THIS MUST HAPPEN NEVER AGAIN!!! We MUST HAVE this all working! Not just for audio but for every single piece of the System!!! Proper full automation when executed with success MUST MEAN that manual testing will be as much positive at least regarding the success results! ... Solution MUST BE universal, generic that solves working flows for all System components and for all future and all existing projects! ... Everything we do MUST BE validated and verified with rock-solid proofs and anti-bluff policy enforcement and fulfillment!”*

Universal generalisation of §11.4.68 (audio-specific) across every user-visible feature class. Every user-visible feature MUST map to one entry in the closed-set §11.4.69 sink-side evidence taxonomy (audio_output, audio_input, video_display, network_throughput, network_connectivity, bluetooth_a2dp, bluetooth_pair, touch_input, sensor, gpu_render, storage_read, storage_write, mediacodec_decode, mediacodec_encode, miracast, cast, boot_service, package_install, permission_grant, wifi_link, wifi_throughput, ethernet_link, display_topology, drm_playback, subtitle_render — open to additions, never contraction). Every PASS for a feature in the taxonomy MUST cite a captured-evidence artefact path matching the required evidence shape. New helper contracts (additive during grace, mandatory after 2026-06-19): `ab_pass_with_evidence <description> <evidence_path>` (verifies path exists + non-empty), `ab_skip_with_reason <description> <closed-set-reason>` (reasons: `geo_restricted`, `operator_attended`, `hardware_not_present`, `topology_unsupported`, `network_unreachable_external`, `feature_disabled_by_config`; forbids `network_unreachable_external` for any taxonomy feature with a sink-side probe); bare `ab_pass` deprecated (WARN pre-grace, FAIL post-grace). Three pre-build gates + paired §1.1 mutations: `CM-SINK-EVIDENCE-PER-FEATURE`, `CM-NO-FAIL-OPEN-SKIP`, `CM-AB-PASS-WITH-EVIDENCE-EVERYWHERE`. No escape hatch — no `--skip-evidence`, `--config-only-pass`, `--allow-fail-open-skip`, `--legacy-ab-pass-permitted` flag.

Cascade requirement: This anchor (verbatim or by §11.4.69 reference) MUST appear in every owned submodule’s `CONSTITUTION.md`, `CLAUDE.md`, and `AGENTS.md`. Propagation gate `CM-COVENANT-114-69-PROPAGATION` enforces the anchor literal across the consumer fleet; paired mutation strips the literal → gate FAILs. Severity-equivalent to a §11.4 PASS-bluff at the sink-side-evidence layer. **Canonical authority:** constitution submodule `Constitution.md` §11.4.69 for the full mandate.

§11.4.85 — Stress + Chaos Test Mandate (cascaded from constitution submodule §11.4.85)

Verbatim user mandate (2026-05-24): “Every fix or improvement you do *MUST BE* covered with full automation stress and chaos tests so we are sure nothing can break the functionality and all edge cases are monitored and polished and additionally fixed if that is needed! Everything must produce rock solid proofs and follow fully no-bluff policy!”

Every fix or improvement landed MUST ship with full-automation **stress** AND **chaos** test suites exercising edge cases, sustained load, concurrent contention, and failure-injection. Happy-path coverage alone is a §11.4 / §107 PASS-bluff at the resilience layer. **Stress** (closed-set): sustained load ($N \geq 100$ iterations OR ≥ 30 s wall-clock, p50/p95/p99 latency recorded) + concurrent contention ($N \geq 10$ parallel invocations, no deadlock/leak) + boundary conditions (empty/max/off-by-one, each categorised). **Chaos** (closed-set, per fix-class appropriateness): process-death injection + network-fault injection (drop/delay/reorder) + input-corruption injection + resource-exhaustion injection (disk full, OOM, FD exhaustion — refuse cleanly OR degrade, NEVER crash) + state-corruption injection (mid-flight lock loss, partial-write). Every stress + chaos PASS MUST cite a captured-evidence artefact path per §11.4.5 + §11.4.69. Helper library `stress_chaos.sh` provides `ab_stress_run`, `ab_stress_concurrent`, `ab_chaos_kill_pid_during`, `ab_chaos_drop_network_during`, `ab_chaos_corrupt_file_during`, `ab_chaos_oom_pressure_during`, `ab_chaos_disk_full_during`, each composing with `ab_pass_with_evidence / ab_skip_with_reason`. Cleanup non-negotiable in `trap '...' EXIT` (cleanup failure = §11.4.14 violation). Four-layer coverage per §11.4.4(b) + paired §1.1 mutation (strip chaos-injection or evidence-capture → gate FAILs). No escape hatch — no `--skip-stress`, `--no-chaos`, `--happy-path-suffices`, `--stress-test-later` flag.

Cascade requirement: This anchor (verbatim or by §11.4.85 reference) MUST appear in every owned submodule’s `CONSTITUTION.md`, `CLAUDE.md`, and `AGENTS.md`. Propagation gate `CM-COVENANT-114-85-PROPAGATION`; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule `Constitution.md` §11.4.85 for the full mandate.

§11.4.87 — Endless-Loop Autonomous Work + Zero-Idle Agent Dispatch + Anti-Bluff Testing Mandate (cascaded from constitution submodule §11.4.87)

Verbatim user mandate (2026-05-26): “*continue in endless loop fully autonomously*” (and any semantically-equivalent phrasing).

When the operator instructs an AI agent to continue in an endless autonomous loop, the agent MUST treat it as a HARD-CONTRACT covenant: (A) continue working until `docs/Issues.md` Status-column has zero non-terminal entries AND `docs/CONTINUATION.md` §3 Active work is empty AND no background subagent is mid-execution AND no external dependency is in-flight; (B) dispatch background subagents for parallelisable work — main + every subagent operate concurrently, “waiting for results” is the ONLY acceptable idle reason; (C) every closure lands four-layer test coverage per §11.4.4(b) with captured-evidence (audio/video/

network/UI/sysfs physical proofs); (D) the §11.4 anti-bluff covenant family (§11.4.1 / §11.4.2 / §11.4.6 / §11.4.7 / §11.4.27 / §11.4.50 / §11.4.52 / §11.4.68 / §11.4.69 / §11.4.83) is the operative truth-discipline — tests AND HelixQA Challenges bound equally; (E) the loop terminates ONLY on all-conditions-met, explicit operator STOP, host-session-safety demand, or scheduled wake on a known-future-actionable signal. No escape hatch — no `--idle-OK`, `--skip-endless-loop`, `--bluff-permitted-for-this-task`, `--metadata-only-test-suffices`, `--no-physical-proof-required` flag.

Cascade requirement: This anchor (verbatim or by §11.4.87 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-87-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.87 for the full mandate.

§11.4.95 — Workable-Items SQLite DB Is TRACKED in Git, NEVER Gitignored (cascaded from constitution submodule §11.4.95)

Verbatim user mandate (2026-05-27): *“We shall not Git ignore our workable items SQLite DB since it is our single source of truth ... workable items SQLite DB regularly committed and pushed to all upstreams!”*

§11.4.93's earlier “gitignored per §11.4.30” clause is AMENDED — the DB at docs/workable_items.db is TRACKED in git, NEVER gitignored. It IS authoritative source data, NOT a build artefact. Every workable-items sync md-to-db that mutates state MUST stage + commit + push the DB alongside the MD regen per §11.4.19 atomic-move + §2.1 multi-upstream push. A WAL-checkpoint (PRAGMA wal_checkpoint(TRUNCATE)) is required before commit-stage so the transient .db-wal + .db-shm sidecars (gitignored per §11.4.30) are safely discardable. The §11.4.77 regeneration mechanism does NOT apply — the DB IS the source. Destructive DB ops require §9.2 hardlinked-backup + operator authorization; §11.4.41 force-push merge-first applies if DB history ever needs rewrite. Gates CM-COVENANT-114-95-PROPAGATION + CM-WORKABLE-ITEMS-DB-TRACKED + paired §1.1 mutation.

Cascade requirement: This anchor (verbatim or by §11.4.95 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-95-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.95 for the full mandate.

§11.4.98 — Full-Automation Anti-Bluff Mandate (cascaded from constitution submodule §11.4.98)

Verbatim user mandate (2026-05-28): *“Make sure we have full automation testing of all scenarios with real bot, main group and users without any manual intervention or contribution of real user! Everything MUST BE fully automatic and autonomous! These tests MUST BE able to rerun endless times*

when needed! ... Make sure there is no false positives in testing! Every test and its results MUST obtain real proofs of everything working! No bluff is allowed!”

Closes the manual-intervention gap (§11.4 / §11.4.2 / §11.4.5 / §11.4.50 / §11.4.85 / §11.4.87 / §11.4.89 / §11.4.94 did not explicitly forbid it). A live/integration/e2e/Challenge test that requires a human action during execution (typing a message, clicking UI, hand-triggering a webhook, attaching a file — anything beyond startup) is by definition a §11.4 PASS-bluff at the automation layer. (A) Every governed test — unit/integration/e2e/Challenge/stress/chaos/live — MUST be fully self-driving end-to-end, reporting PASS/FAIL/SKIP-with-reason without any further human action after startup. (B) Single permissible exception: one-time credential bootstrap performed OUTSIDE test execution (.env from vault, shell exports, OAuth at first install, MTProto session activation) — configuration, not test driving. (C) Live messenger/channel/agent tests: no “operator must type” prompts (drive programmatically via second account / webhook fixture / loopback); no hard-coded session UUIDs that collide with the active dev session (Herald 2026-05-28 c laude --resume silent exit -1 lesson); no 60 s human-response windows (§11.4.50 determinism violation); re-runnability proof — PASS at -count=3 consecutive automated invocations with self-cleaning state; §11.4.98 obsolescence audit classifies every existing test COMPLIANT vs NON-COMPLIANT; no silent-skip-reported-as-PASS or stale-evidence-as-fresh. (D) With §11.4.85 + §11.4.89 + §11.4.87 + §11.4.94 forms a continuously-validated, non-flake, anti-bluff regime. (F) Manual-dependency tests not rewritten within 30 days graduate to §11.4.90 Obsolete citing §11.4.98.

Cascade requirement: This anchor (verbatim or by §11.4.98 reference) MUST appear in every owned submodule’s CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-98-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.98 for the full mandate.

§11.4.99 — Latest-Source Documentation Cross-Reference Mandate (cascaded from constitution submodule §11.4.99)

Verbatim user mandate (2026-05-28): “Make sure we ALWAYS check against latest versions of services we use web / online docs before creating instructions! This situation is illustration of how we can misguide ourselves or get banned! ... These are mandatory rules / constraints and the result is consistency and safety of created instructions, guides and manuals!”

Misguidance-by-stale-docs is the same severity class as a §11.4 PASS-bluff at the documentation layer (Herald 2026-05-28 case: a first-draft MTProto guide recommended VoIP fallback numbers and omitted the recover@telegram.org pre-login email — both contradicted Telegram’s official docs + the gotd/td maintainer guide and could have caused a permanent account ban). Closes the gap §11.4.92 Pass 4 alludes to but does not mandate. (A) Before committing any operator-facing instruction/guide/manual/troubleshooting/setup doc, the author MUST: (1) fetch the LATEST official online documentation of the documented service/library via WebFetch / MCP / direct browsing — NEVER training data, memory, or prior

committed docs; (2) cross-reference every instruction step against that source; (3) seek secondary authoritative sources (maintainer SUPPORT.md, official changelogs, vetted community FAQs) when the official source is sparse/silent; (4) cite source URLs + date in a `## Sources verified` footer in the doc; (5) cite a `Sources verified <date>: <urls>` footer in the commit message. (B) Negative findings (gaps/silences/contradictions) MUST be documented explicitly. (C) Docs older than 6 months are STALE — re-verify before citing as operator authority, at every vN.0.0 release boundary, on service breaking-change announcements, or on operator error reports. (D) Risk-classified services (messengers, cloud APIs, payment systems, AI/LLM providers, code-hosting, package managers) carry a 90-day max staleness + explicit safety warnings. (E) Composes with but is INDEPENDENT of §11.4.92 Pass 4. (G) Commit missing either footer is BLOCKED at release-gate; stale-beyond-grace docs graduate to §11.4.90 Obsolete (Reason=stale-documentation).

Cascade requirement: This anchor (verbatim or by §11.4.99 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-99-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.99 for the full mandate.

§11.4.101 — Autonomous-Decision-Over-Blocking Mandate (cascaded from constitution submodule §11.4.101)

Verbatim user mandate (2026-05-28): *“when working in endless working loop fully autonomously try to decide most properly about points which would block execution and wait for us. If we haven't answered now work would be blocked whole night! If possible and if that will not cause any issues make proper and most reliable and safe decision so we achieve maximal efficiency and work gets fully done!”*

In autonomous / endless-loop mode (per §11.4.87), the agent MUST minimize operator-blocking and make the safe, reliable, reversible decision itself so work is not stalled (e.g. overnight) waiting for input — §11.4.87 says keep working, §11.4.101 says HOW to clear the decision points. **Proceed-autonomously (closed-set, ALL must hold):** (a) the action is reversible OR has a captured pre-op backup per §9.2; (b) the safe choice is determinable from captured evidence per §11.4.6 (no guessing — LIKELY/probably/seems is NOT a determination); (c) a wrong choice's blast radius is bounded AND recoverable; (d) it composes with anti-bluff §11.4, host-safety §12, data-safety §9. **Block-only-when (BLOCK via the §11.4.66 interactive mechanism ONLY when ALL hold):** the action is irreversible AND high-blast-radius AND the safe choice cannot be determined from evidence — e.g. external-account state the agent cannot inspect, hardware it cannot access, destructive ops without backup, force-push (also §9.2 + §11.4.41), spending money or sending data to third parties. Operator-blocked per §11.4.21 is reached only after this rule fires AND the self-resolution-exhaustion audit completes. An unavoidable block parks one work unit — it does NOT pause the loop; the agent keeps progressing every non-blocked item in parallel per §11.4.87 + §11.4.94 (posing the question then going idle is a §11.4.94 + §11.4.97 violation). Classification: universal (§11.4.17).

Cascade requirement: This anchor (verbatim or by §11.4.101 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-101-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.101 for the full mandate.

§11.4.102 — Mandatory systematic-debugging activation + always-loaded skill-discovery + plugin-dependency availability (cascaded from constitution submodule §11.4.102)

Verbatim user mandate (2026-05-29): *“Make sure that we ALWAYS trigger / start the”/superpowers:systematic-debugging” skills when any issues happen! ... we MUST activate the skill(s) and make strongest efforts in full in depth analysis / debugging and determine root causes of all problem ... we MUST make sure that “/using-superpowers” skill is ALWAYS loaded, applied and used! All dependencies (plugins) that Claude Code or other market places are offering MUST BE installed if these are not already available for loading and use!”*

Three cooperating invariants — the difference between guess-and-retry and investigate-to-root-cause-first. **(A) Mandatory systematic-debugging activation.** On ANY spotted issue / bug / test failure / gate failure / regression / misalignment / inconsistency / unexpected behaviour, the agent MUST activate `superpowers:systematic-debugging` (or the platform-equivalent structured-debugging discipline) **BEFORE proposing, writing, or applying any fix** — the **Iron Law: NO FIXES WITHOUT ROOT CAUSE INVESTIGATION FIRST.** Full four-phase arc: root-cause → pattern → hypothesis → implementation (the fix is designed only against the proven root cause). Guess-and-retry, symptom-patching, and re-running a failed test hoping it passes (“probably transient / flaky”) WITHOUT a completed investigation are §11.4.102 violations; calling a failure transient/flaky/intermittent/probably-timing without captured forensic evidence is simultaneously a §11.4.6 (no-guessing) and §11.4.7 (demotion-evidence) violation. **(B) Mandatory always-loaded using-superpowers.** `superpowers:using-superpowers` (or the platform-equivalent skill-discovery / capability-index discipline) MUST be loaded and applied at session start and consulted before any task — survey available skills before acting on ANY request; if ANY skill could apply (even at 1% relevance) it MUST be invoked rather than improvised from memory. **(C) Mandatory plugin / dependency availability.** Every skill plugin / marketplace package / capability dependency the project relies on MUST be installed + loadable BEFORE the dependent work proceeds; a missing plugin that blocks a mandated skill is a release-blocker until installed + confirmed loadable (confirm by observing the skill in the live capability list — install exit 0 ≠ skill loadable, per the §11.4.80 lesson). Composes with §11.4.4 / §11.4.6 / §11.4.7 / §11.4.8 / §11.4.43 / §11.4.70 / §11.4.82(A) / §11.4.92. Classification: universal (§11.4.17). No escape hatch — no `--skip-systematic-debugging`, `--guess-and-retry-OK`, `--symptom-patch-permitted`, `--skip-skill-discovery`, `--plugin-optional`, `--missing-plugin-is-warning` flag.

Cascade requirement: This anchor (verbatim or by §11.4.102 reference) **MUST** appear in every owned submodule’s CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-102-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.102 for the full mandate.

§11.4.122 — No-silent-removal-of-existing-components-without-operator-confirmation mandate (User mandate, 2026-06-03)

Forensic anchor — verbatim user mandate (2026-06-03):

“Never ever remove any application, system component or service from already existing codebase / System without interactively asked question to us! THIS IS MANDATORY RULE / CONSTRAINT!”

Forensic case study (FACT). During the 1.1.8-dev burn-down, two shipped capabilities — F2 (an Apple-TV-class application) and F4 (a Huawei HMS / Mobile-Services component) — were removed from the existing System WITHOUT first asking the operator; the operator reversed both. A removal the operator has to discover and reverse after the fact is a defect of the same severity class as a §11.4 PASS-bluff: the System silently lost a user-facing capability the operator never agreed to drop.

No application, system component, service, package, feature, driver, module, library, prebuilt asset — any already-existing end-user capability of the existing codebase / shipped System — may be removed (deleted, dropped from the package set, disabled-into-non-shipping, un-bundled, de-listed, or otherwise made unavailable to the end user) WITHOUT FIRST interactively asking the operator and receiving an EXPLICIT keep-or-remove decision. The question **MUST** be posed through the platform’s interactive clarification mechanism per §11.4.66 (AskUserQuestion on Claude Code) — NEVER a free-text “should I remove X?” buried in narrative, NEVER a silent removal justified post-hoc, NEVER an autonomous removal decision. A silent removal is a **release blocker** regardless of how well-intentioned the rationale (deduplication, “it was broken anyway”, geo-restricted, incompatible, superseded) — the operator decides, the agent asks.

What counts as a removal (non-exhaustive): deleting an app/APK/binary from the build’s package set (PRODUCT_PACKAGES / device.mk / equivalent), removing a service from the init/boot/service-registry set, dropping a kernel module / driver / config from the shipping configuration, un-bundling a prebuilt asset, deleting a submodule or its shipped output, removing a feature flag that gated a live capability, or any edit whose NET EFFECT is “an end-user capability that shipped before no longer ships.” Adding, replacing-with-operator-approved-equivalent, or fixing a capability is NOT a removal. When uncertain whether an edit constitutes a removal, treat it AS a removal and ask (per §11.4.6 no-guessing + §11.4.101 — removal of an existing user-facing capability is high-blast-radius and **MUST** be operator-confirmed, never autonomously decided). The tracked DROP path: ask → operator approves → mark the item Obsolete (→ Fixed.md) with Obsolete-Details reason feature-removed + an operator-approval citation (§11.4.90) → then remove; the removal never precedes the operator’s yes.

Classification: universal (§11.4.17) — a platform-neutral discipline reusable by ANY project that ships a set of user-facing capabilities; the consuming project supplies its concrete capability-manifest paths per §11.4.35. Composes §11.4.66 /

§11.4.101 / §11.4.90 / §11.4.112 / §11.4.6 / §11.4.40 / §11.4.42. Propagation gate CM-COVENANT-114-122-PROPAGATION (literal 11.4.122) + recommended gate CM-NO-SILENT-COMPONENT-REMOVAL + paired §1.1 meta-test mutation (gate-code = separate work item).

Canonical authority: constitution submodule Constitution.md §11.4.122. Non-compliance is a release blocker. No escape hatch — no `--remove-without-asking`, `--silent-removal`, `--autonomous-removal-OK`, `--dedup-removal-exempt`, `--it-was-broken-anyway` flag.

§11.4.123 — Rock-solid-proof-or-deep-research mandate (User mandate, 2026-06-03)

Forensic anchor — verbatim user mandate (2026-06-03):

“Every single reported issue MUST BE fully and 100% validated with rock solid proofs! Nothing can be considered fixed or completed without hard evidence! No false results or bluff(s) of any kind is allowed! If we are not sure on how to achieve full testing, validation and verification of something we MUST ALWAYS perform deep web research for all possible data (articles, documentation, guides, and other resources) and opensourced codebases which we can use to solve our problems and perform testing with validation and verification which produces rock-solid evidence(s) and leaves no space for false results or any kind of bluff!”

Forensic case study (FACT). In the 1.1.8-dev remediation the validation method for two feature classes was, at first, genuinely unclear: relocating a FLAG_SECURE secure surface to a secondary display (pixel capture returns black) and asserting on-screen content in non-introspectable streaming-app UIs (blank accessibility hierarchy). Rather than declaring them “untestable” or accepting a metadata-only PASS, the cycle performed deep web research (docs/research/testing_frameworks_20260603/) that yielded the CV/OCR/liveness/sink-probe oracle stack (now §11.4.107 + §11.4.112 + §11.4.117) — making rock-solid evidence possible where it had appeared impossible. “Unclear how to validate” is a research trigger, NEVER a bluff licence.

Every single reported issue, every fix, and every claimed completion MUST be fully and 100% validated with rock-solid CAPTURED proof per §11.4.5 / §11.4.69 / §11.4.107 before it may be marked fixed / implemented / completed (§11.4.33 closure vocabulary). Nothing may be considered fixed or complete without hard captured evidence — metadata-only / configuration-only / absence-of-error / grep-without-runtime PASS are all forbidden (§11.4 / §11.4.1); no false results, no bluff of any kind, at any layer.

The research-or-don’t-bluff rule (the operative addition): when the agent is UNSURE how to fully test / validate / verify something — when no obvious evidence-producing method exists OR the candidate method would yield only metadata/config/absence-of-error evidence — it MUST ALWAYS first perform deep web research per §11.4.8 + §11.4.99 (official docs, articles, guides, vendor references, standards, issue trackers, reusable open-source codebases) to DISCOVER or BUILD a validation method that produces rock-solid evidence and leaves no space for a false result. Declaring something “untestable” / “not automatable” / accepting a metadata-only PASS WITHOUT first exhausting this deep-research path is itself a §11.4.123 violation — same severity class as a PASS-

bluff. The research output (cited source URLs + the evidence-producing method, OR the literal “NO external solution found — original work” per §11.4.8) is the captured proof the path was exhausted. Only after that research genuinely fails may the item be classified `PENDING_FORENSICS: / 0operator-blocked` (§11.4.21) / structurally-impossible won’t-fix (§11.4.112) — with the cited research as the evidence the classification is earned, never a convenience.

Classification: universal (§11.4.17) — a platform-neutral discipline reusable by ANY project; the consuming project supplies its concrete capture mechanisms + research corpora per §11.4.35. Composes §11.4.5 / §11.4.6 / §11.4.8 / §11.4.52 / §11.4.69 / §11.4.99 / §11.4.107 / §11.4.118 / §11.4.21 / §11.4.112. Propagation gate `CM-COVENANT-114-123-PROPAGATION` (literal 11.4.123) + recommended gate `CM-ROCK-SOLID-PROOF-OR-RESEARCH` + paired §1.1 meta-test mutation (gate-code = separate work item).

Canonical authority: constitution submodule `Constitution.md` §11.4.123. Non-compliance is a release blocker. No escape hatch — no `--metadata-pass-suffices`, `--skip-proof`, `--untestable-without-research`, `--config-only-closure-OK`, `--bluff-when-unsure` flag.

§11.4.124 — Dead/unwired-code investigate-before-remove mandate (User mandate, 2026-06-04)

Forensic anchor — verbatim user mandate (2026-06-04):

“Before removing any seemingly-dead (zero-importer / unwired) codebase, we MUST investigate via git history where/how it was originally used and how it became dead. Removal is permitted ONLY when we have captured PROOF it is genuinely no longer needed — and that removal MUST be its own separate commit with a proper descriptive message. If there is no such proof, the code MUST be investigated for where/how it should be wired in properly, and any missing or unwired tests MUST be added. We MUST ALWAYS be extra careful with any codebase removal.”

“Zero importers / never called / unwired $\Rightarrow$ dead $\Rightarrow$ delete” is a GUESS (§11.4.6), never a finding — a “no references” result proves only *current* non-reference, not genuinely-unneeded. Before removing ANY seemingly-dead element (zero-importer / never-called / unwired function / method / type / file / module / package / asset / config / build target) the agent MUST FIRST investigate via git history (`git log --follow`, `git log -S/-G` pickaxe across all history, blame on the deleted call-site) and capture as FACT: (1) WHERE/HOW it was originally wired in, (2) WHEN/HOW it became dead — call-site deleted deliberately / by mistake (regression) / never-completed / refactored-unreachable, (3) whether “no references” is real OR a hidden reference the static tool cannot see (reflection / dynamic dispatch / build-tags / codegen / DI / plugin registry / FFI / config-driven wiring). The investigation output (cited commits + determination) is the captured evidence. **Removal is conditional:** permitted ONLY with captured PROOF the element is genuinely no longer needed; that removal MUST be its OWN SEPARATE COMMIT (independently reviewable + revertible, composes §11.4.84 quiescence + §11.4.92 multi-pass) with a descriptive message citing the git-history evidence — plus §11.4.122 operator-confirmation when the element is an end-user capability; the §11.4.90 tracked path marks it `Obsolete` ($\rightarrow$ `Fixed.md`). **No proof $\Rightarrow$ do NOT delete:** investigate WHERE/HOW to wire it in properly (restore a mistakenly-deleted call-site per §11.4.114; finish never-completed wiring) AND add any

missing / unwired tests (§11.4.27 / §11.4.43 / §11.4.115 — the missing test is part of why it drifted into apparent-deadness). **Extra-caution default:** when uncertain whether removal-proof is sufficient, default to NOT removing (investigate + wire + test) per §11.4.6 + §11.4.101 + §11.4.122; “probably dead” is never sufficient — the bar is captured proof. Classification: universal (§11.4.17) — the consuming project supplies its static-analysis / importer-graph tooling + hidden-reference mechanisms per §11.4.35. Composes §11.4.6 / §11.4.8 / §11.4.84 / §11.4.90 / §11.4.92 / §11.4.101 / §11.4.114 / §11.4.122 / §11.4.27 / §11.4.43 / §11.4.115. Propagation gate CM-COVENANT-114-124-PROPAGATION (literal 11.4.124) + recommended gate CM-DEAD-CODE-INVESTIGATE-BEFORE-REMOVE (a net-deletion commit must be removal-only + cite the git-history investigation OR be part of a tracked Obsolete item) + paired §1.1 meta-test mutation (gate-code = separate work item).

Canonical authority: constitution submodule Constitution.md §11.4.124. Non-compliance is a release blocker. No escape hatch — no --zero-importers-means-dead, --delete-unwired-on-sight, --skip-git-history-investigation, --remove-without-proof, --bundle-removal-with-other-work flag.

§11.4.125 — Code-review-agent gate before pre-build + main build (mandatory multi-layer review) (User mandate, 2026-06-04)

Forensic anchor — verbatim user mandate (2026-06-04):

“After all fixes/changes/implementations are done, BEFORE running pre-build tests and the main build, dispatch code-review agent(s) that analyze all work done + all existing data/facts + the existing codebase + current git history to determine quality, safety, and whether the fixes/changes will REALLY work; they MUST validate and verify that every test covering the fixes/changes genuinely validates the work with NO chance of false results or bluff of any kind. Any finding MUST be fixed, polished, improved, and covered with additional tests before the build proceeds. Multiple strong layers of checks.”

After all fixes / changes / implementations in a batch are done, and BEFORE running the pre-build test sweep AND the main (artifact) build (for ANY project), the agent MUST dispatch one or more dedicated code-review agent(s) (subagent-driven by default per §11.4.70/§11.4.20) performing a multi-layer review that: (1) analyzes ALL work done in the batch (every fix/change + its source diff + stated intent); (2) analyzes ALL existing data + facts (captured evidence per §11.4.5/§11.4.69/§11.4.107, tracker entries, prior findings, the §11.4.108 runtime-signature registry); (3) analyzes the existing codebase (blast radius per §11.4.92, cross-feature interaction, contract integrity of every dependency); (4) analyzes current git history (what each change touched, how it composes with concurrent/recent work, whether it reproduces a known-broken pattern per §11.4.114/§11.4.124); (5) determines quality + safety + will-it-REALLY-work (robust + not error-prone — no solve-A-create-B; no host/data/security regression; genuinely delivers the end-user-visible behaviour per §11.4/§107); (6) validates + verifies the tests covering the work — every covering test genuinely exercises the work-under-test and catches its negation, with ZERO chance of a false result or bluff (a test that PASSES on broken-for-the-user work, a metadata-only/config-only/absence-of-error/grep-without-runtime assertion, or a gate whose paired §1.1 mutation does not make it FAIL is a finding). Any finding (defect / error-prone change / safety risk / will-not-really-work / bluff-or-false-result-capable test / missing-coverage gap) MUST be fixed, polished,

improved, and covered with additional tests (four-layer per §11.4.4(b), TDD-RED-first per §11.4.43/§11.4.115) BEFORE the pre-build sweep + main build proceed; the review iterates (re-review after each remediation) until no blocking findings remain. The review is itself anti-bluff (its conclusions are captured evidence per §11.4.5/§11.4.69; a rubber-stamp review of a defective batch = PASS-bluff). It is one of MULTIPLE STRONG LAYERS — complementing, never replacing, the §1 pre-build sweep, §11.4.92 multi-pass (author-side self-review; §11.4.125 adds the structurally-separated reviewer seam per §11.4.70), §11.4.108 four-layer fix-verification, §11.4.110 build-readiness verdict, and the post-build / runtime-on-clean-target / user-visible layers. Composes §11.4 / §11.4.1 / §11.4.4 / §11.4.6 / §11.4.40 / §11.4.43 / §11.4.50 / §11.4.70 / §11.4.20 / §11.4.92 / §11.4.102 / §11.4.107 / §11.4.108 / §11.4.110. Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-125-PROPAGATION (literal 11.4.125) + recommended gate CM-CODE-REVIEW-GATE-BEFORE-BUILD (build starts only with a fresh code-review-completed marker for the current batch, produced after the last fix + before the pre-build sweep + main build) + paired §1.1 mutation (gate-code = separate work item).

Canonical authority: constitution submodule Constitution.md §11.4.125. Non-compliance is a release blocker. No escape hatch — no `--skip-code-review`, `--build-without-review`, `--no-review-gate`, `--review-optional`, `--trust-the-author` flag.

§11.4.126 — Default autonomous-loop working mode from first prompt (User mandate, 2026-06-04)

Forensic anchor — verbatim user mandate (2026-06-04):

“Make sure that you continue work in endless fully autonomous loop, do not stop until new fully validated and verified version (tag) is created and published (all submodules and main repo) or IN A CASE OF some other main stream work until it is fully completed with all side work streams and nothing else is left in our working queue! THIS MUST BE ALWAYS the default working mode without us asking you! We tend to achieve ABSOLUTE EFFICIENCY, with this and all other projects which will incorporate this MANDATORY RULE / CONSTRAINT!!! This way of (your) working will be ALWAYS applied / followed / executed / fully respected, as soon as we assign / send first request (prompt) in the session! This stops only if we explicitly say so or nothing is left to be done in current working scope (release that will come / upcoming version)!!! Any mimicking (imitation) of this behavior / rules / mandatory constraints, false results or any kind of bluff(s) is ABSOLUTELY FORBIDDEN!!!”

The endless fully-autonomous loop is the **DEFAULT working mode**, engaged automatically the moment the operator sends the FIRST request / prompt of a session — the operator MUST NOT have to ask for it, request it, restate it, or re-enable it per session. §11.4.87 framed the endless-loop covenant as an explicit-instruction opt-in (“continue in endless loop fully autonomously” or a semantically-equivalent phrasing); §11.4.126 is the **capstone** that promotes the same covenant to always-on: from the first prompt onward, every agent operates in the §11.4.87 loop discipline as the standing default, with §11.4.94 zero-idle, §11.4.97 maximum-idle-use, §11.4.101 autonomous-decision-over-blocking, and §11.4.103 continuous-parallel-stream all engaged by default — no per-session activation handshake. The continuation contract: the loop continues until ONE of two terminal conditions holds

— (A) **Release scope** — a new, fully-validated-and-verified version (tag) is created AND published across all owned submodules AND the main repo to all configured remotes (per §2.1 multi-upstream push + §11.4.40 full-suite-retest-before-tag + §11.4.113 absolute-no-force-push merge-onto-latest-main); OR (B) **Non-release main-stream scope** — the main-stream goal is fully completed AND every side work stream is done AND the working queue holds nothing left for the current scope. Until (A) or (B) holds, the agent **MUST** keep working (claim the next priority item, dispatch the next parallel stream, progress every non-blocked item per §11.4.42 / §11.4.72 / §11.4.94 / §11.4.103). The loop **STOPS ONLY** on: (1) the operator explicitly saying so (STOP / pause / end); (2) nothing left to do in the current working scope — the upcoming release / current main-stream goal — with the queue genuinely empty per the (A)/(B) terminal conditions; (3) a §12 host-session-safety demand (the loop yields to host safety unconditionally). Idle-while-blocked parks one work unit, it does not stop the loop — the agent keeps progressing every non-blocked item in parallel per §11.4.101 + §11.4.94 + §11.4.97. Goal — **ABSOLUTE EFFICIENCY** (no operator-side restart overhead, no idle gaps, no stop-and-wait round-trips); applies to this project AND every project that incorporates this Constitution. Anti-bluff: mimicking / imitating this loop behaviour, narrating continuation without performing it, fabricating progress, or emitting false / bluff results of ANY kind is **ABSOLUTELY FORBIDDEN** — this composes the entire §11.4 anti-bluff covenant family (§11.4 / §11.4.1 / §11.4.2 / §11.4.5 / §11.4.6 / §11.4.50 / §11.4.69 / §11.4.107); the agent **MUST** genuinely perform the continuous work and capture positive evidence for every closure, and a report claiming the loop ran while no real work / no captured evidence was produced is a §11.4 PASS-bluff at the operating-mode layer. Classification: universal (§11.4.17). Composes with §11.4.87 (the endless-loop covenant — §11.4.126 promotes it from opt-in to always-on default) / §11.4.94 / §11.4.97 / §11.4.101 / §11.4.103 / §11.4.66 / §11.4.6 / §11.4.40 / §11.4.42 / §11.4.72 / §11.4.113 / §2.1 / §12. Propagation gate CM-COVENANT-114-126-PROPAGATION (literal 11.4.126 across the consumer fleet) + paired §1.1 meta-test mutation (strip the literal → propagation gate FAILs; gate-code = separate work item).

Canonical authority: constitution submodule Constitution.md §11.4.126. Non-compliance is a release blocker. No escape hatch — no --ask-before-continuing, --single-turn-only, --not-default-loop, --mimic-OK flag.

§11.4.127 — Session-handoff resumption-prompt mandate (User mandate, 2026-06-06)

Forensic anchor — verbatim user mandate (2026-06-06): “make sure that in situations like this now when new session is needed you **ALWAYS** prepera such sentence - which will be valid for particular moment and the phase of the project and enough for work to continue.”

When the agent determines a fresh session is needed (context-window limits, performance degradation) OR the operator asks whether a new session is needed / requests a handoff, the agent **MUST ALWAYS** prepare + proactively provide a ready-to-paste **resumption prompt valid for that EXACT moment and project phase** — self-contained enough that pasting it into a fresh session resumes work with **ZERO** loss. Two variants on demand: a **SHORT** first-sentence (“Read <handoff docs>, then continue <terminal goal> ...”) AND a **FULL** detailed block. The prompt **MUST**: (1) point to the live handoff doc(s) — .remember / remember.md if present + docs/CONTINUATION.md per §12.10 — read **FIRST** + git fetch --all; (2) state current **PHASE** + immediate **NEXT** action + terminal

goal; (3) embed exact live-state anchors (build IDs / artifact MD5, device/target serials, commit HEAD, in-flight PIDs + log paths, captured-evidence paths); (4) restate binding constraints (anti-bluff §11.4, no-force-push §11.4.113, exact version/naming, hardware/target gotchas); (5) be MOMENT-VALID, NEVER a generic template. Handoff doc(s) MUST be current BEFORE the prompt is given (§12.10). A missing / stale / generic prompt is a §11.4.127 violation. Composes §12.10 / §11.4.6 / §11.4.66 / §11.4.87 / §11.4.103 / §11.4.126. Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-127-PROPAGATION (literal 11.4.127) + paired §1.1 meta-test mutation.

Canonical authority: constitution submodule Constitution.md §11.4.127. Non-compliance is a release blocker. No escape hatch — no `--skip-handoff-prompt`, `--generic-prompt-OK`, `--no-resumption-sentence`, `--handoff-without-state` flag.

§11.4.128 — Always-on device-recording mandate (User mandate, 2026-06-06)

Forensic anchor — direct user mandate (2026-06-06): we MUST ALWAYS live-record all available data from all devices we use for testing (or known to be under manual testing), EXTRA carefully so it never harms the device / its performance / causes side effects; raw recordings are NOT processed without need (token-conscious) and are ALWAYS git-ignored + code-intelligence-excluded; only curated evidence is committed, and only at release prep.

For EVERY test/debug device the project uses + every device under known manual testing, across EVERY reachable transport (USB / wireless ADB / SSH / serial / network introspection API), the project MUST ALWAYS live-record all analysable data: activities, all logs, performance metrics (CPU/memory/I/O/thermal/load), every sink-side report per §11.4.13, and any other live-changeable parameter. (1) **Extra-careful, side-effect-free** — non-invasive read-only probes only, bounded sampling, bounded write-volume, an observer-effect budget; a recorder that perturbs the device-under-test is a §11.4.128 violation, NOT evidence. (2) **Background + parallel + subagent-driven** per §11.4.103 + §11.4.70 — never blocks the main stream. (3) **Token-conscious — record-now, analyse-later** — raw data NOT processed without need; the only standing analyse-trigger is release-tag prep (§11.4.40 / §11.4.42) OR explicit operator ask. (4) **Raw is git-ignored (with a §11.4.77 regen-mechanism declaration) AND code-intelligence-excluded (§11.4.78/§11.4.79)** — only CURATED evidence is committed, and only at release prep under docs/qa/<run-id>/ (§11.4.83). (5) **Deterministic layout** <recording-root>/YYYY-MM-DD/<combined main+submodules state hash>/<DEVICE>_<SERIAL>/recording_NNN/<files>. (6) **Anti-bluff** — a recorder claimed running but with no growing corpus is a §11.4 bluff; every curated finding traces to a real raw-corpus path; recorder health is itself captured evidence per §11.4.5/§11.4.69.

Composes §11.4.2 / §11.4.5 / §11.4.13 / §11.4.69 / §11.4.40 / §11.4.42 / §11.4.70 / §11.4.77 / §11.4.78 / §11.4.79 / §11.4.83 / §11.4.103 / §11.4.119. Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-128-PROPAGATION (literal 11.4.128) + recommended gate CM-DEVICE-RECORDING-ALWAYS-ON + paired §1.1 mutation.

Canonical authority: constitution submodule Constitution.md §11.4.128. Non-compliance is a release blocker. No escape hatch — no `--skip-recording`, `--record-without-layout`, `--commit-raw-corpus`, `--index-raw-corpus`, `--analyse-corpus-always`, `--invasive-probe-OK` flag.

§11.4.129 — Huge-blocker release protocol (User mandate, 2026-06-06)

Forensic anchor — direct user mandate (2026-06-06): when a huge blocker is discovered during release validation we **MUST** stop all testing, fix ALL discovered issues, process all recorded data from the last session, land rock-solid fixes, author NEW validation+verification tests of ALL supported test types, rebuild, reflash, and RESTART the full validation+verification of every fix/change from the last release tag to now — on both devices in parallel, recorded, with real physical captured proofs and no bluff.

On discovery of a HUGE BLOCKER (release-blocking-severity defect: core user-facing capability broken, regression invalidating the in-flight cycle, or blast radius reaching the batch's other fixes) during release validation, execute in order with NO spot-check shortcut: (1) **STOP all testing** on every device (the §11.4.4 test-interrupt STOP at release granularity — continuing past a huge blocker is the §11.4 PASS-bluff). (2) **Fix ALL discovered issues** — not just the blocker; root-cause each per §11.4.102 + isolate regressions against the last known-good tag per §11.4.114. (3) **Process all recorded data from the last session** — analyse the §11.4.128 raw-corpus slice (this IS the §11.4.128(3) release-prep analyse-trigger). (4) **Land rock-solid fixes** per §11.4.123 + §11.4.43/§11.4.115 + §11.4.9. (5) **Author NEW validation+verification tests of ALL supported test types** per §11.4.27 + §11.4.85, each anti-bluff + paired §1.1 mutation. (6) **Rebuild (full, not module-only) + reflash to a CLEAN target** per §11.4.108. (7) **RESTART the full validation+verification from the last release tag to now** per §11.4.40 — RESTART, never resume — on both/all owned devices IN PARALLEL per §11.4.103/§11.4.119, every run RECORDED per §11.4.128, real physical captured proofs per §11.4.5/§11.4.69/§11.4.107, no bluff. This anchor BINDS the existing release anchors for the huge-blocker case (adds STOP→fix-all→process-recordings→new-tests-all-types→rebuild→reflash→full-restart + the restart-not-resume rule), citing them rather than duplicating.

Composes §11.4.4 / §11.4.40 / §11.4.42 / §11.4.9 / §11.4.27 / §11.4.85 / §11.4.102 / §11.4.108 / §11.4.114 / §11.4.115 / §11.4.123 / §11.4.128 / §11.4.103 / §11.4.119. Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-129-PROPAGATION (literal 11.4.129) + recommended gate CM-HUGE-BLOCKER-FULL-RESTART + paired §1.1 mutation.

Canonical authority: constitution submodule Constitution.md §11.4.129. Non-compliance is a release blocker. No escape hatch — no `--resume-after-blocker`, `--spot-validate-after-fix`, `--skip-recording-analysis`, `--skip-new-tests`, `--module-only-after-blocker`, `--single-device-restart` flag.

§11.4.130 — Post-remediation validate-the-fix-FIRST-after-redeploy (User mandate, 2026-06-06)

Forensic anchor — direct user mandate (2026-06-06): when a blocker discovered during release validation is fixed and a new artifact (rebuild / new flashing image / redeploy) is produced + the target reflashed, we **MUST** first re-test the **SPECIFIC** last-failing features + validate the just-incorporated fixes **BEFORE** the broader / full validation.

When a blocker / critical failure found during release validation is **FIXED** and a new artifact is produced + the target reflashed / redistributed / updated, the agent **MUST**:

- (1) **re-test the SPECIFIC last-failing features FIRST** (targeted guard tests for exactly the defects this fix addressed) **BEFORE** any broader / full-suite validation;
- (2) **validate the just-incorporated fixes with real captured evidence** — the §11.4.115 RED test flips GREEN at RED_MODE=0 on the new artifact **AND** the §11.4.108 runtime-signature verifies on the **CLEAN** target the redeploy produced (metadata-only / config-only / absence-of-error / grep-without-runtime **PASS** forbidden per §11.4 / §11.4.1; proof per §11.4.5/§11.4.69/§11.4.107/§11.4.123);
- (3) **only after the targeted fix is CONFIRMED working** proceed to the §11.4.40 full retest from the last tag to now. Rationale: a first fix attempt may not work / may be incomplete / may regress again under the new artifact — confirming the targeted fix **FIRST** catches a fix-did-not-take case immediately instead of hours later at the **END** of a full cycle (then restarting per §11.4.129); cheap-confirmation-first is §11.4.82 applied to the post-blocker reflash. This is the §11.4.46 recent-work-validation gate specialised for the post-blocker-reflash case + the targeted-confirmation phase that GATES §11.4.129's step-7 full-restart. Honest boundary (§11.4.6): “the fix probably took” ≠ “the fix took” — the RED→GREEN flip + runtime-signature on the new artifact is the proof; a still-FAILing targeted re-test re-enters the §11.4.114/§11.4.115 isolate→RED→fix loop, never proceeds to the full cycle on a still-broken fix. Composes §11.4.4 / §11.4.40 / §11.4.46 / §11.4.108 / §11.4.114 / §11.4.115 / §11.4.123 / §11.4.129 / §11.4.82. Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-130-PROPAGATION (literal 11.4.130) + recommended gate CM-FIX-FIRST-AFTER-REDEPLOY + paired §1.1 mutation.

Canonical authority: constitution submodule Constitution.md §11.4.130. Non-compliance is a release blocker. No escape hatch — no `--skip-targeted-retest`, `--full-cycle-first`, `--assume-fix-took`, `--validate-fix-at-end`, `--skip-red-green-flip-on-new-artifact` flag.

§11.4.131 — Standing session-resumption file mandate (User mandate, 2026-06-07)

Forensic anchor — verbatim user mandate (2026-06-07): “Make this markdown a standard file which will be written **EVERY TIME** when we need fresh session out of the box! It **MUST BE** always up to date and in sync so whenever new session is created all we have to do is just point to it!”

Every project **MUST** maintain a **SINGLE** canonical, always-current **session-resumption file** at a fixed, project-declared standard path (declared once per §11.4.35, never moved without a §11.4.66 operator decision). This file is the **OUT-OF-THE-BOX** entry point for any fresh session: creating a new session requires **ONLY** pointing the new agent at this one file. §11.4.131 promotes §11.4.127 (PREPARE a resumption prompt on demand) into a **STANDING**, version-controlled

ARTIFACT — ALWAYS present, ALWAYS in sync. (A) **Existence + fixed path** — exists at the declared path at all times, encoded as a literal path in the project-layer instantiation (§11.4.35), never silently moved. (B) **Always written + always synced** — (re)written whenever a fresh session is needed OR the live state materially changes (new HEAD, build/artifact id, phase, device/target state, in-flight job, blocking decision) — the §12.10 trigger set; a stale resumption file is a §11.4.131 violation of the same severity class as a §12.10 stale-CONTINUATION violation. (C) **Content (composes §11.4.127)** — both SHORT + FULL variants; points to `.remember/remember.md + docs/CONTINUATION.md` read FIRST + `git fetch`; embeds exact live-state anchors (HEAD, build/artifact ids + checksums, device serials, in-flight PIDs + log paths, captured-evidence paths); states PHASE + immediate NEXT + terminal goal; restates binding constraints (anti-bluff §11.4, no-force-push §11.4.113, exact version/naming, hardware gotchas); MOMENT-VALID, never a generic template (§11.4.6). (D) **Export + freshness** — §11.4.65 scope (synchronized `.html/.pdf` siblings) + §11.4.44 revision header. (E) **Out-of-the-box resumption** — a fresh session, given ONLY this file's path, fully resumes with zero additional context. Composes §12.10 / §11.4.127 / §11.4.65 / §11.4.44 / §11.4.6 / §11.4.66 / §11.4.126. Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-131-PROPAGATION (literal 11.4.131) + recommended gate CM-SESSION-RESUMPTION-FILE-PRESENT + paired §1.1 meta-test mutation.

Canonical authority: constitution submodule `Constitution.md` §11.4.131. Non-compliance is a release blocker. No escape hatch — no `--skip-resumption-file`, `--ephemeral-prompt-only`, `--stale-resumption-OK`, `--generic-template-OK` flag.

§11.4.132 — Risk-ordered validation priority mandate (User mandate, 2026-06-07)

Forensic anchor — verbatim user mandate (2026-06-07): “We MUST ALWAYS first test and validate features, functionalities and fixes/changes that have been worked most recently, the ones which were most problematic, which have the most chance to crash or break again, the ones which have been re-opened the most times! Then, after we validate and verify all this with real (physical) proofs and hard evidence, with no false results and bluffs of any kind, we continue with all other existing tests in the test suites! This IS MANDATORY.”

Tests / validations / verifications MUST run in **RISK-DESCENDING order** — the highest-risk set FIRST, and ONLY AFTER that set is fully GREEN with real (physical) captured evidence does the remainder of the suite run. Risk ranking is computed from a CLOSED set of factors, highest-risk first: (a) **most-recently-worked** features / fixes / changes; (b) **historically most-problematic** (longest defect history, most prior fixes/failures); (c) **highest crash/break/regress likelihood** (greatest blast radius / complexity / dependency surface); (d) **most-reopened** per §11.4.55 reopens-count (a high reopen count is the strongest empirical fragility signal). Each item in the highest-risk set MUST pass with real (physical) captured evidence per §11.4.5/§11.4.69/§11.4.107 — no metadata-only / config-only / absence-of-error / grep-without-runtime PASS (§11.4/§11.4.1), no false results, no bluff (§11.4.6). ONLY AFTER the entire highest-risk set is GREEN with captured proof does the rest of the suite run; running the suite in arbitrary order, or running lower-risk tests before the highest-risk set is GREEN, is a §11.4.132 violation. §11.4.132 REFINES/STRENGTHENS §11.4.130 (generalises “validate the just-fixed items first” to the full risk-ordered set) + §11.4.46 (adds explicit risk-ordering within the recent/high-risk set) + §11.4.42 (applies the implementation-layer priority

discipline to VALIDATION ordering). Classification: universal (§11.4.17) — the consuming project supplies its recency / problematic-history / reopen-count sources (e.g. §11.4.93 workable-items DB reopens_count+last_modified) per §11.4.35. Composes §11.4.4/.5/.6/.7/.40/.42/.46/.50/.55/.69/.107/.130. Propagation gate CM-COVENANT-114-132-PROPAGATION (literal 11.4.132) + recommended gate CM-RISK-ORDERED-VALIDATION-PRIORITY + paired §1.1 meta-test mutation.

Canonical authority: constitution submodule Constitution.md §11.4.132. Non-compliance is a release blocker. No escape hatch — no --skip-risk-ordering, --any-order-OK, --suite-order-fixed flag.

§11.4.133 — Target-System + hardware safety mandate (User mandate, 2026-06-08)

Forensic anchor — verbatim user mandate (2026-06-08): “Make sure that all changes we do to the System are ALWAYS safe for the System itself and for the hardware the system runs on! This is MANDATORY.”

Every change to the TARGET system (firmware, kernel, init/boot scripts, drivers, sysfs/devfreq/voltage/clock/thermal/regulator register writes, partition/bootloader/U-Boot, HAL, framework, prebuilts, device config) MUST ALWAYS be safe for BOTH (a) the target System itself — MUST NOT brick, boot-loop, corrupt data, or render the device unrecoverable — AND (b) the hardware it runs on — MUST NOT exceed safe electrical/thermal/voltage/clock limits or damage panels/storage/radios/regulators. Concrete obligations: (1) reversible-first — verify irreversible high-blast-radius changes (bootloader/U-Boot MD5, partition layout) against known-good values + capture a pre-op backup (§9.2) BEFORE applying; (2) NO unverified hardware-control writes — never write an unverified value to a voltage/clock/regulator/thermal-throttle/current-limit sysfs node or register that could exceed datasheet limits, the safe range established as FACT (§11.4.6), never guessed; (3) thermal/perf changes (forcing a performance governor, pinning the top OPP, disabling thermal management) MUST respect the device’s cooling design, validated by captured thermal evidence; (4) flashing MUST use the sanctioned tool + a freshly-built integrity-verified image — never an ad-hoc partition write or stale/unverified artifact; (5) unprovable-safety ⇒ blocked — a change whose target/hardware safety cannot be established from captured evidence is treated as UNSAFE and blocked (§11.4.6 + §11.4.101 reversible-first + §11.4.123 rock-solid-proof). DISTINCT from §12 host-session safety: §12 protects the DEVELOPER’s HOST + session; §11.4.133 protects the TARGET device + its hardware — both apply, neither weakens the other. Classification: universal (§11.4.17) — the consuming project supplies its concrete hardware-control surfaces, datasheet-safe ranges, known-good bootloader/image hashes, and sanctioned flashing tool per §11.4.35. Composes §12 / §11.4.6 / §11.4.101 / §11.4.108 / §11.4.123. Propagation gate CM-COVENANT-114-133-PROPAGATION (literal 11.4.133) + recommended gate CM-TARGET-HARDWARE-SAFETY + paired §1.1 meta-test mutation.

Canonical authority: constitution submodule Constitution.md §11.4.133. Non-compliance is a release blocker. No escape hatch — no --unsafe-hardware-write, --skip-system-safety, --brick-risk-accepted flag.

§11.4.134 — Code-review iterate-until-GO + rock-solid-evidence mandate (User mandate, 2026-06-08)

Forensic anchor — verbatim user mandate (2026-06-08): “For any fixes/changes given back to us for re-work by the code-review process, once we fix/improve everything per the code-review’s requests, we **MUST RE-RUN** code-review **AGAIN** until we get a **GO** from it with **NO** new issues reported or warnings of any kind! All results produced by this whole process **MUST ALWAYS** give us rock-solid **PHYSICAL** evidence that the fixed/improved codebase really works now as expected, with no false results and no bluff(s) of any kind.”

When the §11.4.125 code-review returns ANY finding — **BLOCKING**, nit, or warning — and the author fixes/improves the batch per that review, the code review **MUST BE RE-RUN**, and **MUST KEEP** being re-run after each remediation round, until it returns a clean **GO** with **ZERO** new issues **AND ZERO** warnings of any kind. A single pass that “addressed the findings” is **NOT** sufficient: the corrected batch **MUST** pass a **FRESH** adversarial review (a re-review can surface **NEW** findings introduced by the very fixes that closed the prior ones — the §11.4.1 fix-A-creates-B failure mode). The loop terminates **ONLY** on a clean **GO** (no new findings, no warnings); a residual warning is itself a finding that re-arms the loop. Every round’s verdict **AND** every fix’s validation **MUST** carry rock-solid **PHYSICAL** captured evidence per §11.4.5 / §11.4.69 / §11.4.107 (captured audio / video / sysfs / dumpsys / sink-side / runtime-signature) proving the fixed/improved codebase **REALLY** works as expected — never metadata-only / configuration-only / absence-of-error / grep-without-runtime; no false results, no bluff at any round; a reported **GO** unbacked by captured physical evidence is itself a §11.4 **PASS-bluff** at the review-loop layer. §11.4.134 **REFINES / STRENGTHENS** §11.4.125 (iterate “until no blocking findings remain”): it makes the loop **EXPLICIT** (re-run after every remediation round, not once), raises termination to **ZERO** findings **AND ZERO** warnings (not merely zero-blocking), and **BINDS** rock-solid physical evidence to every round. Classification: universal (§11.4.17). Composes §11.4.125 / §11.4.1 / §11.4.4 / §11.4.5 / §11.4.6 / §11.4.69 / §11.4.107 / §11.4.50 / §11.4.108 / §11.4.123. Propagation gate **CM-COVENANT-114-134-PROPAGATION** (literal **11.4.134**) + recommended gate **CM-CODE-REVIEW-ITERATE-UNTIL-GO** + paired §1.1 meta-test mutation (gate-code = separate work item).

Canonical authority: constitution submodule Constitution.md §11.4.134. Non-compliance is a release blocker. No escape hatch — no **--skip-rereview**, **--single-review-pass**, **--warnings-ok**, **--evidence-optional** flag.

§11.4.135 — Standing regression-guard suite + every-fixed-defect-gets-a-permanent-regression-test (User mandate, 2026-06-08). Every project **MUST** maintain a **STANDING** regression-guard suite that runs on **EVERY** build+deploy and **BLOCKS** the release tag on any failure. Every closed defect (stable ticket id, e.g. ATM-NNN) **MUST**, in the **SAME** commit as its fix (extending the §11.4.43 **DOCUMENT** step), register a permanent §11.4.115 **RED-on-broken-artifact** regression test into the suite — **RED_MODE=1** capturing the historical defect on a pre-fix artifact (the proof the guard is real), **RED_MODE=0** the standing **GREEN** guard asserting the defect is **ABSENT**. A closure without a registered guard is a §11.4.123 violation. The suite runs **FIRST** in the post-deploy cycle (highest-risk set per §11.4.132) and is a §11.4.40 release-gate blocker. Forensic anchor (**FACT**): the wrong-subtitle-on-2nd-display defect was “fixed” via a source-side **CONTROL_MENU_LABEL_DENYLIST** that **NO** test mirrored or re-ran, so the **NEXT** chrome class recurred silently while the **GREEN** suite passed. Industry-standard

bug-driven testing (Google content-driven testing; AOSP CTS/Tradefed) made mechanical + enforced. Composes §11.4.4 / §11.4.40 / §11.4.43 / §11.4.46 / §11.4.50 / §11.4.107 / §11.4.108 / §11.4.115 / §11.4.118 / §11.4.123 / §11.4.124 / §11.4.130 / §11.4.132. Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-135-PROPAGATION (literal 11.4.135) + recommended gates CM-REGRESSION-GUARD-REGISTERED / CM-REGRESSION-GUARD-SUITE-WIRED + paired §1.1 mutation. **Canonical authority:** constitution submodule Constitution.md §11.4.135. Non-compliance is a release blocker. No escape hatch — no --skip-regression-guard, --no-guard-on-close, --guard-optional flag.

§11.4.136 — Real-content end-to-end playback-test mandate (User mandate, 2026-06-08). Refines/strengthens §11.4.107. Any test asserting media playback works MUST drive REAL content (catalog stream or offline reference clip) through the user's path (§11.4.48 UI-driven → §11.4.117 CV/OCR fallback) and assert it genuinely PLAYS via the §11.4.107 liveness battery PLUS a decoder-health census — a numeric drop-buffer budget, no buffer-timestamp re-order/discard, no codec-reject (cite Android/Media3 ExoPlayer OEM pre-OTA playback-test mandate: “too many dropped buffers” >25, “unexpected presentation timestamp”, “test timed out”). Metadata-only / launch-only / registration-only / single-frame / config-only PASS is forbidden (§11.4 / §11.4.1). A golden/reference clip corpus (BBC ExoPlayer testing samples) is the offline ground-truth. Composes §11.4.5 / §11.4.48 / §11.4.50 / §11.4.107 / §11.4.117 / §11.4.123 / §11.4.13 / §11.4.69. Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-136-PROPAGATION (literal 11.4.136) + recommended gate CM-REAL-CONTENT-PLAYBACK-TEST + paired §1.1 mutation. **Canonical authority:** constitution submodule Constitution.md §11.4.136. Non-compliance is a release blocker. No escape hatch — no --launch-proves-playback, --skip-decoder-health, --metadata-playback-pass-suffices flag.

§11.4.137 — Subtitle/caption content-correctness oracle + secure-display-proxy-honesty mandate (User mandate, 2026-06-08). Refines §11.4.117 + §11.4.107 + §11.4.112. Forensic anchor (FACT): tests tasked to “physically verify the 2nd-display subtitle” PASSED GREEN while subtitles did NOT show / showed WRONG — the streaming player surface is FLAG_SECURE so screencap -d <secondary> returns BLACK (autonomous PIXEL verification structurally impossible per §11.4.112), so the test fell back to the accessibility-scraped/persist.atmosphere.subdebug proxy, and the proxy accepted a chrome/menu LABEL (Аудио и субтитры) as a valid subtitle because the prose floor accepted any multibyte prose and NO menu-label denylist + NO position/cadence check existed. The mandate: a subtitle-correctness test MUST classify the cue's *content class* — a present cue is NOT a correct cue. CHROME (FAIL) if a known control/menu label (closed multilingual deny-list MIRRORED from source, case-folded incl. non-ASCII), time/numeric chrome, not prose, OUTSIDE the lower safe-title band (CEA-708 9-anchor grid), OR STATIC across the window (real subtitle changes → ≥2 distinct prose cues, a metamorphic relation). DIALOGUE (PASS) only when prose + not-denied + not-chrome + position-ok + cadence ≥2 OR fuzzy-matches the SOURCE-extracted cue via normalized edit distance (§11.4.123 host ground truth). The oracle MUST be self-validated golden-good/golden-bad (§11.4.107(10)) and the deny-list MUST be verified present in the SHIPPED artifact (§11.4.108) — a source-green denylist with no test mirror + no artifact check is the exact recurrence pattern forbidden here. Secure-display honesty (§11.4.112): where FLAG_SECURE makes pixel verification impossible, the rock-solid autonomous proof is the player's caption telemetry + source-track presence + content-class oracle — NEVER a faked pixel “physical” pass; human-eye pixel confirmation is

operator_attended (§11.4.52) with a tracked migration item. App-agnostic (keys off content class). Composes §11.4.3 / §11.4.5 / §11.4.6 / §11.4.107 / §11.4.108 / §11.4.112 / §11.4.115 / §11.4.117 / §11.4.123 / §11.4.13 / §11.4.69. Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-137-PROPAGATION (literal 11.4.137) + recommended gate CM-SUBTITLE-CONTENT-CORRECTNESS-ORACLE + paired §1.1 mutation (strip the denylist/position/cadence check → golden-bad Аудио и субтитры PASSes → gate FAILs). **Canonical authority:** constitution submodule Constitution.md §11.4.137. Non-compliance is a release blocker. No escape hatch — no --present-cue-is-correct, --skip-chrome-oracle, --length-heuristic-suffices, --pixel-pass-on-secure-display, --skip-position-check, --skip-cadence-check flag.

§11.4.138 — Operator-escape => mandatory bluff-audit + permanent guard (User mandate, 2026-06-08). When the operator (or any out-of-band channel) finds a defect that the GREEN test suite passed, this is by definition a §11.4 PASS-bluff — it MUST trigger, before the fix is closed: (1) a §11.4.102 systematic-debugging pass to FACT-root-cause; (2) a bluff-audit identifying the EXACT assertion that should have caught it but didn't, cited to file:line (canonical example: lib/subtitle_content_validation.sh:sub_is_prose() returning TRUE for Аудио и субтитры); (3) a permanent §11.4.135 regression guard registered in the SAME commit as the fix, with its §11.4.115 RED capturing the operator-found defect; (4) the bluff-audit committed under docs/research/<scope>/<defect>_bluff_audit/. Closing an operator-found defect WITHOUT the bluff-audit + permanent guard is itself a §11.4 violation (the bluff that let it through is still live and the defect will recur). Composes §11.4 / §11.4.1 / §11.4.102 / §11.4.108 / §11.4.115 / §11.4.118 / §11.4.123 / §11.4.135. Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-138-PROPAGATION (literal 11.4.138) + recommended gate CM-OPERATOR-ESCAPE-BLUFF-AUDIT + paired §1.1 mutation. **Canonical authority:** constitution submodule Constitution.md §11.4.138. Non-compliance is a release blocker. No escape hatch — no --close-without-bluff-audit, --operator-find-is-just-a-bug, --skip-permanent-guard flag.

§11.4.139 — Fresh-process clean-artifact runtime-signature mandate (User mandate, 2026-06-08). Refines §11.4.108. Before any post-deploy validation — ESPECIALLY a non-pixel proxy verification (the subdebug/accessibility-cue channel used for FLAG_SECURE displays) — the harness MUST assert running-artifact == built-artifact: the deploy yielded a CLEAN target (mutable-overlay/userdata wiped) OR a pre-validation check proves no stale overlay shadows the deployed code (e.g. every guarded package — incl. the Presenter that emits the subtitle cue — resolves to the system partition, no per-user override). A stale shadow of the cue-emitting component (e.g. a Presenter APK predating the denylist) makes the proxy report on code that was never deployed — any PASS is a §11.4 PASS-bluff. Each fix declares ONE machine-checkable runtime signature verified on the clean target (the §11.4.108 registry IS the definition of done); for the subtitle class the signature is “the shipped Presenter APK contains the denylist literal (case-insensitive) AND the subdebug channel emits candidate REJECTED reason=chrome-label for a menu label.” Composes §11.4.46 / §11.4.108 / §11.4.130 / §11.4.135 / §11.4.137. Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-139-PROPAGATION (literal 11.4.139) + recommended gate CM-CLEAN-ARTIFACT-RUNTIME-SIGNATURE + paired §1.1 mutation. **Canonical authority:** constitution submodule Constitution.md §11.4.139. Non-compliance is a release blocker. No escape hatch — no --validate-against-running-state, --skip-clean-precondition, --shadow-OK flag.

§11.4.140 & §11.4.141 — action-prefix system + token-efficiency (cascaded from constitution submodule 60e2d66, CONST-047/049)

§11.4.140 — **Universal action-prefix system (ACTION_NAME ::)** (User mandate, 2026-06-09; GRAMMAR_ADDENDUM 2026-06-09). When a user prompt's FIRST non-blank line starts with a recognised action prefix, you MUST: (1) look the action token up in the action registry constitution/actions/registry.yaml (or \$HELIX_ACTION_REGISTRY); (2) if it is a registered action, REPLACE the prefix with that action's expansion text and apply its rules; (3) execute the remainder of the prompt under the expanded instruction. **Four EQUIVALENT forms** — same action, same expansion, same execution: (1) ACTION_NAME :: <rest> (bare ::), (2) PREFIX::ACTION_NAME :: <rest> (namespaced ::), (3) / ACTION_NAME <rest> (bare slash), (4) /PREFIX::ACTION_NAME <rest> (namespaced slash). Thus BACKGROUND :: x ≡ DEFAULT::BACKGROUND :: x ≡ /BACKGROUND x ≡ /DEFAULT::BACKGROUND x. PREFIX is an action NAMESPACE; the reserved default namespace is **DEFAULT**, and an action runs WITH or WITHOUT the prefix. Grammar (all hold): anchored at the FIRST non-blank line only (mid-prose tokens never match); the action token AND the namespace are UPPERCASE-only [A-Z] [A-Z0-9_]* (lowercase never matches); the namespace separator :: inside the token carries NO surrounding spaces (PREFIX::ACTION_NAME), DISTINCT from the action-body separator " :: " (one ASCII space on each side of :: — avoids C++ Foo::Bar, YAML key: value, URLs) in forms 1/2 and the slash-body separator (one space) in forms 3/4; stacked prefixes (A :: B :: rest) apply outer-to-inner, left-to-right (expand A, re-scan, expand B, then the residual is the task); a leading \ escapes the prefix for BOTH the :: and the slash form (\BACKGROUND :: x, \BACKGROUND x — treat literally, strip the backslash, NO expansion) so action names can be discussed. **Conflict rule (slash form):** /ACTION_NAME (form 3) is honored as the action ONLY when ACTION_NAME (case-folded) does not collide with a built-in/host slash command (registry slash_bare: auto + slash_conflicts: [.]); form 4 (/PREFIX::ACTION_NAME) is ALWAYS unambiguous and always honored. An unknown token that matches the grammar shape (any of the 4 forms) but is NOT registered is NEVER silently expanded or silently dropped — ask which registered action was meant (§11.4.66 / §11.4.105) or treat it as a literal prompt, NEVER invent an expansion (§11.4.6); any prompt not satisfying the grammar is an ordinary prompt and the system is a no-op. The registered action **BACKGROUND** expands to: *“The following prompt that we will provide MUST BE executed in background in parallel with all main work streams using the subagents-driven development approach! All work done MUST PRODUCE rock solid evidence covered with hard physical proof(s) that all done is working as expected and as specified without any false results and without any bluff!”* (composes §11.4.20 / §11.4.70 subagent-driven, §11.4.58 / §11.4.103 parallel streams, §11.4.89 background execution, §11.4.5 / §11.4.69 / §11.4.107 captured physical evidence, §11.4 anti-bluff). The system is UNIVERSAL (every CLI agent reads this block via its context carrier per §11.4.35), extensible (new action = new registry row), decoupled + reusable (§11.4.28), and loads out-of-the-box. Classification: universal (§11.4.17). **Canonical authority:** constitution submodule Constitution.md

§11.4.140. Non-compliance is a release blocker. No escape hatch — no `--skip-action-prefix`, `--ignore-prefix`, `--no-registry`, `--invent-expansion-OK`, `--single-layer-only` flag.

§11.4.141 — Token-efficiency mandate (research-derived + operator mandate, 2026-06-09). Every project worked on by AI coding agents **MUST** cut token spend (input AND output) toward **30–40% of current (a 60–70% reduction)** WITHOUT degrading quality/performance/safety or breaking any existing mechanism, via a composable, safety-ranked measure set: (1) **prompt-cache the static governance prefix** — the always-loaded governance forms a byte-stable cache-breakpointed prefix with no volatile bytes ahead of it; cache reads cost $\sim 0.1 \times$ base input (the dominant cost driver — measured $\sim 170\text{K}$ tokens of governance re-sent every turn, externally corroborated by Claude Code issue #24147); caching is transparent so it removes no rule, weakens no gate, changes no verdict — only billing (PRIMARY, biggest + safest lever); (2) **subagent model-tiering + output-to-file** — mechanical non-judgment work (search/grep/status/doc-export/read-only probes) to a Haiku-class model, the strong model RESERVED for all reasoning/verdicts/fix-design (§11.4.102)/code-review (§11.4.125)/demotion (§11.4.7), large output persisted to a file not an inline 350–520K-token transcript; the cheap model never emits a PASS so §11.4.50 + anti-bluff are untouched; (3) **thin always-loaded INDEX + on-demand detail** — concise index (one line per fix/anchor, EACH carrying the literal 11.4.N token so propagation gates pass) with the canonical full text kept gate-scanned in constitution/Constitution.md and reachable in one hop — a de-duplication realising §11.4.35, never a deletion; (4) **CodeGraph/retrieval-first over full-file loading** (§11.4.78/§11.4.79); (5) **output-token reduction** — terse status + effort: "low" on the mechanical allowlist only; (6) **tool-call batching + no re-reads**; (7) **compaction/context-editing for long sessions. Mandatory measured proof:** a token-accounting harness measures tokens-per-development-cycle BEFORE vs AFTER on a frozen deterministic workload from the authoritative usage object (input/cache_read/cache_creation/output split; NEVER tiktoken, NEVER the client-side cost estimate), reproduced N times (§11.4.50), $\text{pass} = \text{AFTER} \leq 40\%$ of BEFORE OR the measured best-safe reduction with a cited cold-cache reason; the AFTER run **MUST** show ZERO regression on the pre-build sweep + meta-test mutation sweep + propagation gates + a strong-model reasoning probe + a cache-warm proof (`cache_read_input_tokens > 0`) — cost reduction with quality regression is a §11.4 FAIL. The headline number is the *measured* reduction, never the design estimate (§11.4.6/§11.4.123). No measure may break/degrade any existing mechanism, and the rule is structured so none can. Composes §11.4.5/.6/.20/.40/.50/.58/.69/.70/.78/.79/.80/.103/.106/.123/.125/.128/§12.6/§1.1. Classification: universal (§11.4.17). Propagation gate CM-COVENANT-114-141-PROPAGATION (literal 11.4.141) + recommended gate CM-TOKEN-EFFICIENCY + paired §1.1 mutation (inject a pre-breakpoint volatile token → cache collapses → measured reduction falls below bar → gate FAILs). **Canonical authority:** constitution submodule Constitution.md §11.4.141. Non-compliance is a release blocker. No escape hatch — no `--skip-token-efficiency`, `--no-cache-governance`, `--assert-reduction-without-measuring`, `--tier-down-reasoning`, `--inline-all-governance`, `--tiktoken-estimate-OK` flag.

§11.4.104 — Participant identity, attribution & notification-tagging (User mandate, 2026-05-31)

Cascade reference — see constitution submodule `Constitution.md` §11.4.104 for the full mandate. Every consumer that ships a messenger/notification surface **MUST** model participants as logical subscribers with per-channel aliases, carry `created_by` + `assigned_to` on workable items, apply the tagging matrix (tag assigned-to/created-by humans who are not the Operator and not the system agent Claude), and anti-bluff the full implementation per §11.4. Propagation gate CM-COVENANT-114-104-PROPAGATION (literal 11.4.104). Non-compliance is a release blocker.

Canonical authority: constitution submodule `Constitution.md` §11.4.104.

§11.4.106 — Docs Chain — mechanical documentation/DB sync engine (Operator mandate, 2026-05-31)

Cascade reference — see constitution submodule `Constitution.md` §11.4.106 for the full mandate. Every consumer **MUST** use the `vasic-digital/docs_chain` engine (inherited by reference, never copied) as the canonical mechanical enforcer of all documentation-sync mandates; register chains via per-context YAML; never accept a faked transform. Ad-hoc sync scripts are superseded. Propagation gate CM-COVENANT-114-106-PROPAGATION (literal 11.4.106). Non-compliance is a release blocker.

Canonical authority: constitution submodule `Constitution.md` §11.4.106.

§11.4.109 — Mandatory Anti-Forgetting Enforcement: PreToolUse Guard Hook + Subagent Constitutional Preamble + Orchestrator Pre-Action Checklist (Operator mandate)

Cascade reference — see constitution submodule `Constitution.md` §11.4.109 for the full mandate. Every consumer **MUST** wire `constitution/scripts/hooks/guard-forbidden-commands.sh` as a `PreToolUse` hook, maintain `docs/AGENT_GUARDRAILS.md` with the subagent constitutional preamble and orchestrator pre-action checklist, and provide a hermetic hook test suite. The hook is inherited by reference, never copied. Propagation gate CM-COVENANT-114-109-PROPAGATION (literal 11.4.109). Non-compliance is a release blocker.

Canonical authority: constitution submodule `Constitution.md` §11.4.109.

§11.4.111 — Resolve-by-stable-name-not-by-enumeration-index mandate (research-derived, 2026-06-03)

Cascade reference — see constitution submodule `Constitution.md` §11.4.111 for the full mandate. Any binding to a hardware device / resource handle / enumerated entity MUST resolve by a stable identifier (name / UUID / serial / label) and MUST NOT bind by enumeration index / ordinal / slot unless the platform documents that ordinal as deterministically pinned and the pin is captured + asserted. Propagation gate `CM-COVENANT-114-111-PROPAGATION` (literal `11.4.111`). Non-compliance is a release blocker.

Canonical authority: constitution submodule `Constitution.md` §11.4.111.

§11.4.116 — Real-time conductor[?] autonomous-test-framework sync channel mandate (1.1.8-dev remediation, 2026-06-03)

Cascade reference — see constitution submodule `Constitution.md` §11.4.116 for the full mandate. Any autonomous long-running test/QA/validation framework MUST expose a structured append-only event stream and an atomically-rewritten status snapshot. Every verdict event MUST carry the evidence path that backs it. A PASS event with no evidence path is a channel-layer PASS-bluff. Propagation gate `CM-COVENANT-114-116-PROPAGATION` (literal `11.4.116`). Non-compliance is a release blocker.

Canonical authority: constitution submodule `Constitution.md` §11.4.116.

§11.4.120 — Fix-breaks-its-own-gate reconciliation mandate (1.1.8-dev remediation, 2026-06-03)

Cascade reference — see constitution submodule `Constitution.md` §11.4.120 for the full mandate. When a correct fix causes a pre-existing gate/test to FAIL because it asserted old behaviour, the gate MUST be reconciled (rewritten to assert the new mechanism + paired §1.1 mutation updated) — never fake-passed, weakened to a tautology, or deleted; and the correct fix MUST NOT be reverted to satisfy the stale gate. Propagation gate `CM-COVENANT-114-120-PROPAGATION` (literal `11.4.120`). Non-compliance is a release blocker.

Canonical authority: constitution submodule `Constitution.md` §11.4.120.

§11.4.121 — No-commit-while-build-writes-tracked-artifacts mandate (1.1.8-dev remediation, 2026-06-03)

Cascade reference — see constitution submodule `Constitution.md` §11.4.121 for the full mandate. A commit MUST NOT run while a build/packaging/generation step is actively writing artifacts into tracked (version-controlled) directories. The commit MUST be deferred until the writing step completes so the tree is quiescent and committed artifacts are the fresh, whole outputs. Propagation gate CM-COVENANT-114-121-PROPAGATION (literal 11.4.121). Non-compliance is a release blocker.

Canonical authority: constitution submodule `Constitution.md` §11.4.121.

§11.4.103 — Continuous parallel-stream working routine (User mandate, 2026-05-29)

Cascaded from constitution submodule §11.4.103. Promotes the multi-stream operating pattern into the project's standing default working routine. The main work stream MUST always stay FREE; ALL commit AND push operations run detached. At least three parallel background subagent streams MUST run at all times alongside the main stream whenever three-plus non-contending actionable items exist; the moment any stream finishes a new stream MUST immediately start. Most-critical + most-visible first; audio always top per §11.4.72. Safe-during-build scope only (§11.4.96 SAFE catalogue). Heavy anti-bluff on every closure. Idle ONLY when genuinely externally blocked OR operator STOP OR §12 host-safety.

Cascade requirement: This anchor (verbatim or by §11.4.103 reference) MUST appear in every owned submodule's `CONSTITUTION.md`, `CLAUDE.md`, and `AGENTS.md`. Propagation gate CM-COVENANT-114-103-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule `Constitution.md` §11.4.103 for the full mandate.

§11.4.105 — Natural-language intent recognition & clarification (User mandate, 2026-05-31)

Cascaded from constitution submodule §11.4.105. Users MUST NOT be required to know command syntax. Three-tier resolution: TIER 1 — recognize existing commands from natural language; TIER 2 — infer exact intent via LLM dispatch; TIER 3 — reply, tag sender (@username), and ask a precise clarifying question. Never guess, never drop a message silently; only genuine ambiguity reaches Tier 3, which always replies-tags-and-asks.

Cascade requirement: This anchor (verbatim or by §11.4.105 reference) MUST appear in every owned submodule's `CONSTITUTION.md`, `CLAUDE.md`, and `AGENTS.md`. Propagation gate CM-COVENANT-114-105-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule `Constitution.md` §11.4.105 for the full mandate.

§11.4.107 — Anti-bluff AV/test-validation techniques mandate (User-driven research, 2026-06-02)

Cascaded from constitution submodule §11.4.107. Every test asserting audio/video output is genuinely playing MUST satisfy: single captured frame NOT proof — prove LIVE ADVANCING frames via freeze-detection oracle; independent frame-advance counter from compositor/decoder telemetry; loading/buffering is a distinct state; not-stale-from-previous cross-check; measured FPS / no-lost-frames; no-flash-on-wrong-output; drive through realistic feed/UI path; metamorphic relations; full-reference quality metrics vs golden source; mutation-test every analyzer with golden-good + golden-bad fixture pair; per-channel audio RMS/loudness + XRUN census; OCR confidence floor + ROI; thresholds calibrated on project's own fixtures.

Cascade requirement: This anchor (verbatim or by §11.4.107 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-107-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.107 for the full mandate.

§11.4.108 — Four-layer fix-verification + runtime-signature-as-definition-of-done mandate (systematic-debugging Phase 4.5, 2026-06-03)

Cascaded from constitution submodule §11.4.108. A fix crosses FOUR distinct layers: (1) SOURCE, (2) ARTIFACT, (3) RUNTIME-ON-CLEAN-TARGET, (4) USER-VISIBLE. Green at layer 1 says nothing about layers 2–4. Every fix declares ONE machine-checkable runtime signature verified on a CLEAN/fresh deployment. Gates span all four layers. Deployment MUST yield a CLEAN state OR a pre-validation assertion proves running-artifact == built-artifact. On ≥3 “fixed-but-not-working” discoveries in one cycle: STOP patching symptoms, fix the VERIFICATION pipeline.

Cascade requirement: This anchor (verbatim or by §11.4.108 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-108-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.108 for the full mandate.

§11.4.110 — Pre-build build-readiness verdict + change-impact clash detection mandate (operator mandate, 2026-06-03)

Cascaded from constitution submodule §11.4.110. A single deterministic READY-FOR-BUILD verdict gates every rebuild. A diff-driven change-impact + clash detector cross-checks every newly-introduced second-artifact dependency (new property read ⇔ property-context type + read-grant; new service ⇔ service-context

entry; etc.). Coverage-completeness is a gate — every changed file maps to ≥ 1 gate + ≥ 1 deployed-target test + ≥ 1 paired §1.1 mutation. Two-speed honesty: grep-speed always-on gates vs REQUIRES_BUILD heavy gates as diff-gated opt-in stages.

Cascade requirement: This anchor (verbatim or by §11.4.110 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-110-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.110 for the full mandate.

§11.4.112 — Structural-impossibility won't-fix classification mandate (research-derived, 2026-06-03)

Cascaded from constitution submodule §11.4.112. When deep research per §11.4.8 PROVES a goal is structurally impossible on the target platform (forbidden by platform design / hardware-protocol constraint / documented kernel-or-API limitation), the goal MUST be: classified Won't-fix + closed per §11.4.90 with closure reason structurally-impossible; documented with impossibility evidence; NOT re-attempted without NEW evidence the platform constraint changed. structurally-impossible is reserved for PROVEN platform/hardware/protocol impossibility — “could not find a way” is Operator-blocked, not won't-fix.

Cascade requirement: This anchor (verbatim or by §11.4.112 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-112-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.112 for the full mandate.

§11.4.113 — Absolute no-force-push + merge-onto-latest-main mandate (User mandate, 2026-06-03)

Cascaded from constitution submodule §11.4.113. Force-push is STRICTLY FORBIDDEN with NO exception — `git push --force`, `--force-with-lease`, `<ref>`, or any history-rewriting overwrite of a remote ref, against EVERY repository. The mandated 6-step integration procedure: (1) `git fetch --all --prune --tags`; (2) set base to LATEST commit on canonical main/master; (3) carefully MERGE every change on top; (4) resolve every conflict carefully; (5) commit the merge (stage only intended files); (6) push to ALL upstreams as fast-forward. REMOVES the force-push escape hatch from §11.4.41/§11.4.71/§9.2/CONST-043.

Cascade requirement: This anchor (verbatim or by §11.4.113 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-113-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.113 for the full mandate.

§11.4.114 — Last-known-good-tag regression isolation mandate (1.1.8-dev remediation, 2026-06-03)

Cascaded from constitution submodule §11.4.114. When a previously-working feature is observed broken, the FIRST diagnostic action MUST be to identify the last release tag at which it was KNOWN-GOOD and diff/bisect the broken state against it — BEFORE any open-ended root-cause hunt or speculative fix. The known-good revision is the regression oracle. Default to a SURGICAL forward-fix (keep post-good-tag features, revert ONLY the broken sub-part) over a wholesale revert. “It worked before” is a HYPOTHESIS until the known-good tag is identified and confirmed.

Cascade requirement: This anchor (verbatim or by §11.4.114 reference) MUST appear in every owned submodule’s CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-114-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.114 for the full mandate.

§11.4.115 — RED-baseline-on-the-broken-artifact + polarity-switch mandate (1.1.8-dev remediation, 2026-06-03)

Cascaded from constitution submodule §11.4.115. Every RED test MUST be authored to REPRODUCE the defect on the CURRENT pre-fix artifact, capturing positive evidence that the defect is genuinely present. The SAME test source carries a single polarity switch (env flag RED_MODE, default 1 = reproduce-and-assert-defect-present; flipped to 0 post-fix = standing GREEN regression-guard). One source, two roles: the bug-catcher IS the regression-guard. A RED test that passes on the known-broken artifact is a blind test — a finding, not evidence.

Cascade requirement: This anchor (verbatim or by §11.4.115 reference) MUST appear in every owned submodule’s CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-115-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.115 for the full mandate.

§11.4.117 — Computer-vision / OCR pixel-oracle fallback for non-introspectable UIs mandate (1.1.8-dev remediation, 2026-06-03)

Cascaded from constitution submodule §11.4.117. Any test needing to drive a UI control OR assert on-screen content MUST NOT assume the accessibility/semantic/DOM hierarchy is the source of truth. When the hierarchy is blank/partial/known-unreliable, the test MUST fall back to a PIXEL ORACLE: drive input by computer-vision template-match; assert content by ROI OCR with per-word confidence floor + region-of-interest. The CV/OCR analyzer is self-validated — golden-good fixture PASSES, golden-bad fixture FAILs, wired into meta-test.

Cascade requirement: This anchor (verbatim or by §11.4.117 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-117-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.117 for the full mandate.

§11.4.118 — Discovery-pressure to confirm known-issue-set completeness mandate (1.1.8-dev remediation, 2026-06-03)

Cascaded from constitution submodule §11.4.118. A remediation/release cycle MUST NOT treat “every reported defect is fixed” as “the build is good.” After/ alongside fixing the reported set, the cycle MUST run a discovery + stress pass across ALL target devices/environments that deliberately exercises subsystems, journeys, and edge cases BEYOND the reported defects. The pass MUST produce an enumerated list of subsystems/user-journeys/stress scenarios actually exercised, each with its outcome. “We found no other issues” is a bluff unless accompanied by “here is the enumerated set we exercised.”

Cascade requirement: This anchor (verbatim or by §11.4.118 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-118-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.118 for the full mandate.

§11.4.119 — Single-resource-owner partitioning for parallel hardware testing mandate (1.1.8-dev remediation, 2026-06-03)

Cascaded from constitution submodule §11.4.119. When multiple parallel streams exercise SHARED hardware or any exclusive-access resource, exactly ONE stream MUST own each such resource at a time. The exclusive owner drives it; every other concurrent stream targeting the same resource MUST be READ-ONLY (passive probes only). Parallelism is partitioned by resource: distinct devices/sinks run fully concurrent, but the same device's exclusive resource is single-owner. Ownership enforced by advisory lock/token. Concurrent drivers of one exclusive resource produce cross-contaminated evidence — a PASS under contention is a §11.4 evidence-integrity bluff.

Cascade requirement: This anchor (verbatim or by §11.4.119 reference) MUST appear in every owned submodule's CONSTITUTION.md, CLAUDE.md, and AGENTS.md. Propagation gate CM-COVENANT-114-119-PROPAGATION; paired mutation strips the literal → gate FAILs. Release blocker. **Canonical authority:** constitution submodule Constitution.md §11.4.119 for the full mandate.